

Code Rundown & Pipeline Field Guide

A mentor-ready explanation of how the current PDF-to-knowledge-graph code works, how every production definition fits, and where to look when something goes wrong.



CURRENT SNAPSHOT	2026-08-05
CANONICAL ENTRY	Pipeline/START_PIPELINE.cmd
PRODUCTION	285 Python definitions + 32 JavaScript functions
SUPPORT	240 test/tool definitions indexed separately
OUTPUT ROOT	PipelineAudits/runs/

Prepared from the code itself. Line references and counts are regenerated from this workspace snapshot.

Contents

1. What this repository does	5
The one-sentence mental model	5
Why so many artifacts?	5
2. How to run it and where execution begins	6
Normal local launch	6
Public entry points	6
Naming gotcha	6
3. The data model: what changes at each boundary	7
The crucial distinction	7
4. Stage 1 - Paper parsing	8
Goal	8
Core path	8
Key behavior	8
Failure boundary	8
5. Stage 2 - Grammatical parsing and SciBERT typing	9
Goal	9
Core path	9
Key behavior	9
Failure boundary	9
6. Stage 3 - Node candidate generation and NER boundary	10
Goal	10
Trusted channels	10
Review channels	10
Audit invariant	10
7. Stage 4 - Node adjudication and reference resolution	11
Node judgment	11
Fallback	11
References	11
Human loop	11
8. Stage 5 - Relationship extraction	12
Goal	12
Context	12
Quantities	12

Safety	12
9. Stage 6 - Canonicalization and duplicate review	13
Exact grouping	13
Fuzzy comparison	13
Reviewed merges	13
Cross-paper identity	13
10. Stage 7 - Questions, validation, and diagnostics	14
Question answerability	14
Semantic completeness	14
Generalization diagnostics	14
Run validation	14
11. Stage 8 - Neo4j persistence	15
Goal	15
Before writing	15
Write model	15
Operational note	15
12. Artifacts and review loops	16
Three controlled human interventions	16
13. Configuration and external dependencies	17
Fail-open versus fail-closed	17
14. Two pipelines you should not confuse	18
15. Mentor walkthrough script	19
1. Start with the boundary	19
2. Trace one sentence	19
3. Separate occurrence from identity	19
4. Trace one edge	19
5. Show the safeguards	19
6. Finish with reproducibility	19
A candid answer if asked 'Did you write this?'	19
16. Likely mentor questions	20
Why not ask one LLM to extract the whole graph?	20
Why SciBERT if an LLM also judges nodes?	20
What prevents hallucinated nodes?	20
What gets auto-merged?	20
How is traceability preserved?	20

What happens without Ollama?	20
What is the biggest technical risk?	20
Why is node_only_server serving edges?	20
17. Production function and class reference	21
Public API	21
Paper Parsing	21
Grammatical Parsing	28
Node Pipeline	31
Edge Extraction	42
Question Evaluation	43
Validation	43
Graph Persistence	45
GENERAL	47
Sector Discovery	50
Knowledge Graph Core	52
Top-level	68
18. Browser JavaScript function reference	70
19. Tests and maintenance tools index	73
20. Fast index: module ownership	85

Reading path for tomorrow: pages 1-20 explain the system; the later reference pages are for answering line-specific questions.

1. What this repository does

The project reads research-paper PDFs about engineered water systems and constructs a traceable semantic graph. Its central design choice is evidence preservation: a node or edge is never just a label. It carries the exact sentence, page, span, parser context, decision history, and canonical identity needed to explain where it came from.



The production workbench processes one uploaded paper per run. A separate shared-core batch runner exists for directory-scale experiments; it is documented later so you can distinguish the path your mentor is likely to execute from reusable older infrastructure.

The one-sentence mental model

Parse the paper into evidence units; propose exact-span entities; adjudicate uncertain entities; resolve vague references; infer ontology-constrained relationships; canonicalize duplicates; validate completeness and traceability; then upsert the verified evidence graph into Neo4j.

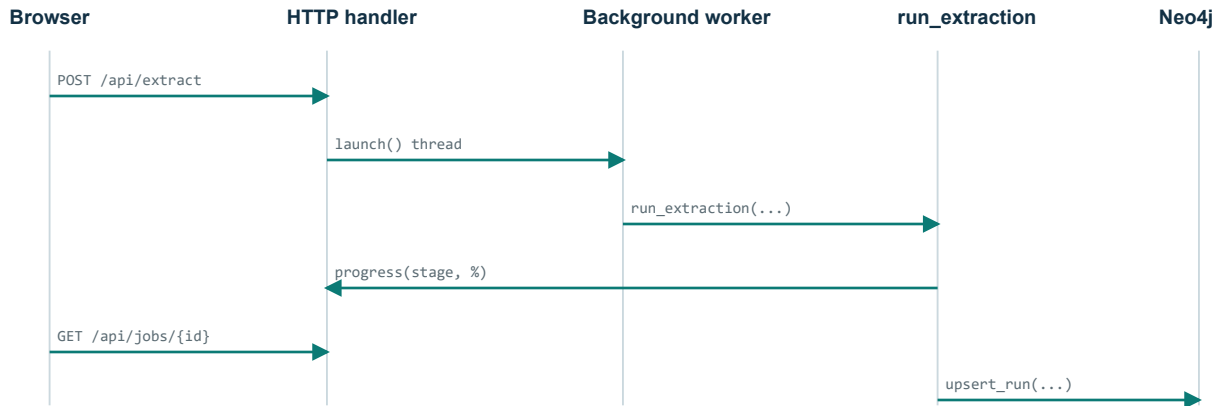
Why so many artifacts?

Each stage writes durable JSON/JSONL before the next stage. That makes the run auditable and lets human decisions rebuild downstream state without asking an LLM to recreate history.

2. How to run it and where execution begins

Normal local launch

1. Start the bundled Neo4j instance. 2. Start Ollama with deepseek-r1:7b. 3. Run `.\Pipeline\START_PIPELINE.cmd`. 4. Open `http://127.0.0.1:8767/`. 5. Upload a paper and monitor the stage cards.



The browser does not run the scientific pipeline directly. It uploads bytes to the Python HTTP server. `launch()` stores the PDF, creates a job record, and starts `run_extraction()` on a daemon thread. Progress callbacks update the in-memory job state and `stage_outputs.json`; the browser polls the job endpoint.

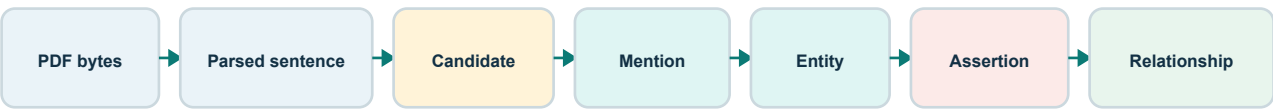
Public entry points

Pipeline/START_PIPELINE.cmd	Starts dependencies/workbench on Windows	Pipeline.GENERAL.node_only_server.main
Pipeline.GENERAL.node_only_server.main	Serves the complete paper graph workbench despite its historical filename	ThreadingHTTPServer + NodeOnlyHandler
Pipeline.run_extraction	Canonical full single-paper API	Pipeline.Node_Pipeline.orchestrator.run_extraction
Pipeline.run_node_only_extraction	Programmatic node-only API	Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction
Knowledge_Graph_Core.run_pipeline	Alternate directory/batch API	Shared-core NodeExtractor, EdgeExtractor, EntityResolver

Naming gotcha

`node_only_server.py` is a historical name. Its status payload says `mode: paper_graph`, and it exposes the full node-and-relationship workbench through the inherited handler. The truly node-only behavior lives in `Node_Pipeline/node_only_runner.py`.

3. The data model: what changes at each boundary



Occurrence-level provenance remains attached through every transformation.

Layer	Definition	Provenance
Parsed sentence	Narrative sentence recovered from PDF layout	Stable document/section/paragraph/sentence IDs, pages, exact text, context
Candidate	A possible ontology-typed node span	Exact offsets; generator channel; SciBERT alternatives; trusted flag
Judgment	Accept/review/reject decision for one candidate	Rubric version, validator, model audit, source-grounding checks
Mention	One accepted occurrence of a real thing	Mention ID, canonical ID, evidence quote, page and sentence
Entity	Canonical thing supported by one or more mentions	Canonical name/type plus complete mention list
Assertion	One evidence occurrence of a predicate between endpoints	Subject/object IDs, predicate, confidence, exact context
Relationship	Canonical edge aggregating compatible assertions	Merged endpoints and all supporting assertion IDs

The crucial distinction

Candidates are hypotheses. Mentions are accepted occurrences. Entities are canonical identities. Assertions are evidence occurrences. Relationships are canonical graph edges. Most review and audit bugs come from accidentally treating one layer as another.

4. Stage 1 - Paper parsing

Primary source: Pipeline/Paper_Parsing

Goal

Recover narrative reading order and provenance from embedded PDF text, while excluding tables, figures, captions, formulas, references, and publisher furniture.

Core path

pdf_parser.extract_lines -> pipeline.parse_pdf -> text_utils normalization/sentence splitting -> parsed.json

Key behavior

Layout-aware word grouping supports multi-column papers. Publication profiles influence headings and front matter. Sections receive hierarchy; paragraphs and sentences inherit page evidence and stable IDs.

Failure boundary

If narrative extraction is suspiciously short or the PDF has no usable embedded text, required downstream NER checks stop the run rather than manufacturing content.

5. Stage 2 - Grammatical parsing and SciBERT typing

Primary source: Pipeline/Grammatical_Parsing/candidates.py

Goal

Create high-recall candidates from complete grammatical subject/object arguments instead of searching only a fixed vocabulary.

Core path

SpacyPredicateTripleExtractor.load/analyze_records -> predicate-first triples -> SciBERTOntologyTyper.type_arguments -> generate_grammatical_candidates

Key behavior

spaCy supplies tokens, POS tags, dependencies, subjects, predicates, and objects. SciBERT embeds argument phrases and ontology definitions; threshold and margin rules decide whether a provisional type is strong enough.

Failure boundary

Grammar parsing and SciBERT are configured as required. Missing models or suspicious early cutoff are hard failures.

6. Stage 3 - Node candidate generation and NER boundary

Primary source: Pipeline/Node_Pipeline/candidate_generation.py + ner_boundary.py

Goal

Unify trusted metadata/lexicon hits with untrusted SVO-derived spans without letting a generator bypass adjudication.

Trusted channels

Structural metadata and unambiguous persistent lexicon mappings can be accepted immediately.

Review channels

Ambiguous aliases, SciBERT-typed arguments, and document discovery remain candidates until validated.

Audit invariant

Every candidate must point to an exact sentence span; duplicate trusted and review candidates are collapsed without erasing distinct source occurrences.

7. Stage 4 - Node adjudication and reference resolution

Primary source: Pipeline/Node_Pipeline/adjudication.py + reference_resolution.py

Node judgment

The local LLM applies a versioned rubric. Its structured response is validated against candidate IDs, offsets, ontology labels, allowed decisions, and grounded canonical names.

Fallback

In node-only-compatible behavior, an unavailable node judge does not auto-accept: untrusted candidates remain in the review queue and are audited.

References

Generic phrases such as 'the model' are never promoted into canonical nodes. They are compared with compatible antecedent candidates inside an exact 2-3 sentence context.

Human loop

SAME materializes a traceable mention of an existing node; DIFFERENT is ignored; UNCERTAIN or ambiguity enters a review artifact.

8. Stage 5 - Relationship extraction

Primary source: Pipeline/Edge_Extraction + Knowledge_Graph_Core/edge_extractor.py

Goal

Turn accepted mentions into ontology-valid assertions using the mandatory relationship LLM.

Context

The extractor creates target-sentence windows with the previous and next sentence in the same section when available and supplies only current-paper accepted nodes as legal endpoints.

Quantities

Explicit quantitative results become occurrence-scoped Observation mentions so conditions and values do not collapse into reusable domain objects.

Safety

Unknown endpoints are audited but not materialized; invalid or incomplete model output follows reject-and-audit; there is no semantic fallback in full mode.

9. Stage 6 - Canonicalization and duplicate review

Primary source: Pipeline/Node_Pipeline/canonicalization.py

Exact grouping

Mentions sharing a canonical identity become one entity with retained provenance.

Fuzzy comparison

Same-type pairs above the configured similarity threshold become review candidates. Fuzzy similarity never auto-merges.

Reviewed merges

Accepted pair decisions feed a union-find structure; materialization rewrites merged entities and relationship endpoints deterministically.

Cross-paper identity

Neo4j persistence can align reusable canonical identities already present in the graph while keeping observations occurrence-scoped.

10. Stage 7 - Questions, validation, and diagnostics

Primary source: Pipeline/Question_Evaluation + Pipeline/Validation

Question answerability

A question is answerable only from a materialized graph fact with exact evidence. Parsed prose is evidence, not a shortcut around graph construction.

Semantic completeness

Domain signals in the paper are compared with grounded, connected core nodes and assertions to report likely recall gaps separately from schema validity.

Generalization diagnostics

Read-only profiling reports document characteristics, extraction distribution, missing signals, and LLM usage without creating facts.

Run validation

The final report summarizes traceability, NER, reference resolution, relationships, completeness, timing, and failure counts.

11. Stage 8 - Neo4j persistence

Primary source: Pipeline/Graph_Persistence/neo4j_writer.py

Goal

Write a paper-only semantic graph that is validated against the authoritative ontology.

Before writing

Entity classes must exist; duplicate real-world IDs are rejected; predicates must exist; subject/object types must satisfy domain and range.

Write model

The upsert is idempotent and stores documents, evidence, mention provenance, canonical nodes, assertions, and canonical relationships.

Operational note

The checked-in local Neo4j disables authentication and binds to localhost. It is development configuration, not deployment configuration.

12. Artifacts and review loops

A run is a directory under `PipelineAudits/runs/<run_id>/`. The manifest is the index; JSON files hold stage summaries; JSONL files hold independently reviewable records.

Artifact	Produced by	Contents
<code>parsed.json</code>	Paper parsing	Canonical evidence hierarchy and source metadata
<code>grammatical_analysis.jsonl</code>	Grammar	Tokens, POS tags, and dependencies for every sentence
<code>grammatical_triples.jsonl</code>	Grammar	Predicate-first subject/object spans
<code>scibert_typings.jsonl</code>	Grammar	Ontology ranking and margins for argument phrases
<code>node_candidates.jsonl</code>	Candidate generation	Trusted and untrusted candidate contract
<code>candidate_judgments.jsonl</code>	Adjudication	Durable decision per candidate
<code>mentions.jsonl</code>	Node/reference/observation stages	Accepted occurrence-level nodes
<code>reference_resolution_review.jsonl</code>	Reference resolution	Ambiguous antecedent comparisons
<code>relationship_candidates.jsonl</code>	Relationship extraction	Every proposed edge and final status
<code>assertions.jsonl</code>	Relationship extraction	Accepted evidence occurrences
<code>canonical_entities_merged.jsonl</code>	Merge materialization	Final nodes after human pair decisions
<code>canonical_relationships.jsonl</code>	Relationship materialization	Final edges with aggregated support
<code>question_answerability.json</code>	Question evaluation	Which graph facts answer which research questions
<code>semantic_completeness.json</code>	Validation	Coverage gaps and score
<code>neo4j_upsert.json</code>	Persistence	Database write outcome/counts
<code>manifest.json</code>	Orchestrator	Run-level summary, timings, counts, and downloads

Three controlled human interventions

- Node review: accept or reject a source-grounded untrusted candidate. Acceptance rebuilds mentions, entities, similarity pairs, validation, and graph state.
- Reference review: select which existing specific node a vague mention refers to. The chosen target gains a traceable mention; the vague surface does not pollute the lexicon.
- Duplicate review: accept or reject a possible same-type entity pair. Accepted decisions feed union-find merge materialization and rewire relationships.

13. Configuration and external dependencies

config/node_extraction.json	Required grammar/NER, SciBERT, node/reference LLM, context, similarity	Authoritative
config/relationship_extraction.json	Relationship LLM, endpoint policy, thresholds	Authoritative in full mode
config/node_validity_rubric.json	Accept/review/reject decision rules	Versioned audit input
ontology/ontology.json	Node classes, hierarchy, predicates, domain/range	All writes validated against it
lexicon/lexicon.json	Persistent aliases and canonical identities	Trusted only when unambiguous
en_core_web_sm	spaCy dependency parser	Required
allenai/scibert_scivocab_uncased	Candidate-to-ontology embedding similarity	Required
Ollama + deepseek-r1:7b	Node, reference, relationship structured judgments	Relationship judge required in full mode
Neo4j + Java	Graph persistence	Required by current orchestrators

Fail-open versus fail-closed

The node and reference LLMs can degrade into review/audit in node-only-compatible behavior. The relationship LLM cannot: full semantic edge extraction has no fallback, so its absence stops the full run. Schema and provenance validation are always fail-closed.

14. Two pipelines you should not confuse

	Node Pipeline	Knowledge Graph Core
Entry	Node_Pipeline.orchestrator.run_extraction	Knowledge_Graph_Core.pipeline.run_pipeline
Input	One uploaded PDF	Directory of PDFs
Candidate system	Grammar + SciBERT + staged node package	NodeExtractor class
Resolution	Exact canonicalization + explicit pair review	EntityResolver embedding decisions; never auto-merge
Persistence	Neo4j upsert inside run	Writes portable artifacts; persistence is separate
Workbench uses it?	Yes	No, unless called programmatically

The shared core is not dead code: the production relationship stage imports its `EdgeExtractor` and judges, and tests exercise its contracts. But its directory-level `run_pipeline` orchestration is an alternate interface, not the browser's route.

15. Mentor walkthrough script

1. Start with the boundary

Open Pipeline/Node_Pipeline/orchestrator.py and show run_extraction. Explain that every stage writes an audit artifact before handing data forward.

2. Trace one sentence

Follow parsed.jsonl -> grammatical_triples.jsonl -> node_candidates.jsonl -> candidate_judgments.jsonl -> mentions.jsonl.

3. Separate occurrence from identity

Show that repeated mentions support one canonical entity while retaining their own sentence/page provenance.

4. Trace one edge

Follow a three-sentence window into relationship_candidates.jsonl, then an accepted assertion, then a canonical relationship.

5. Show the safeguards

Point to exact-span checks, ontology type checks, unknown-endpoint audit, domain/range validation, and no fuzzy auto-merge.

6. Finish with reproducibility

Open manifest.json, validation_report.json, llm usage audits, and neo4j_upsert.json.

A candid answer if asked 'Did you write this?'

No - I am learning and taking ownership of it now. I can explain the pipeline boundaries, the evidence model, and the safeguards, and I have this source-indexed guide to help me trace any function we open.

16. Likely mentor questions

Why not ask one LLM to extract the whole graph?

Because candidates, judgments, mentions, assertions, and canonical records need separate evidence and validation boundaries. The staged design makes hallucinations visible and reviewable.

Why SciBERT if an LLM also judges nodes?

SciBERT provides reproducible high-recall ontology ranking for grammatical arguments. The LLM applies the validity rubric and contextual cleanup; neither is allowed to erase source grounding.

What prevents hallucinated nodes?

Candidates must match exact source spans; returned canonical names must be grounded cleanup or an exact allowed prior identity; unseen relationship endpoints are audited rather than created.

What gets auto-merged?

Only exact canonical identity and reviewed aliases. Fuzzy same-type similarity creates a review pair; it does not merge automatically.

How is traceability preserved?

Stable IDs and source objects flow from sentence spans into mentions and assertions, then canonical entities and relationships aggregate rather than replace that evidence.

What happens without Ollama?

Node-only-compatible stages leave untrusted nodes and ambiguous references for review. Full relationship extraction is required and stops because there is no semantic fallback.

What is the biggest technical risk?

PDF layout diversity and semantic recall. The code addresses these separately with publication profiles, required grammar/NER completeness checks, and post-run semantic diagnostics.

Why is `node_only_server` serving edges?

It is a historical filename retained as the launcher. Its handler advertises `paper_graph` mode and delegates uploads to the full `run_extraction` path.

17. Production function and class reference

This appendix covers all 285 Python class/function definitions outside tests and maintenance tools. Nested helpers and class methods are included. 'Called by' is a static name-based aid, so treat ambiguous common names as navigation hints rather than a formal runtime trace.

Public API

Pipeline/Node_Pipeline/orchestrator.py

def run_extraction(pdf_path: Path, run_dir: Path, progress: Callable[..., None]) -> dict

Purpose. The canonical single-paper production orchestrator. It runs parsing, mandatory grammar/NER, optional node/reference adjudication, required relationship extraction, canonicalization, review preparation, validation, and Neo4j persistence.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to lexicon, add, aggregate_llm_usage, append; returns a derived value or artifact.

Calls. Lexicon, add, aggregate_llm_usage, append, attach_prior_nodes_same_type, build_generalization_diagnostics, build_required_ner_artifacts, candidate_artifact, canonical_entities, copy2

Called by. Pipeline.GENERAL.run.launch, Pipeline.GENERAL.run.launch.worker

SOURCE Pipeline/Node_Pipeline/orchestrator.py:31-307

def emit(stage: str, message: str, value: int, status: str='running', output: dict | None=None) -> None

Purpose. Implements the internal emit operation.

Mechanics. Branches on validation or availability conditions; delegates to json_write, perf_counter, pop, progress.

Calls. json_write, perf_counter, pop, progress, round, setdefault, values

Called by. Pipeline.Node_Pipeline.orchestrator.run_extraction

SOURCE Pipeline/Node_Pipeline/orchestrator.py:38-53

Paper Parsing

Pipeline/Paper_Parsing/models.py

def stable_id(prefix: str, *parts: object) -> str

Purpose. Implements the internal stable id operation.

Mechanics. Iterates over input records; delegates to encode, hexdigest, join, sha1; returns a derived value or artifact.

Calls. encode, hexdigest, join, sha1

Called by. Pipeline.Edge_Extraction.materialization.materialize_canonical_relationships, Pipeline.Edge_Extraction.observation_mentions.assemble_observation_mentions, Pipeline.GENERAL.run.launch, Pipeline.Graph_Persistence.neo4j_writer.build_entity_rows, Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract

SOURCE Pipeline/Paper_Parsing/models.py:6-9

def chunk_record(*, chunk_id: str, chunk_type: str, text: str, ordinal: int, pages: list[int], parent_id: str | None, section_id: str | None) -> dict

Purpose. Implements the internal chunk record operation.

Mechanics. Returns a derived value or artifact.

Called by. Pipeline.Paper_Parsing.pipeline.parse_pdf

SOURCE Pipeline/Paper_Parsing/models.py:12-25

Pipeline/Paper_Parsing/pdf_parser.py

def _infer_ruled_table_bboxes(page) -> list[tuple[float, float, float, float]]

Purpose. Find column-width ruled tables missed by pdfplumber's cell detector.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, get, round, values; returns a derived value or artifact.

Calls. append, get, round, values

Called by. Pipeline.Paper_Parsing.pdf_parser.extract_lines, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_infers_ruled_column_table

SOURCE Pipeline/Paper_Parsing/pdf_parser.py:14-49

class Line

Purpose. Groups the state and behavior for Line.

Mechanics. Encapsulates one small, reusable operation.

Called by. Pipeline.Paper_Parsing.pdf_parser._words_to_lines, Pipeline.Paper_Parsing.pdf_parser._reading_order, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_layout_change_does_not_cut_unfinished_sentence, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_reference_table_header_before_midpoint_is_not_back_matter, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_late_bold_references_is_back_matter, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_bold_preprint_front_matter_fallback_and_reference_stop

SOURCE Pipeline/Paper_Parsing/pdf_parser.py:53-62

def _inside_bbox(word: dict, bbox: tuple[float, float, float, float]) -> bool

Purpose. Implements the internal inside bbox operation.

Mechanics. Returns a derived value or artifact.

Called by. Pipeline.Paper_Parsing.pdf_parser.extract_lines

SOURCE Pipeline/Paper_Parsing/pdf_parser.py:65-69

def _words_to_lines(words: list[dict], page_number: int) -> list[Line]

Purpose. Implements the internal words to lines operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to line, abs, append, get; returns a derived value or artifact.

Calls. Line, abs, append, get, join, lower, median, next, normalize_text, reversed

Called by. Pipeline.Paper_Parsing.pdf_parser.extract_lines, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_column_gap_splits_words_before_line_interleaving

SOURCE Pipeline/Paper_Parsing/pdf_parser.py:72-115

def _reading_order(lines: list[Line], width: float) -> list[Line]

Purpose. Implements the internal reading order operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to line, append, extend; returns a derived value or artifact.

Calls. Line, append, extend

Called by. Pipeline.Paper_Parsing.pdf_parser.extract_lines

SOURCE Pipeline/Paper_Parsing/pdf_parser.py:118-133

def extract_lines(pdf_bytes: bytes) -> tuple[list[Line], dict]

Purpose. Locates and materializes evidence-backed records.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to bytesio, counter, runtimeerror, _infer_ruled_table_bboxes; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. BytesIO, Counter, RuntimeError, _infer_ruled_table_bboxes, _inside_bbox, _reading_order, _words_to_lines, abs, any, append

Called by. Pipeline.Paper_Parsing.pipeline.parse_pdf, Pipeline.Paper_Parsing.tools.inspect_pdf_lines.main

SOURCE Pipeline/Paper_Parsing/pdf_parser.py:136-249

Pipeline/Paper_Parsing/pipeline.py

def _canonical_heading(text: str) -> str

Purpose. Implements the internal canonical heading operation.

Mechanics. Delegates to get, join, lower, split; returns a derived value or artifact.

Calls. get, join, lower, split, strip, sub, visible_heading

Called by. Pipeline.Paper_Parsing.pipeline._paragraph_groups

SOURCE Pipeline/Paper_Parsing/pipeline.py:21-34

def _back_matter_label(text: str) -> str

Purpose. Normalize both 'References' and numbered forms such as '8.

Mechanics. Delegates to compact_label, strip, sub, visible_heading; returns a derived value or artifact.

Calls. compact_label, strip, sub, visible_heading

Called by. Pipeline.Paper_Parsing.pipeline._is_back_matter_heading, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_numbered_references_heading_is_back_matter

SOURCE Pipeline/Paper_Parsing/pipeline.py:37-40

def _is_back_matter_heading(line: Line, profile: ParsingProfile, page_count: int, body_size: float) -> bool

Purpose. Reject table headers such as a lone 'Reference' cell.

Mechanics. Branches on validation or availability conditions; delegates to _back_matter_label, ceil, looks_like_heading, match; returns a derived value or artifact.

Calls. _back_matter_label, ceil, looks_like_heading, match, visible_heading

Called by. Pipeline.Paper_Parsing.pipeline._front_matter, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_reference_table_header_before_midpoint_is_not_back_matter, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_late_bold_references_is_back_matter

SOURCE Pipeline/Paper_Parsing/pipeline.py:43-58

def _clean_authors_text(text: str) -> str

Purpose. Remove common preprint affiliation markers without changing author names.

Mechanics. Branches on validation or availability conditions; delegates to __import__, decode, encode, normalize; returns a derived value or artifact.

Calls. __import__, decode, encode, normalize, replace, strip, sub

Called by. Pipeline.Paper_Parsing.pipeline._front_matter, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_cleans_preprint_author_markers

SOURCE Pipeline/Paper_Parsing/pipeline.py:61-79

def _person_like(line: Line) -> bool

Purpose. Implements the internal person like operation.

Mechanics. Delegates to search; returns a derived value or artifact.

Calls. search

Called by. Pipeline.Paper_Parsing.pipeline._author_continuation, Pipeline.Paper_Parsing.pipeline._front_matter

SOURCE Pipeline/Paper_Parsing/pipeline.py:82-83

def _affiliation_like(line: Line) -> bool

Purpose. Implements the internal affiliation like operation.

Mechanics. Delegates to search; returns a derived value or artifact.

Calls. search

Called by. Pipeline.Paper_Parsing.pipeline._front_matter

SOURCE Pipeline/Paper_Parsing/pipeline.py:86-90

def _author_furniture(line: Line) -> bool

Purpose. Implements the internal author furniture operation.

Mechanics. Delegates to search; returns a derived value or artifact.

Calls. search

Called by. Pipeline.Paper_Parsing.pipeline._front_matter

SOURCE Pipeline/Paper_Parsing/pipeline.py:93-101

def _author_continuation(line: Line) -> bool

Purpose. Implements the internal author continuation operation.

Mechanics. Delegates to _person_like, fullmatch, lstrip, startswith; returns a derived value or artifact.

Calls. _person_like, fullmatch, lstrip, startswith, strip

Called by. Pipeline.Paper_Parsing.pipeline._front_matter

SOURCE Pipeline/Paper_Parsing/pipeline.py:104-110

def _is_introduction_heading(text: str) -> bool

Purpose. Implements the internal is introduction heading operation.

Mechanics. Delegates to fullmatch, strip, visible_heading; returns a derived value or artifact.

Calls. fullmatch, strip, visible_heading

Called by. Pipeline.Paper_Parsing.pipeline._front_matter, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_recognizes_numbered_introduction_heading

SOURCE Pipeline/Paper_Parsing/pipeline.py:113-117

def _is_narrative_text(text: str) -> bool

Purpose. Implements the internal is narrative text operation.

Mechanics. Branches on validation or availability conditions; delegates to findall; returns a derived value or artifact.

Calls. findall

Called by. Pipeline.Paper_Parsing.pipeline.parse_pdf, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_equation_fragment_is_not_narrative, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_short_scientific_sentence_is_narrative

SOURCE Pipeline/Paper_Parsing/pipeline.py:120-124

def _section_level(title: str) -> int

Purpose. Implements the internal section level operation.

Mechanics. Branches on validation or availability conditions; delegates to group, match, split; returns a derived value or artifact.

Calls. group, match, split

Called by. Pipeline.Paper_Parsing.pipeline._apply_section_hierarchy

SOURCE Pipeline/Paper_Parsing/pipeline.py:127-133

def _apply_section_hierarchy(sections: list[dict]) -> None

Purpose. Applies a reviewed decision to downstream state.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _section_level, append, extend, pop.

Calls. _section_level, append, extend, pop, reversed

Called by. Pipeline.Paper_Parsing.pipeline.parse_pdf, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_builds_section_hierarchy_and_inherits_pages

SOURCE Pipeline/Paper_Parsing/pipeline.py:136-155

def _paragraph_groups(lines: list[Line], body_size: float, profile: ParsingProfile) -> list[tuple[str | None, list[Line]]]

Purpose. Implements the internal paragraph groups operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _canonical_heading, abs, append, begins_list_item; returns a derived value or artifact.

Calls. _canonical_heading, abs, append, begins_list_item, compile, flush, fullmatch, group, has_terminal_punctuation, is_profile_heading

Called by. Pipeline.Paper_Parsing.pipeline.parse_pdf, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_layout_change_does_not_cut_unfinished_sentence, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_wrapped_styled_heading_is_joined, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_body_line_with_changed_style_does_not_extend_heading, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_acs_inline_subsection_body_is_split_and_retained

SOURCE Pipeline/Paper_Parsing/pipeline.py:158-246

def flush() -> None

Purpose. Implements the internal flush operation.

Mechanics. Branches on validation or availability conditions; delegates to append.

Calls. append

Called by. Pipeline.Paper_Parsing.pipeline._paragraph_groups

SOURCE Pipeline/Paper_Parsing/pipeline.py:181-185

def _is_publisher_furniture(text: str, profile: ParsingProfile) -> bool

Purpose. Implements the internal is publisher furniture operation.

Mechanics. Branches on validation or availability conditions; delegates to join, match, split; returns a derived value or artifact.

Calls. join, match, split

Called by. Pipeline.Paper_Parsing.pipeline._reconstruct_paragraphs

SOURCE Pipeline/Paper_Parsing/pipeline.py:249-255

def _has_explicit_paragraph_start(lines: list[Line], body_size: float) -> bool

Purpose. Detect a publisher's first-line indent without treating its return as a split.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to abs; returns a derived value or artifact.

Calls. abs

SOURCE Pipeline/Paper_Parsing/pipeline.py:258-269

def _reconstruct_paragraphs(groups: list[tuple[str | None, list[Line]]], profile: ParsingProfile, body_size: float) -> tuple[list[tuple[str | None, list[Line]]], int]

Purpose. Implements the internal reconstruct paragraphs operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _is_publisher_furniture, append, endswith, enumerate; returns a derived value or artifact.

Calls. _is_publisher_furniture, append, endswith, enumerate, fullmatch, has_terminal_punctuation, join_lines, match, next, range

Called by. Pipeline.Paper_Parsing.pipeline.parse_pdf

SOURCE Pipeline/Paper_Parsing/pipeline.py:272-333

def _clean_profile_lines(lines: list[Line], profile: ParsingProfile) -> list[Line]

Purpose. Removes known formatting noise while preserving meaning.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, fullmatch, match, replace; returns a derived value or artifact.

Calls. append, fullmatch, match, replace, strip, sub

Called by. Pipeline.Paper_Parsing.pipeline._front_matter

SOURCE Pipeline/Paper_Parsing/pipeline.py:336-349

def _front_matter(lines: list[Line], body_size: float, profile: ParsingProfile) -> tuple[dict, list[Line]]

Purpose. Implements the internal front matter operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _affiliation_like, _author_continuation, _author_furniture, _clean_authors_text; returns a derived value or artifact.

Calls. _affiliation_like, _author_continuation, _author_furniture, _clean_authors_text, _clean_profile_lines, _is_back_matter_heading, _is_introduction_heading, _person_like, abs, append

Called by. Pipeline.Paper_Parsing.pipeline.parse_pdf, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_bold_preprint_front_matter_fallback_and_reference_stop, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_wrapped_smaller_font_author_line_is_retained, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_same_baseline_author_fragments_are_joined, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_affiliation_lines_are_not_appended_to_authors, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_first_page_introduction_is_retained_after_abstract

SOURCE Pipeline/Paper_Parsing/pipeline.py:352-566

def parse_pdf(pdf_bytes: bytes, filename: str='document.pdf') -> dict

Purpose. The narrative parser entry point. It converts PDF bytes into document metadata, hierarchical sections, paragraphs, sentences, page provenance, and stable IDs while excluding non-narrative material.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to counter, path, valueerror, _apply_section_hierarchy; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. Counter, Path, ValueError, _apply_section_hierarchy, _front_matter, _is_narrative_text, _paragraph_groups, _reconstruct_paragraphs, append, ceil

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline, Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Paper_Parsing.tools.audit_corpus.main

SOURCE Pipeline/Paper_Parsing/pipeline.py:569-697

def as_plain_text(result: dict) -> str

Purpose. Implements the internal as plain text operation.

Mechanics. Iterates over input records; delegates to append, extend, join, strip; returns a derived value or artifact.

Calls. append, extend, join, strip

SOURCE Pipeline/Paper_Parsing/pipeline.py:700-705

Pipeline/Paper_Parsing/profiles.py

class ParsingProfile

Purpose. Groups the state and behavior for ParsingProfile.

Mechanics. Delegates to dataclass, frozenset.

Calls. dataclass, frozenset

SOURCE Pipeline/Paper_Parsing/profiles.py:8-16

def compact_label(text: str) -> str

Purpose. Implements the internal compact label operation.

Mechanics. Delegates to lower, sub; returns a derived value or artifact.

Calls. lower, sub

Called by. Pipeline.Paper_Parsing.pipeline._back_matter_label, Pipeline.Paper_Parsing.pipeline._front_matter,

Pipeline.Paper_Parsing.profiles.detect_profile

SOURCE Pipeline/Paper_Parsing/profiles.py:71-72

def visible_heading(text: str) -> str

Purpose. Implements the internal visible heading operation.

Mechanics. Delegates to strip, sub; returns a derived value or artifact.

Calls. strip, sub

Called by. Pipeline.Paper_Parsing.pipeline._canonical_heading, Pipeline.Paper_Parsing.pipeline._back_matter_label,

Pipeline.Paper_Parsing.pipeline._is_back_matter_heading, Pipeline.Paper_Parsing.pipeline._is_introduction_heading,

Pipeline.Paper_Parsing.pipeline._paragraph_groups, Pipeline.Paper_Parsing.profiles.is_profile_heading,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest

SOURCE Pipeline/Paper_Parsing/profiles.py:75-79

def detect_profile(lines: list) -> ParsingProfile

Purpose. Implements the internal detect profile operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to any, compact_label, findall, join; returns a derived value or artifact.

Calls. any, compact_label, findall, join, lower, match, search

Called by. Pipeline.Paper_Parsing.pipeline.parse_pdf, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_detects_acs_est_profile,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_detects_water_research_profile,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_detects_frontiers_profile,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_detects_ieee_arxiv_profile_and_repairs_small_caps,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_detects_all_caps_conference_preprint_profile

SOURCE Pipeline/Paper_Parsing/profiles.py:82-105

def is_profile_heading(text: str, profile: ParsingProfile, *, size: float | None=None, body_size: float | None=None, bold: bool=False, italic: bool=False) -> bool

Purpose. Implements the internal is profile heading operation.

Mechanics. Branches on validation or availability conditions; delegates to endswith, findall, match, search; returns a derived value or artifact.

Calls. endswith, findall, match, search, strip, upper, visible_heading

Called by. Pipeline.Paper_Parsing.pipeline._paragraph_groups, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_bare_number_is_not_a_profile_heading

SOURCE Pipeline/Paper_Parsing/profiles.py:108-125

Pipeline/Paper_Parsing/text_utils.py

def normalize_text(text: str) -> str

Purpose. Converts text into a stable comparison form.

Mechanics. Delegates to group, normalize, replace, strip; returns a derived value or artifact.

Calls. group, normalize, replace, strip, sub

Called by. Pipeline.Paper_Parsing.pdf_parser._words_to_lines, Pipeline.Paper_Parsing.pdf_parser.extract_lines,

Pipeline.Paper_Parsing.text_utils.join_lines, Pipeline.Paper_Parsing.text_utils.has_terminal_punctuation,

Pipeline.Paper_Parsing.text_utils.starts_like_continuation, Pipeline.Paper_Parsing.text_utils.begins_list_item,

Pipeline.Paper_Parsing.text_utils.split_sentences

SOURCE Pipeline/Paper_Parsing/text_utils.py:35-53

def join_lines(lines: list[str]) -> str

Purpose. Implements the internal join lines operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `endswith`, `group`, `islower`, `lower`; returns a derived value or artifact.

Calls. `endswith`, `group`, `islower`, `lower`, `normalize_text`, `search`, `strip`

Called by. `Pipeline.Paper_Parsing.pipeline._paragraph_groups`, `Pipeline.Paper_Parsing.pipeline._reconstruct_paragraphs`, `Pipeline.Paper_Parsing.pipeline._front_matter`, `Pipeline.Paper_Parsing.pipeline.parse_pdf`, `Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest`, `Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_dehyphenates_line_break`, `Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_preserves_real_compound_at_line_break`

SOURCE `Pipeline/Paper_Parsing/text_utils.py:56-71`

def has_terminal_punctuation(text: str) -> bool

Purpose. Implements the internal has terminal punctuation operation.

Mechanics. Delegates to `endswith`, `normalize_text`, `search`; returns a derived value or artifact.

Calls. `endswith`, `normalize_text`, `search`

Called by. `Pipeline.Paper_Parsing.pipeline._paragraph_groups`, `Pipeline.Paper_Parsing.pipeline._reconstruct_paragraphs`, `Pipeline.Paper_Parsing.text_utils.split_sentences`, `Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest`, `Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_citation_after_period_is_terminal`, `Pipeline.Paper_Parsing.tools.audit_json.main`

SOURCE `Pipeline/Paper_Parsing/text_utils.py:74-76`

def starts_like_continuation(text: str) -> bool

Purpose. Implements the internal starts like continuation operation.

Mechanics. Delegates to `islower`, `normalize_text`, `sub`; returns a derived value or artifact.

Calls. `islower`, `normalize_text`, `sub`

Called by. `Pipeline.Paper_Parsing.pipeline._reconstruct_paragraphs`

SOURCE `Pipeline/Paper_Parsing/text_utils.py:79-82`

def begins_list_item(text: str) -> bool

Purpose. Implements the internal begins list item operation.

Mechanics. Delegates to `match`, `normalize_text`; returns a derived value or artifact.

Calls. `match`, `normalize_text`

Called by. `Pipeline.Paper_Parsing.pipeline._paragraph_groups`

SOURCE `Pipeline/Paper_Parsing/text_utils.py:85-86`

def split_sentences(text: str) -> list[str]

Purpose. Divides input into traceable units.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `append`, `compile`, `escape`, `fullmatch`; returns a derived value or artifact.

Calls. `append`, `compile`, `escape`, `fullmatch`, `group`, `has_terminal_punctuation`, `normalize_text`, `replace`, `split`, `strip`

Called by. `Pipeline.Paper_Parsing.pipeline.parse_pdf`, `Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest`, `Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_preserves_scientific_abbreviation`, `Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_preserves_abbreviation_capitalization`, `Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_incomplete_fragment_is_not_a_sentence`, `Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_citation_fragment_remains_in_sentence_coverage`

SOURCE `Pipeline/Paper_Parsing/text_utils.py:89-113`

def looks_like_caption(text: str) -> bool

Purpose. Implements the internal looks like caption operation.

Mechanics. Delegates to `match`, `normalize_text`; returns a derived value or artifact.

Calls. `match`, `normalize_text`

Called by. `Pipeline.Paper_Parsing.pdf_parser.extract_lines`

SOURCE `Pipeline/Paper_Parsing/text_utils.py:116-117`

def looks_like_formula(text: str) -> bool

Purpose. Implements the internal looks like formula operation.

Mechanics. Iterates over input records; delegates to isalnum, isspace, match, normalize_text; returns a derived value or artifact.

Calls. isalnum, isspace, match, normalize_text

Called by. Pipeline.Paper_Parsing.pdf_parser.extract_lines, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest, Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_cid_formula_encoding_is_detected

SOURCE Pipeline/Paper_Parsing/text_utils.py:120-123

def looks_like_heading(text: str, size: float, body_size: float, *, bold: bool=False, italic: bool=False) -> bool

Purpose. Implements the internal looks like heading operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to endswith, findall, group, isupper; returns a derived value or artifact.

Calls. endswith, findall, group, isupper, lower, match, normalize_text, search, strip, sub

Called by. Pipeline.Paper_Parsing.pipeline_is_back_matter_heading, Pipeline.Paper_Parsing.pipeline_paragraph_groups,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_numbered_prose_is_not_a_heading,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_styled_numbered_section_is_a_heading,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_plain_numbered_section_is_a_heading,

Pipeline.Paper_Parsing.tests.test_text_utils.TextUtilitiesTest.test_numbered_prose_and_chart_labels_are_not_headings

SOURCE Pipeline/Paper_Parsing/text_utils.py:126-165

Grammatical Parsing

Pipeline/Grammatical_Parsing/candidates.py

def _stable_id(prefix: str, *values: object) -> str

Purpose. Implements the internal stable id operation.

Mechanics. Iterates over input records; delegates to encode, hexdigest, join, sha256; returns a derived value or artifact.

Calls. encode, hexdigest, join, sha256

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,

Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._analyze_doc,

Pipeline.Sector_Discovery.discovery._cluster_system_candidates, Pipeline.Sector_Discovery.discovery._agent_candidates,

Pipeline.Sector_Discovery.discovery._model_candidates, Pipeline.Sector_Discovery.discovery._canonicalize_counted_datasets,

Pipeline.Sector_Discovery.discovery._profile_pattern_candidates

SOURCE Pipeline/Grammatical_Parsing/candidates.py:31-33

class ArgumentSpan

Purpose. Groups the state and behavior for ArgumentSpan.

Mechanics. Delegates to dataclass.

Calls. dataclass

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,

Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._argument_span

SOURCE Pipeline/Grammatical_Parsing/candidates.py:37-41

class PredicateSpan

Purpose. Groups the state and behavior for PredicateSpan.

Mechanics. Delegates to dataclass.

Calls. dataclass

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,

Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._predicate_span

SOURCE Pipeline/Grammatical_Parsing/candidates.py:45-51

class GrammarRuntimeError(RuntimeError)

Purpose. Groups the state and behavior for GrammarRuntimeError.

Mechanics. Encapsulates one small, reusable operation.

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,

Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor.load, Pipeline.Grammatical_Parsing.candidates.SciBERTOntologyTyper,

Pipeline.Grammatical_Parsing.candidates.SciBERTOntologyTyper.__init__, Pipeline.Grammatical_Parsing.candidates.generate_grammatical_candidates

SOURCE Pipeline/Grammatical_Parsing/candidates.py:54-55

def clean_argument_phrase(sentence_text: str, start_char: int, end_char: int, tokens: list[dict]) -> dict

Purpose. Remove only syntactically certain excess while preserving source truth.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, casefold, get, pop; returns a derived value or artifact.

Calls. append, casefold, get, pop

Called by. Pipeline.Grammatical_Parsing.candidates.generate_grammatical_candidates

SOURCE Pipeline/Grammatical_Parsing/candidates.py:58-109

class SpacyPredicateTripleExtractor

Purpose. Extract grammatical triples from a trained spaCy dependency parse.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to argumentspan, grammarruntimeerror, predicatespan, _analyze_doc; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. ArgumentSpan, GrammarRuntimeError, PredicateSpan, _analyze_doc, _argument_span, _expand_conjunctions, _expand_objects, _objects, _predicate_span, _stable_id

SOURCE Pipeline/Grammatical_Parsing/candidates.py:112-320

def __init__(self, nlp) -> None

Purpose. Implements the internal init operation.

Mechanics. Encapsulates one small, reusable operation.

Called by. Pipeline.Knowledge_Graph_Core.Ilm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.Ilm_reference_judge.MandatoryLLMReferenceJudge.__init__, Pipeline.Knowledge_Graph_Core.Ilm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.Ilm_relationship_judge.MandatoryLLMRelationshipJudge.__init__, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases

SOURCE Pipeline/Grammatical_Parsing/candidates.py:115-116

def load(cls, model_name: str)

Purpose. Reads and normalizes persisted state.

Mechanics. Branches on validation or availability conditions; delegates to grammarruntimeerror, cls, join; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. GrammarRuntimeError, cls, join, load

Called by. Pipeline.check_environment.main, Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor, Pipeline.Grammatical_Parsing.candidates.generate_grammatical_candidates, Pipeline.GENERAL.tests.test_grammatical_candidates.GrammaticalCandidateTests, Pipeline.GENERAL.tests.test_grammatical_candidates.GrammaticalCandidateTests.parser

SOURCE Pipeline/Grammatical_Parsing/candidates.py:119-138

def analyze_records(self, records: list[dict]) -> tuple[list[dict], list[dict]]

Purpose. Implements the internal analyze records operation.

Mechanics. Iterates over input records; delegates to _analyze_doc, append, extend, pipe; returns a derived value or artifact.

Calls. _analyze_doc, append, extend, pipe, zip

Called by. Pipeline.Grammatical_Parsing.candidates.generate_grammatical_candidates

SOURCE Pipeline/Grammatical_Parsing/candidates.py:140-148

def _analyze_doc(self, record: dict, doc) -> tuple[dict, list[dict]]

Purpose. Implements the internal analyze doc operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _argument_span, _expand_conjunctions, _expand_objects, _objects; returns a derived value or artifact.

Calls. _argument_span, _expand_conjunctions, _expand_objects, _objects, _predicate_span, _stable_id, _subjects, add, any, append

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor, Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor.analyze_records

SOURCE Pipeline/Grammatical_Parsing/candidates.py:150-225

def _subjects(predicate) -> list

Purpose. Implements the internal subjects operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; returns a derived value or artifact.

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,
Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._analyze_doc

SOURCE Pipeline/Grammatical_Parsing/candidates.py:228-232

def _objects(predicate) -> list[tuple[object, object | None]]

Purpose. Implements the internal objects operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to extend; returns a derived value or artifact.

Calls. _objects, extend

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,
Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._analyze_doc

SOURCE Pipeline/Grammatical_Parsing/candidates.py:235-262

def _expand_conjunctions(tokens: Iterable) -> list

Purpose. Implements the internal expand conjunctions operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to add, append; returns a derived value or artifact.

Calls. add, append

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,
Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._analyze_doc

SOURCE Pipeline/Grammatical_Parsing/candidates.py:265-273

def _expand_objects(objects: Iterable[tuple[object, object | None]]) -> list[tuple[object, object | None]]

Purpose. Implements the internal expand objects operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to add, append; returns a derived value or artifact.

Calls. add, append

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,
Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._analyze_doc

SOURCE Pipeline/Grammatical_Parsing/candidates.py:276-285

def _argument_span(root, noun_chunks: dict[int, object], doc) -> ArgumentSpan

Purpose. Implements the internal argument span operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to argumentspan, get; returns a derived value or artifact.

Calls. ArgumentSpan, get

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,
Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._analyze_doc

SOURCE Pipeline/Grammatical_Parsing/candidates.py:288-302

def _predicate_span(predicate, preposition, doc) -> PredicateSpan

Purpose. Implements the internal predicate span operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to predicatespan, append, join, values; returns a derived value or artifact.

Calls. PredicateSpan, append, join, values

Called by. Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,
Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._analyze_doc

SOURCE Pipeline/Grammatical_Parsing/candidates.py:305-320

class SciBERTOntologyTyper

Purpose. Rank ontology definitions using an unfine-tuned SciBERT encoder.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to grammarruntimeerror, sentencetransformer, _ontology_text, append; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. GrammarRuntimeError, SentenceTransformer, _ontology_text, append, argsort, asarray, encode, filter, get, join

Called by. Pipeline.Grammatical_Parsing.candidates.generate_grammatical_candidates,
Pipeline.GENERAL.tests.test_grammatical_candidates.GrammaticalCandidateTests,
Pipeline.GENERAL.tests.test_grammatical_candidates.GrammaticalCandidateTests.test_pretrained_scibert_similarity_assigns_provisional_type

SOURCE Pipeline/Grammatical_Parsing/candidates.py:323-400

```
def __init__(self, ontology: dict, model_name: str, *, threshold: float, top_k: int, cache_dir: Path, min_margin: float=0.0, encoder=None) -> None
```

Purpose. Implements the internal init operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `grammarruntimeerror`, `sentencetransformer`, `_ontology_text`, `asarray`; fails explicitly when an invariant is broken.

Calls. `GrammarRuntimeError`, `SentenceTransformer`, `_ontology_text`, `asarray`, `encode`

Called by. `Pipeline.Knowledge_Graph_Core.Ilm_reference_judge.MandatoryLLMReferenceJudge`, `Pipeline.Knowledge_Graph_Core.Ilm_reference_judge.MandatoryLLMReferenceJudge.__init__`, `Pipeline.Knowledge_Graph_Core.Ilm_relationship_judge.MandatoryLLMRelationshipJudge`, `Pipeline.Knowledge_Graph_Core.Ilm_relationship_judge.MandatoryLLMRelationshipJudge.__init__`, `Pipeline.Node_Pipeline.lexicon.Lexicon`, `Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases`

SOURCE `Pipeline/Grammatical_Parsing/candidates.py`:326-354

```
def _ontology_text(label: str, spec: dict) -> str
```

Purpose. Implements the internal ontology text operation.

Mechanics. Delegates to `filter`, `get`, `join`; returns a derived value or artifact.

Calls. `filter`, `get`, `join`

Called by. `Pipeline.Grammatical_Parsing.candidates.SciBERTOntologyTyper`, `Pipeline.Grammatical_Parsing.candidates.SciBERTOntologyTyper.__init__`

SOURCE `Pipeline/Grammatical_Parsing/candidates.py`:357-363

```
def type_arguments(self, arguments: list[dict]) -> list[dict]
```

Purpose. Implements the internal type arguments operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `append`, `argsort`, `asarray`, `encode`; returns a derived value or artifact.

Calls. `append`, `argsort`, `asarray`, `encode`, `get`, `round`, `zip`

Called by. `Pipeline.Grammatical_Parsing.candidates.generate_grammatical_candidates`, `Pipeline.GENERAL.tests.test_grammatical_candidates.GrammaticalCandidateTests`, `Pipeline.GENERAL.tests.test_grammatical_candidates.GrammaticalCandidateTests.test_pretrained_scibert_similarity_assigns_provisional_type`

SOURCE `Pipeline/Grammatical_Parsing/candidates.py`:365-400

```
def _sentence_records(parsed: dict) -> list[dict]
```

Purpose. Implements the internal sentence records operation.

Mechanics. Iterates over input records; delegates to `get`; returns a derived value or artifact.

Calls. `get`

Called by. `Pipeline.Grammatical_Parsing.candidates.generate_grammatical_candidates`

SOURCE `Pipeline/Grammatical_Parsing/candidates.py`:403-417

```
def generate_grammatical_candidates(parsed: dict, ontology_path: Path, config: dict, *, parser=None, typer=None) -> dict
```

Purpose. Loads the required spaCy dependency parser and SciBERT typer, records token analysis and predicate-first triples, then returns typed subject/object candidates grouped by sentence.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `grammarruntimeerror`, `path`, `scibertontologytyper`, `_sentence_records`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `GrammarRuntimeError`, `Path`, `SciBERTOntologyTyper`, `_sentence_records`, `add`, `analyze_records`, `append`, `clean_argument_phrase`, `defaultdict`, `get`
Called by. `Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction`, `Pipeline.Node_Pipeline.orchestrator.run_extraction`, `Pipeline.GENERAL.tests.test_grammatical_candidates.GrammaticalCandidateTests`, `Pipeline.GENERAL.tests.test_grammatical_candidates.GrammaticalCandidateTests.test_complete_pos_parse_and_predicate_first_triples`, `Pipeline.GENERAL.tests.test_grammatical_candidates.GrammaticalCandidateTests.test_conjoined_predicates_inherit_the_subject`, `Pipeline.GENERAL.tests.test_grammatical_candidates.GrammaticalCandidateTests.test_ambiguous_scibert_type_does_not_remove_grammar_candidate`

SOURCE `Pipeline/Grammatical_Parsing/candidates.py`:420-537

Node Pipeline

Pipeline/Node_Pipeline/adjudication.py

def _prior_node_comparison(candidate: dict, ontology_label: str, canonical_name: str) -> dict

Purpose. Implements the internal prior node comparison operation.

Mechanics. Iterates over input records; delegates to get, normalized; returns a derived value or artifact.

Calls. get, normalized

Called by. Pipeline.Node_Pipeline.adjudication.optional_llm_review, Pipeline.Node_Pipeline.review_workflows.apply_node_review

SOURCE Pipeline/Node_Pipeline/adjudication.py:20-41

def _canonical_name_is_grounded(candidate: dict, canonical_name: str, comparison: dict) -> bool

Purpose. Allow source-span cleanup or an exact same-type prior mapping, never a new identity.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to any, get, normalized; returns a derived value or artifact.

Calls. any, get, normalized

Called by. Pipeline.Node_Pipeline.adjudication.optional_llm_review

SOURCE Pipeline/Node_Pipeline/adjudication.py:44-63

def _ontology_label_is_scibert_supported(candidate: dict, ontology_label: str) -> bool

Purpose. Implements the internal ontology label is scibert supported operation.

Mechanics. Iterates over input records; delegates to get; returns a derived value or artifact.

Calls. get

Called by. Pipeline.Node_Pipeline.adjudication.optional_llm_review

SOURCE Pipeline/Node_Pipeline/adjudication.py:66-72

def optional_llm_review(review: list[dict], progress: Callable[[str, str, int], None], *, node_judge=None) -> tuple[list[dict], list[dict], list[dict], list[dict], dict]

Purpose. Runs the configured node judge when available, validates every model response against source spans and ontology rules, and otherwise leaves candidates reviewable instead of silently accepting them.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to mandatoryllmnodejudge, ontology, _canonical_name_is_grounded, _ontology_label_is_scibert_supported; returns a derived value or artifact.

Calls. MandatoryLLMNodeJudge, Ontology, _canonical_name_is_grounded, _ontology_label_is_scibert_supported, _prior_node_comparison, append, extend, get, iter, json_read

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_llm_accept_review_and_reject_are_all_audited, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_missing_sentence_id_is_rejected_before_llm, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_hallucinated_canonical_name_is_rejected, Pipeline.GENERAL.tools.rematerialize_node_run.main

SOURCE Pipeline/Node_Pipeline/adjudication.py:75-312

Pipeline/Node_Pipeline/candidate_generation.py

def flatten_sentences(parsed: dict) -> list[dict]

Purpose. Implements the internal flatten sentences operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, enumerate, get, range; returns a derived value or artifact.

Calls. append, enumerate, get, range

Called by. Pipeline.Node_Pipeline.candidate_generation.generate_candidates, Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.reference_resolution.resolve_references, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_context_crosses_paragraphs_but_not_sections

SOURCE Pipeline/Node_Pipeline/candidate_generation.py:21-51

def trace_source(row: dict, span: Span) -> dict

Purpose. Implements the internal trace source operation.

Mechanics. Iterates over input records; delegates to get; returns a derived value or artifact.

Calls. get

Called by. Pipeline.Node_Pipeline.candidate_generation.generate_candidates

SOURCE Pipeline/Node_Pipeline/candidate_generation.py:54-71

def metadata_mentions(parsed: dict) -> list[dict]

Purpose. Implements the internal metadata mentions operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to any, append, combining, enumerate; returns a derived value or artifact.

Calls. any, append, combining, enumerate, findall, get, isdigit, join, normalize, normalized

Called by. Pipeline.Node_Pipeline.candidate_generation.generate_candidates, Pipeline.GENERAL.tests.test_author_metadata.AuthorMetadataTests, Pipeline.GENERAL.tests.test_author_metadata.AuthorMetadataTests.test_affiliation_markers_do_not_become_people

SOURCE Pipeline/Node_Pipeline/candidate_generation.py:74-120

def mention_dedup_key(row: dict) -> tuple

Purpose. Keep distinct metadata values while deduplicating repeated source spans.

Mechanics. Branches on validation or availability conditions; delegates to get, normalized; returns a derived value or artifact.

Calls. get, normalized

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_distinct_authors_in_one_metadata_field_are_not_deduplicated, Pipeline.GENERAL.tools.rematerialize_node_run.main

SOURCE Pipeline/Node_Pipeline/candidate_generation.py:123-134

def remove_redundant_review_candidates(accepted: list[dict], review: list[dict]) -> list[dict]

Purpose. Do not re-adjudicate an exact sentence span already accepted by a trusted channel.

Mechanics. Iterates over input records; delegates to get, normalized; returns a derived value or artifact.

Calls. get, normalized

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_trusted_exact_span_removes_redundant_review_candidate, Pipeline.GENERAL.tools.rematerialize_node_run.main

SOURCE Pipeline/Node_Pipeline/candidate_generation.py:137-156

def generate_candidates(parsed: dict, lexicon: Lexicon, grammatical_candidates: dict[str, list[dict]] | None=None) -> tuple[list[dict], list[dict]]

Purpose. Combines trusted metadata and lexicon mentions with untrusted grammatical candidates, preserving exact source spans and separating automatic acceptance from review.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to span, append, flatten_sentences, get; returns a derived value or artifact.

Calls. Span, append, flatten_sentences, get, lexicon_spans, metadata_mentions, select_spans, stable_id, trace_source

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_scibert_svo_arguments_are_reviewable_and_lexicon_stays_trusted, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_only_metadata_and_unambiguous_lexicon_matches_are_trusted

SOURCE Pipeline/Node_Pipeline/candidate_generation.py:159-225

Pipeline/Node_Pipeline/canonicalization.py

def canonical_entities(mentions: list[dict]) -> list[dict]

Purpose. Groups accepted mention occurrences by stable canonical identity and produces one entity record with all supporting mentions.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, casefold, defaultdict, get; returns a derived value or artifact.

Calls. append, casefold, defaultdict, get, items, next, normalized, stable_id

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.review_workflows.apply_node_review, Pipeline.Node_Pipeline.review_workflows.apply_reference_resolution, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_equal_looking_statistics_remain_occurrence_scoped, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_same_named_entity_from_lexicon_and_discovery_is_one_node

SOURCE Pipeline/Node_Pipeline/canonicalization.py:19-64

def similarity_features(left: dict, right: dict) -> tuple[float, list[str], dict]

Purpose. Implements the internal similarity features operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to sequencematcher, append, findall, normalized; returns a derived value or artifact.

Calls. SequenceMatcher, append, findall, normalized, ratio, round, split

Called by. Pipeline.Node_Pipeline.canonicalization.similarity_candidates, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_different_explicit_versions_are_not_duplicate_candidates

SOURCE Pipeline/Node_Pipeline/canonicalization.py:67-98

def similarity_candidates(entities: list[dict], threshold: float=0.64) -> list[dict]

Purpose. Scores same-type entity pairs as possible duplicates; it creates a human-review queue and does not auto-merge fuzzy matches.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, casefold, enumerate, round; returns a derived value or artifact.

Calls. append, casefold, enumerate, round, similarity_features, stable_id

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.review_workflows.apply_node_review, Pipeline.Node_Pipeline.review_workflows.apply_reference_resolution, Pipeline.GENERAL.tools.rematerialize_node_run.main

SOURCE Pipeline/Node_Pipeline/canonicalization.py:101-121

class UnionFind

Purpose. Groups the state and behavior for UnionFind.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to find; returns a derived value or artifact.

Calls. find

Called by. Pipeline.Node_Pipeline.canonicalization.materialize_merges

SOURCE Pipeline/Node_Pipeline/canonicalization.py:124-137

def __init__(self, values: Iterable[str]) -> None

Purpose. Implements the internal init operation.

Mechanics. Iterates over input records.

Called by. Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge.__init__, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.__init__, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases

SOURCE Pipeline/Node_Pipeline/canonicalization.py:125-126

def find(self, value: str) -> str

Purpose. Searches indexed or source records for matches.

Mechanics. Iterates over input records; returns a derived value or artifact.

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor,

Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor._candidate_from_proposal,

Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._sentence_candidates,

Pipeline.Node_Pipeline.canonicalization.UnionFind, Pipeline.Node_Pipeline.canonicalization.UnionFind.union,

Pipeline.Node_Pipeline.canonicalization.materialize_merges

SOURCE Pipeline/Node_Pipeline/canonicalization.py:128-132

def union(self, left: str, right: str) -> None

Purpose. Implements the internal union operation.

Mechanics. Branches on validation or availability conditions; delegates to find.

Calls. find

Called by. Pipeline.Node_Pipeline.canonicalization.materialize_merges, Pipeline.Sector_Discovery.discovery_cluster_system_candidates

SOURCE Pipeline/Node_Pipeline/canonicalization.py:134-137

def materialize_merges(run_dir: Path) -> list[dict]

Purpose. Reads accepted duplicate-review decisions, uses union-find to form merge groups, and emits the final canonical entity set.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to unionfind, append, defaultdict, find; returns a derived value or artifact.

Calls. UnionFind, append, defaultdict, find, get, items, jsonl_read, jsonl_write, next, normalized

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.review_workflows.apply_node_review, Pipeline.Node_Pipeline.review_workflows.apply_reference_resolution, Pipeline.Node_Pipeline.review_workflows.apply_resolution, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_unmerged_entity_keeps_its_persistent_identifier

SOURCE Pipeline/Node_Pipeline/canonicalization.py:140-174

Pipeline/Node_Pipeline/common.py

def now_iso() -> str

Purpose. Implements the internal now iso operation.

Mechanics. Delegates to isoformat, now; returns a derived value or artifact.

Calls. isoformat, now

Called by. Pipeline.Graph_Persistence.neo4j_writer.build_entity_rows, Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases, Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.review_workflows.apply_node_review

SOURCE Pipeline/Node_Pipeline/common.py:50-51

def stable_id(prefix: str, *values: object) -> str

Purpose. Implements the internal stable id operation.

Mechanics. Iterates over input records; delegates to encode, hexdigest, join, sha256; returns a derived value or artifact.

Calls. encode, hexdigest, join, sha256

Called by. Pipeline.Edge_Extraction.materialization.materialize_canonical_relationships, Pipeline.Edge_Extraction.observation_mentions.assemble_observation_mentions, Pipeline.GENERAL.run.launch, Pipeline.Graph_Persistence.neo4j_writer.build_entity_rows, Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract

SOURCE Pipeline/Node_Pipeline/common.py:54-56

def normalized(text: str) -> str

Purpose. Implements the internal normalized operation.

Mechanics. Delegates to casefold, join, replace, split; returns a derived value or artifact.

Calls. casefold, join, replace, split, sub

Called by. Pipeline.Node_Pipeline.adjudication._prior_node_comparison, Pipeline.Node_Pipeline.adjudication._canonical_name_is_grounded, Pipeline.Node_Pipeline.adjudication.optional_ilm_review, Pipeline.Node_Pipeline.candidate_generation.metadata_mentions, Pipeline.Node_Pipeline.candidate_generation.mention_dedup_key, Pipeline.Node_Pipeline.candidate_generation.remove_redundant_review_candidates, Pipeline.Node_Pipeline.canonicalization.canonical_entities

SOURCE Pipeline/Node_Pipeline/common.py:59-62

def json_read(path: Path) -> dict

Purpose. Implements the internal json read operation.

Mechanics. Delegates to loads, read_text; returns a derived value or artifact.

Calls. loads, read_text

Called by. Pipeline.Edge_Extraction.extractor.run_relationship_extraction, Pipeline.GENERAL.node_only_server.NodeOnlyHandler, Pipeline.GENERAL.node_only_server.NodeOnlyHandler.do_GET, Pipeline.GENERAL.run.load_existing_jobs, Pipeline.GENERAL.run.job_payload, Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_GET

SOURCE Pipeline/Node_Pipeline/common.py:65-66

def json_write(path: Path, value: object) -> None

Purpose. Implements the internal json write operation.

Mechanics. Delegates to dumps, mkdir, write_text.

Calls. dumps, mkdir, write_text

Called by. Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_POST, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases, Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction.emit

SOURCE Pipeline/Node_Pipeline/common.py:69-71

def jsonl_read(path: Path) -> list[dict]

Purpose. Implements the internal jsonl read operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, is_file, loads, read_text; returns a derived value or artifact.

Calls. append, is_file, loads, read_text, splitlines, strip

Called by. Pipeline.GENERAL.run.job_payload, Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_POST, Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run, Pipeline.Graph_Persistence.neo4j_writer.upsert_run, Pipeline.Node_Pipeline.canonicalization.materialize_merges, Pipeline.Node_Pipeline.lexicon.load_prior_node_catalog

SOURCE Pipeline/Node_Pipeline/common.py:74-81

def jsonl_write(path: Path, rows: Iterable[dict]) -> None

Purpose. Implements the internal jsonl write operation.

Mechanics. Iterates over input records; delegates to dumps, join, mkdir, write_text.

Calls. dumps, join, mkdir, write_text

Called by. Pipeline.Edge_Extraction.extractor.run_relationship_extraction, Pipeline.Edge_Extraction.materialization.materialize_canonical_relationships, Pipeline.Node_Pipeline.canonicalization.materialize_merges, Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.review_workflows.apply_node_review, Pipeline.Node_Pipeline.review_workflows.apply_reference_resolution

SOURCE Pipeline/Node_Pipeline/common.py:84-87

def append_jsonl(path: Path, row: dict) -> None

Purpose. Implements the internal append jsonl operation.

Mechanics. Delegates to dumps, mkdir, open, write.

Calls. dumps, mkdir, open, write

Called by. Pipeline.Node_Pipeline.review_workflows.apply_resolution

SOURCE Pipeline/Node_Pipeline/common.py:90-93

def copy_sample(sample_path: Path, target: Path) -> Path

Purpose. Implements the internal copy sample operation.

Mechanics. Delegates to copy2, mkdir; returns a derived value or artifact.

Calls. copy2, mkdir

SOURCE Pipeline/Node_Pipeline/common.py:96-99

Pipeline/Node_Pipeline/contracts.py

def rubric_metadata(rubric: dict) -> dict

Purpose. Implements the internal rubric metadata operation.

Mechanics. Delegates to dumps, encode, get, hexdigest; returns a derived value or artifact.

Calls. dumps, encode, get, hexdigest, sha256

Called by. Pipeline.Node_Pipeline.adjudication.optional_IIIm_review, Pipeline.Node_Pipeline.contracts.judgment_artifact

SOURCE Pipeline/Node_Pipeline/contracts.py:12-17

def candidate_artifact(candidate: dict, *, trusted: bool) -> dict

Purpose. Return the public, generator-facing candidate representation.

Mechanics. Delegates to get; returns a derived value or artifact.

Calls. get

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction

SOURCE Pipeline/Node_Pipeline/contracts.py:20-38

```
def judgment_artifact(candidate: dict, *, decision: str, validation: dict, rubric: dict) -> dict
```

Purpose. Return one durable decision for one supplied candidate.

Mechanics. Delegates to get, rubric_metadata; returns a derived value or artifact.

Calls. get, rubric_metadata

Called by. Pipeline.Node_Pipeline.adjudication.optional_llm_review, Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.review_workflows.apply_node_review, Pipeline.GENERAL.tools.rematerialize_node_run.main

SOURCE Pipeline/Node_Pipeline/contracts.py:41-63

Pipeline/Node_Pipeline/lexicon.py

```
class Lexicon
```

Purpose. Groups the state and behavior for Lexicon.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to __init__, all, append, casefold; returns a derived value or artifact.

Calls. __init__, all, append, casefold, defaultdict, get, json_read, json_write, next, normalized

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.review_workflows.apply_node_review, Pipeline.Node_Pipeline.review_workflows.apply_resolution, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_scibert_svo_arguments_are_reviewable_and_lexicon_stays_trusted, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_only_metadata_and_unambiguous_lexicon_matches_are_trusted

SOURCE Pipeline/Node_Pipeline/lexicon.py:23-86

```
def __init__(self, path: Path=LEXICON_PATH) -> None
```

Purpose. Implements the internal init operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to all, append, defaultdict, get.

Calls. all, append, defaultdict, get, json_read, normalized

Called by. Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge.__init__, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.__init__, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases

SOURCE Pipeline/Node_Pipeline/lexicon.py:24-35

```
def find(self, value: str, label: str | None=None) -> dict | None
```

Purpose. Searches indexed or source records for matches.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to get, normalized; returns a derived value or artifact.

Calls. get, normalized

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor._candidate_from_proposal, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._sentence_candidates, Pipeline.Node_Pipeline.canonicalization.UnionFind, Pipeline.Node_Pipeline.canonicalization.UnionFind.union, Pipeline.Node_Pipeline.canonicalization.materialize_merges

SOURCE Pipeline/Node_Pipeline/lexicon.py:37-41

```
def alias_is_unambiguous(self, value: str, canonical_id: str) -> bool
```

Purpose. Implements the internal alias is unambiguous operation.

Mechanics. Delegates to get, normalized; returns a derived value or artifact.

Calls. get, normalized

Called by. Pipeline.Node_Pipeline.lexicon.lexicon_spans

SOURCE Pipeline/Node_Pipeline/lexicon.py:43-45

```
def add_reviewed_aliases(self, label: str, canonical_name: str, aliases: list[str], provenance: dict) -> dict
```

Purpose. Implements the internal add reviewed aliases operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to __init__, append, casefold, get; returns a derived value or artifact.

Calls. __init__, append, casefold, get, json_write, next, normalized, now_iso, setdefault, stable_id

Called by. Pipeline.Node_Pipeline.review_workflows.apply_node_review, Pipeline.Node_Pipeline.review_workflows.apply_resolution

SOURCE Pipeline/Node_Pipeline/lexicon.py:47-86

```
def load_prior_node_catalog(lexicon: Lexicon, *, exclude_run_dir: Path | None=None) -> list[dict]
```

Purpose. Load reusable nodes from the lexicon and completed earlier runs.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to add, append, casefold, get; returns a derived value or artifact.

Calls. add, append, casefold, get, is_dir, is_file, iterdir, json_read, jsonl_read, normalized

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction

SOURCE Pipeline/Node_Pipeline/lexicon.py:89-159

```
def add(row: dict) -> None
```

Purpose. Implements the internal add operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, get, normalized, setdefault.

Calls. append, get, normalized, setdefault, stable_id, strip

Called by. Pipeline.Edge_Extraction.observation_mentions.assemble_observation_mentions,
Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor,
Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._analyze_doc,
Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._expand_conjunctions,
Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._expand_objects,
Pipeline.Grammatical_Parsing.candidates.generate_grammatical_candidates, Pipeline.Graph_Persistence.neo4j_writer.validate_entities_against_ontology

SOURCE Pipeline/Node_Pipeline/lexicon.py:95-124

```
def attach_prior_nodes_same_type(candidates: list[dict], catalog: list[dict], *, maximum: int=12) -> list[dict]
```

Purpose. Attach ranked earlier nodes, never comparing across ontology types.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to sequencematcher, append, casefold, discard; returns a derived value or artifact.

Calls. SequenceMatcher, append, casefold, discard, get, normalized, ratio, round, setdefault, update

Called by. Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction

SOURCE Pipeline/Node_Pipeline/lexicon.py:162-191

```
def lexicon_spans(text: str, lexicon: Lexicon) -> list[Span]
```

Purpose. Implements the internal lexicon spans operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to span, alias_is_unambiguous, append, compile; returns a derived value or artifact.

Calls. Span, alias_is_unambiguous, append, compile, end, escape, finditer, get, map, match

Called by. Pipeline.Node_Pipeline.candidate_generation.generate_candidates, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests,
Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_ambiguous_persistent_alias_requires_adjudication,
Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_model_variant_suffix_is_not_swallowed_by_base_lexicon_entry

SOURCE Pipeline/Node_Pipeline/lexicon.py:194-218

Pipeline/Node_Pipeline/ner_boundary.py

```
def build_required_ner_artifacts(parsed: dict, accepted: list[dict], review: list[dict], config: dict) -> tuple[list[dict], dict]
```

Purpose. Materialize and validate the mandatory multi-channel NER boundary.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to runtimeerror, append, defaultdict, dumps; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. RuntimeError, append, defaultdict, dumps, encode, get, hexdigest, isinstance, items, normalized

Called by. Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.GENERAL.tests.test_required_ner.RequiredNERTests,
Pipeline.GENERAL.tests.test_required_ner.RequiredNERTests.test_required_ner_materializes_exact_span_audit,
Pipeline.GENERAL.tests.test_required_ner.RequiredNERTests.test_untrusted_generator_cannot_bypass_adjudication,
Pipeline.GENERAL.tests.test_required_ner.RequiredNERTests.test_required_ner_cannot_be_disabled,
Pipeline.GENERAL.tests.test_required_ner.RequiredNERTests.test_required_ner_blocks_suspicious_early_parse_cutoff

SOURCE Pipeline/Node_Pipeline/ner_boundary.py:19-114

Pipeline/Node_Pipeline/node_only_runner.py

```
def run_node_only_extraction(pdf_path: Path, run_dir: Path, progress: Callable[[str, str, int], None], *, include_section_titles: set[str] | None=None) -> dict
```

Purpose. A deliberately bounded paper-to-node path. It stops before relationship extraction but still canonicalizes nodes and persists them to Neo4j.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to lexicon, add, append, attach_prior_nodes_same_type; returns a derived value or artifact.

Calls. Lexicon, add, append, attach_prior_nodes_same_type, candidate_artifact, canonical_entities, casefold, copy2, deepcopy, extend

Called by. Pipeline.GENERAL.tools.run_waterrag_quick_test.main

SOURCE Pipeline/Node_Pipeline/node_only_runner.py:32-192

Pipeline/Node_Pipeline/reference_resolution.py

```
def _reference_head(match: re.Match[str]) -> str
```

Purpose. Implements the internal reference head operation.

Mechanics. Branches on validation or availability conditions; delegates to casefold, endswith, group; returns a derived value or artifact.

Calls. casefold, endswith, group

Called by. Pipeline.Node_Pipeline.reference_resolution.resolve_references

SOURCE Pipeline/Node_Pipeline/reference_resolution.py:50-55

```
def _reference_context(sentence_rows: list[dict], index: int, maximum_previous: int=2) -> list[dict]
```

Purpose. Return an antecedent-oriented, exact 2-3 sentence window.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, get, reverse; returns a derived value or artifact.

Calls. append, get, reverse

Called by. Pipeline.Node_Pipeline.reference_resolution.resolve_references

SOURCE Pipeline/Node_Pipeline/reference_resolution.py:58-78

```
def _reference_source(row: dict, context: list[dict], match: re.Match[str]) -> dict
```

Purpose. Implements the internal reference source operation.

Mechanics. Iterates over input records; delegates to end, get, start; returns a derived value or artifact.

Calls. end, get, start

Called by. Pipeline.Node_Pipeline.reference_resolution.resolve_references

SOURCE Pipeline/Node_Pipeline/reference_resolution.py:81-97

```
def _antecedent_candidate(mention: dict, distance: int) -> dict
```

Purpose. Implements the internal antecedent candidate operation.

Mechanics. Delegates to get; returns a derived value or artifact.

Calls. get

Called by. Pipeline.Node_Pipeline.reference_resolution.resolve_references

SOURCE Pipeline/Node_Pipeline/reference_resolution.py:100-117

```
def _candidate_identity(candidate: dict) -> tuple[str, str]
```

Purpose. Implements the internal candidate identity operation.

Mechanics. Delegates to get, normalized; returns a derived value or artifact.

Calls. get, normalized

Called by. Pipeline.Node_Pipeline.reference_resolution.resolve_references

SOURCE Pipeline/Node_Pipeline/reference_resolution.py:120-122

```
def resolve_references(parsed: dict, mentions: list[dict], *, allow_deterministic_auto: bool=True) -> tuple[list[dict], list[dict], list[dict]]
```

Purpose. Resolve generic surface references without ever making them canonical nodes.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `_antecedent_candidate`, `_candidate_identity`, `_reference_context`, `_reference_head`; returns a derived value or artifact.

Calls. `_antecedent_candidate`, `_candidate_identity`, `_reference_context`, `_reference_head`, `_reference_source`, `append`, `casefold`, `defaultdict`, `end`, `endswith`

Called by. `Pipeline.Node_Pipeline.reference_resolution.optional_llm_reference_resolution`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_definite_framework_after_explicit_introduction_auto_resolves`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_that_complementizer_is_not_resolved_as_that_model`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_unique_compatible_vague_reference_is_resolved_to_existing_node`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_ambiguous_vague_reference_requires_human_review`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_weak_definite_reference_is_not_automatically_resolved`

SOURCE `Pipeline/Node_Pipeline/reference_resolution.py:125-300`

```
def llm_resolved_reference(item: dict, candidate: dict, judgment: dict) -> dict
```

Purpose. Implements the internal llm resolved reference operation.

Mechanics. Delegates to `get`, `stable_id`; returns a derived value or artifact.

Calls. `get`, `stable_id`

Called by. `Pipeline.Node_Pipeline.reference_resolution.optional_llm_reference_resolution`

SOURCE `Pipeline/Node_Pipeline/reference_resolution.py:303-345`

```
def optional_llm_reference_resolution(parsed: dict, mentions: list[dict], progress: Callable[[str, str, int], None], *, reference_judge=None) -> tuple[list[dict], list[dict], list[dict], list[dict], dict]
```

Purpose. Compares vague mentions to compatible specific nodes in local context, materializing only grounded SAME decisions and routing ambiguity to audit/review.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `mandatoryllmreferencejudge`, `ontology`, `_llm_resolved_reference`, `all`; returns a derived value or artifact.

Calls. `MandatoryLLMReferenceJudge`, `Ontology`, `_llm_resolved_reference`, `all`, `append`, `defaultdict`, `get`, `items`, `json_read`, `judge`

Called by. `Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction`, `Pipeline.Node_Pipeline.orchestrator.run_extraction`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_contextual_llm_same_maps_vague_mention_to_specific_node`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_contextual_llm_uncertain_keeps_pair_for_human_review`

SOURCE `Pipeline/Node_Pipeline/reference_resolution.py:348-485`

Pipeline/Node_Pipeline/review_workflows.py

```
def apply_node_review(run_dir: Path, candidate_id: str, decision: str, *, ontology_label: str="", canonical_name: str="", reason: str=") -> dict
```

Purpose. Apply a human accept/reject decision and rebuild downstream node artifacts.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `keyerror`, `lexicon`, `valueerror`, `_prior_node_comparison`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `KeyError`, `Lexicon`, `ValueError`, `_prior_node_comparison`, `add_reviewed_aliases`, `all`, `append`, `canonical_entities`, `get`, `json_read`

Called by. `Pipeline.GENERAL.run.Handler`, `Pipeline.GENERAL.run.Handler.do_POST`, `Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_human_accept_adds_only_sentence_grounding_review_node`

SOURCE `Pipeline/Node_Pipeline/review_workflows.py:25-167`

```
def apply_reference_resolution(run_dir: Path, resolution_id: str, decision: str, target_mention_id: str=") -> dict
```

Purpose. Applies a reviewed decision to downstream state.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `keyerror`, `valueerror`, `append`, `canonical_entities`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `KeyError`, `ValueError`, `append`, `canonical_entities`, `get`, `is_file`, `json_read`, `json_write`, `jsonl_read`, `jsonl_write`

Called by. `Pipeline.GENERAL.run.Handler`, `Pipeline.GENERAL.run.Handler.do_POST`, `Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests`,

`Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_human_reference_choice_adds_traceable_mention_without_alias_pollution`

SOURCE `Pipeline/Node_Pipeline/review_workflows.py:170-263`

def apply_resolution(run_dir: Path, resolution_id: str, decision: str, chosen_name: str="") -> dict

Purpose. Applies a reviewed decision to downstream state.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to keyerror, lexicon, valueerror, add_reviewed_aliases; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. KeyError, Lexicon, ValueError, add_reviewed_aliases, append_jsonl, evaluate_question_answerability, json_read, jsonl_read, jsonl_write, materialize_canonical_relationships

Called by. Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_POST

SOURCE Pipeline/Node_Pipeline/review_workflows.py:266-309

Pipeline/Node_Pipeline/span_detection.py

class Span

Purpose. Groups the state and behavior for Span.

Mechanics. Delegates to dataclass.

Calls. dataclass

Called by. Pipeline.Node_Pipeline.candidate_generation.generate_candidates, Pipeline.Node_Pipeline.lexicon.lexicon_spans, Pipeline.Node_Pipeline.span_detection.introduced_system_spans, Pipeline.Node_Pipeline.span_detection.document_system_alias_spans, Pipeline.Node_Pipeline.span_detection.document_model_alias_spans, Pipeline.Node_Pipeline.span_detection.rule_spans, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests

SOURCE Pipeline/Node_Pipeline/span_detection.py:20-28

def _specific_system_name(value: str) -> bool

Purpose. Implements the internal specific system name operation.

Mechanics. Iterates over input records; delegates to any, isalpha, isupper, normalized; returns a derived value or artifact.

Calls. any, isalpha, isupper, normalized, strip

Called by. Pipeline.Node_Pipeline.span_detection.introduced_system_spans

SOURCE Pipeline/Node_Pipeline/span_detection.py:128-138

def introduced_system_spans(text: str) -> list[Span]

Purpose. Discover a named AI system from an explicit introduce/propose construction.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to span, _specific_system_name, append, end; returns a derived value or artifact.

Calls. Span, _specific_system_name, append, end, finditer, group, match, normalized, stable_id, start

Called by. Pipeline.Node_Pipeline.span_detection.rule_spans, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_explicit_introduction_discovers_unseen_system_names

SOURCE Pipeline/Node_Pipeline/span_detection.py:141-157

def document_system_alias_spans(text: str, names: Iterable[str]) -> list[Span]

Purpose. Implements the internal document system alias spans operation.

Mechanics. Iterates over input records; delegates to span, append, end, escape; returns a derived value or artifact.

Calls. Span, append, end, escape, finditer, normalized, replace, stable_id, start

SOURCE Pipeline/Node_Pipeline/span_detection.py:160-170

def document_model_alias_spans(text: str, names: Iterable[str]) -> list[Span]

Purpose. Propagate only model names first grounded by ontology construction.

Mechanics. Iterates over input records; delegates to span, append, end, escape; returns a derived value or artifact.

Calls. Span, append, end, escape, finditer, normalized, replace, stable_id, start

SOURCE Pipeline/Node_Pipeline/span_detection.py:173-184

def rule_spans(text: str) -> list[Span]

Purpose. Implements the internal rule spans operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to span, append, end, extend; returns a derived value or artifact.

Calls. Span, append, end, extend, finditer, get, group, introduced_system_spans, items, match

Called by. Pipeline.GENERAL.tests.test_ner_generalization.OntologyGuidedNERGeneralizationTests,

Pipeline.GENERAL.tests.test_ner_generalization.OntologyGuidedNERGeneralizationTests.test_author_name_ending_in_rag_is_not_an_ai_system,

Pipeline.GENERAL.tests.test_ner_generalization.OntologyGuidedNERGeneralizationTests.test_true_uppercase_rag_product_is_retained,

Pipeline.GENERAL.tests.test_ner_generalization.OntologyGuidedNERGeneralizationTests.test_water_assets_and_named_role_compounds_are_typed,

Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests,

Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_mas_suffix_does_not_turn_thomas_into_ai_system,

Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_generic_the_benchmark_is_not_a_dataset

SOURCE Pipeline/Node_Pipeline/span_detection.py:187-281

def select_spans(spans: list[Span], text: str) -> list[Span]

Purpose. Implements the internal select spans operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to get, strip, values; returns a derived value or artifact.

Calls. get, strip, values

Called by. Pipeline.Node_Pipeline.candidate_generation.generate_candidates, Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests,

Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_nested_candidates_are_retained_for_high_recall_adjudication,

Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_llm_based_adjective_is_not_an_ai_system,

Pipeline.GENERAL.tests.test_node_pipeline.NodePipelineTests.test_vague_model_is_not_a_node_candidate

SOURCE Pipeline/Node_Pipeline/span_detection.py:284-301

Edge Extraction

Pipeline/Edge_Extraction/common.py

def _sentence_rows(parsed: dict) -> list[tuple[dict, dict, dict]]

Purpose. Implements the internal sentence rows operation.

Mechanics. Iterates over input records; delegates to get; returns a derived value or artifact.

Calls. get

Called by. Pipeline.Edge_Extraction.observation_mentions.assemble_observation_mentions

SOURCE Pipeline/Edge_Extraction/common.py:56-62

def _is_current_study_sentence(text: str, section_title: str="") -> bool

Purpose. Conservatively avoid creating quantitative observations for plans/prior work.

Mechanics. Branches on validation or availability conditions; delegates to search; returns a derived value or artifact.

Calls. search

Called by. Pipeline.Edge_Extraction.observation_mentions.assemble_observation_mentions

SOURCE Pipeline/Edge_Extraction/common.py:65-90

Pipeline/Edge_Extraction/extractor.py

def run_relationship_extraction(parsed: dict, mentions: list[dict], run_dir: Path, progress: Callable[[str, str, int], None]) -> dict

Purpose. Builds three-sentence context windows around accepted mentions, invokes the mandatory relationship judge, audits unknown endpoints, and returns assertions plus occurrence-scoped observation mentions.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to edgeextractor, mandatoryllmrelationshipjudge, ontology, assemble_observation_mentions; returns a derived value or artifact.

Calls. EdgeExtractor, MandatoryLLMRelationshipJudge, Ontology, assemble_observation_mentions, extract, get, json_read, jsonl_write, progress, splittiles

Called by. Pipeline.Node_Pipeline.orchestrator.run_extraction

SOURCE Pipeline/Edge_Extraction/extractor.py:21-96

Pipeline/Edge_Extraction/materialization.py

```
def materialize_canonical_relationships(run_dir: Path, entities: list[dict], assertions: list[dict]) -> list[dict]
```

Purpose. Rewires assertion endpoints through reviewed entity merges and aggregates evidence into one canonical relationship per endpoint-predicate tuple.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, defaultdict, get, items; returns a derived value or artifact.

Calls. append, defaultdict, get, items, jsonl_write, stable_id, values

Called by. Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_POST, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.review_workflows.apply_resolution, Pipeline.GENERAL.tests.test_relationship_pipeline.RelationshipPipelineTests, Pipeline.GENERAL.tests.test_relationship_pipeline.RelationshipPipelineTests.test_canonical_relationship_aggregates_distinct_assertion_support

SOURCE Pipeline/Edge_Extraction/materialization.py:19-64

Pipeline/Edge_Extraction/observation_mentions.py

```
def assemble_observation_mentions(parsed: dict, mentions: list[dict]) -> list[dict]
```

Purpose. Create occurrence-scoped Observation nodes for explicit quantitative results.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _is_current_study_sentence, _sentence_rows, add, append; returns a derived value or artifact.

Calls. _is_current_study_sentence, _sentence_rows, add, append, defaultdict, extend, finditer, get, group, search

Called by. Pipeline.Edge_Extraction.extractor.run_relationship_extraction, Pipeline.GENERAL.tests.test_relationship_pipeline.RelationshipPipelineTests, Pipeline.GENERAL.tests.test_relationship_pipeline.RelationshipPipelineTests.test_quantitative_result_creates_condition_aware_observation, Pipeline.GENERAL.tests.test_relationship_pipeline.RelationshipPipelineTests.test_figure_number_is_not_misread_as_hour_measurement, Pipeline.GENERAL.tests.test_relationship_pipeline.RelationshipPipelineTests.test_existing_observation_is_not_duplicated

SOURCE Pipeline/Edge_Extraction/observation_mentions.py:20-101

Question Evaluation

Pipeline/Question_Evaluation/evaluator.py

```
def _answer_rows(key: str, relationships: list[dict], entities: dict[str, dict]) -> list[str]
```

Purpose. Implements the internal answer rows operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, get; returns a derived value or artifact.

Calls. append, get

Called by. Pipeline.Question_Evaluation.evaluator.evaluate_question_answerability

SOURCE Pipeline/Question_Evaluation/evaluator.py:23-41

```
def evaluate_question_answerability(run_dir: Path, entities: list[dict], relationships: list[dict], assertions: list[dict]) -> dict
```

Purpose. Checks whether configured research questions can be answered using materialized graph facts with exact supporting evidence; prose alone cannot count as an answer.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _answer_rows, add, append, defaultdict; returns a derived value or artifact.

Calls. _answer_rows, add, append, defaultdict, get, json_write, round

Called by. Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_POST, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.review_workflows.apply_resolution, Pipeline.GENERAL.tests.test_question_evaluation.QuestionEvaluationTests, Pipeline.GENERAL.tests.test_question_evaluation.QuestionEvaluationTests.test_question_requires_graph_fact_and_exact_evidence

SOURCE Pipeline/Question_Evaluation/evaluator.py:44-99

Validation

Pipeline/Validation/generalization_diagnostics.py

```
def _sentences(parsed: dict) -> list[dict]
```

Purpose. Implements the internal sentences operation.

Mechanics. Iterates over input records; delegates to get; returns a derived value or artifact.

Calls. get

Called by. Pipeline.Validation.generalization_diagnostics.build_document_profile, Pipeline.Validation.semantic_completeness.evaluate_semantic_completeness

SOURCE Pipeline/Validation/generalization_diagnostics.py:26-32

def _normalized(value: str) -> str

Purpose. Implements the internal normalized operation.

Mechanics. Delegates to casefold, join, split, sub; returns a derived value or artifact.

Calls. casefold, join, split, sub

Called by. Pipeline.Sector_Discovery.discovery._descriptor_tokens, Pipeline.Sector_Discovery.discovery._cluster_system_candidates, Pipeline.Sector_Discovery.discovery._agent_candidates, Pipeline.Sector_Discovery.discovery._valid_model_surface, Pipeline.Sector_Discovery.discovery._model_candidates, Pipeline.Sector_Discovery.discovery._profile_canonical, Pipeline.Sector_Discovery.discovery._canonicalize_counted_datasets

SOURCE Pipeline/Validation/generalization_diagnostics.py:35-36

def build_document_profile(parsed: dict) -> dict

Purpose. Infer a read-only extraction profile; it never creates graph facts.

Mechanics. Iterates over input records; delegates to _sentences, get, items, join; returns a derived value or artifact.

Calls. _sentences, get, items, join, search

Called by. Pipeline.Validation.generalization_diagnostics.build_generalization_diagnostics

SOURCE Pipeline/Validation/generalization_diagnostics.py:39-70

def build_generalization_diagnostics(parsed: dict, mentions: list[dict], entities: list[dict], relationships: list[dict], assertions: list[dict], question_evaluation: dict) -> dict

Purpose. Assembles a derived artifact.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to counter, _normalized, append, build_document_profile; returns a derived value or artifact.

Calls. Counter, _normalized, append, build_document_profile, casefold, get, items, round, rsplit

Called by. Pipeline.Node_Pipeline.orchestrator.run_extraction

SOURCE Pipeline/Validation/generalization_diagnostics.py:73-161

def aggregate_llm_usage(node_calls: list[dict], reference_calls: list[dict], relationship_calls: list[dict], *, accepted_mentions: int, assertions: int) -> dict

Purpose. Implements the internal aggregate llm usage operation.

Mechanics. Iterates over input records; delegates to get, items, round; returns a derived value or artifact.

Calls. get, items, round

Called by. Pipeline.Node_Pipeline.orchestrator.run_extraction

SOURCE Pipeline/Validation/generalization_diagnostics.py:164-214

Pipeline/Validation/semantic_completeness.py

def _sentences(parsed: dict) -> list[dict]

Purpose. Implements the internal sentences operation.

Mechanics. Iterates over input records; delegates to get; returns a derived value or artifact.

Calls. get

Called by. Pipeline.Validation.generalization_diagnostics.build_document_profile, Pipeline.Validation.semantic_completeness.evaluate_semantic_completeness

SOURCE Pipeline/Validation/semantic_completeness.py:9-15

def _signal(sentences: list[dict], pattern: str) -> list[dict]

Purpose. Implements the internal signal operation.

Mechanics. Iterates over input records; delegates to compile, get, search; returns a derived value or artifact.

Calls. compile, get, search

Called by. Pipeline.Validation.semantic_completeness.evaluate_semantic_completeness

SOURCE Pipeline/Validation/semantic_completeness.py:18-24

```
def evaluate_semantic_completeness(parsed: dict, entities: list[dict], relationships: list[dict], assertions: list[dict], question_evaluation: dict) -> dict
```

Purpose. Reports likely domain-recall gaps separately from structural validity, producing an interpretable coverage status and score.

Mechanics. Iterates over input records; delegates to `_sentences`, `_signal`, `get`, `round`; returns a derived value or artifact.

Calls. `_sentences`, `_signal`, `get`, `round`

Called by. `Pipeline.Node_Pipeline.orchestrator.run_extraction`, `Pipeline.GENERAL.tests.test_semantic_completeness.SemanticCompletenessTests`, `Pipeline.GENERAL.tests.test_semantic_completeness.SemanticCompletenessTests.test_missing_expected_core_nodes_is_reported_as_incomplete`, `Pipeline.GENERAL.tests.test_semantic_completeness.SemanticCompletenessTests.test_grounded_connected_core_with_traceability_is_complete`

SOURCE `Pipeline/Validation/semantic_completeness.py:27-132`

Graph Persistence

Pipeline/Graph_Persistence/neo4j_writer.py

```
def _post(statements: list[dict], *, timeout: int=45) -> dict
```

Purpose. Implements the internal post operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `request`, `runtimeerror`, `decode`, `dumps`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `Request`, `RuntimeError`, `decode`, `dumps`, `encode`, `get`, `join`, `loads`, `read`, `urlopen`

Called by. `Pipeline.Graph_Persistence.neo4j_writer.ensure_ontology_foundation`, `Pipeline.Graph_Persistence.neo4j_writer.apply_ontology_labels`, `Pipeline.Graph_Persistence.neo4j_writer.graph_summary`, `Pipeline.Graph_Persistence.neo4j_writer.refresh_display_properties`, `Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run`, `Pipeline.Graph_Persistence.neo4j_writer.align_reusable_entity_ids`, `Pipeline.Graph_Persistence.neo4j_writer.upsert_run`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py:17-36`

```
def _statement(source: str, rows: list[dict] | None=None) -> dict
```

Purpose. Implements the internal statement operation.

Mechanics. Returns a derived value or artifact.

Called by. `Pipeline.Graph_Persistence.neo4j_writer.ensure_ontology_foundation`, `Pipeline.Graph_Persistence.neo4j_writer.apply_ontology_labels`, `Pipeline.Graph_Persistence.neo4j_writer.graph_summary`, `Pipeline.Graph_Persistence.neo4j_writer.refresh_display_properties`, `Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run`, `Pipeline.Graph_Persistence.neo4j_writer.align_reusable_entity_ids`, `Pipeline.Graph_Persistence.neo4j_writer.upsert_run`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py:39-40`

```
def ensure_ontology_foundation() -> dict
```

Purpose. Idempotently materialize the authoritative ontology before instance data.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `runtimeerror`, `valueerror`, `_post`, `_statement`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `RuntimeError`, `ValueError`, `_post`, `_statement`, `append`, `dumps`, `fullmatch`, `get`, `items`, `json_read`

Called by. `Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py:43-116`

```
def apply_ontology_labels(entity_rows: list[dict]) -> dict
```

Purpose. Apply exactly one concrete ontology label to every extracted entity.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `valueerror`, `_post`, `_statement`, `append`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `ValueError`, `_post`, `_statement`, `append`, `defaultdict`, `fullmatch`, `items`, `json_read`

Called by. `Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run`, `Pipeline.Graph_Persistence.neo4j_writer.upsert_run`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests.test_concrete_ontology_labels_are_applied_to_canonical_entities`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests.test_unknown_entity_class_is_rejected_before_neo4j_write`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py:119-149`

def validate_entities_against_ontology(entities: list[dict]) -> dict

Purpose. Reject unknown classes and duplicate real-world nodes before any graph write.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `valueerror`, `add`, `append`, `casefold`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `ValueError`, `add`, `append`, `casefold`, `fullmatch`, `get`, `join`, `json_read`, `strip`, `sub`

Called by. `Pipeline.Graph_Persistence.neo4j_writer.upsert_run`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests.test_duplicate_canonical_nodes_are_rejected`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests.test_scoped_observations_may_have_the_same_display_value`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests.test_provenance_classes_cannot_become_instance_nodes`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py`:152-182

def build_entity_rows(run_id: str, document: dict, entities: list[dict], mentions: list[dict]) -> list[dict]

Purpose. Aggregate mentions into canonical nodes with exact parsed-source provenance.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `valueerror`, `add`, `append`, `dumps`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `ValueError`, `add`, `append`, `dumps`, `get`, `join`, `loads`, `next`, `now_iso`, `stable_id`

Called by. `Pipeline.Graph_Persistence.neo4j_writer.upsert_run`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests.test_mentions_are_embedded_on_one_canonical_ontology_node`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests.test_document_metadata_mentions_use_exact_field_provenance`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py`:185-305

def validate_relationships_against_ontology(entities: list[dict], relationships: list[dict]) -> dict

Purpose. Reject predicates and endpoints that violate the ontology domain/range.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `valueerror`, `append`, `get`, `join`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `ValueError`, `append`, `get`, `join`, `json_read`

Called by. `Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run`, `Pipeline.Graph_Persistence.neo4j_writer.upsert_run`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests.test_relationship_domain_and_range_are_enforced`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests.test_undefined_relationship_is_rejected`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py`:308-332

def graph_summary() -> dict

Purpose. Return health counts for the paper-only semantic graph.

Mechanics. Iterates over input records; delegates to `_post`, `_statement`; returns a derived value or artifact.

Calls. `_post`, `_statement`

Called by. `Pipeline.GENERAL.node_only_server.NodeOnlyHandler`, `Pipeline.GENERAL.node_only_server.NodeOnlyHandler.do_GET`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py`:335-401

def refresh_display_properties() -> None

Purpose. Make the ontology type visible while retaining extracted content separately.

Mechanics. Delegates to `_post`, `_statement`.

Calls. `_post`, `_statement`

Called by. `Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run`, `Pipeline.Graph_Persistence.neo4j_writer.upsert_run`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py`:404-443

def _legacy_upsert_run(run_dir, parsed: dict, entities: list[dict], relationships: list[dict]) -> dict

Purpose. Idempotently add one extraction run and its evidence graph to local Neo4j.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `runtimeerror`, `valueerror`, `_post`, `_statement`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `RuntimeError`, `ValueError`, `_post`, `_statement`, `append`, `apply_ontology_labels`, `casefold`, `defaultdict`, `ensure_ontology_foundation`, `fullmatch`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py`:446-769

```
def align_reusable_entity_ids(entities: list[dict], relationships: list[dict], document_id: str) -> tuple[list[dict], list[dict], dict[str, str]]
```

Purpose. Reuse an existing graph ID for the same canonical ontology identity across papers.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `_post`, `_statement`, `all`, `append`; returns a derived value or artifact.

Calls. `_post`, `_statement`, `all`, `append`, `casefold`, `defaultdict`, `get`, `strip`

Called by. `Pipeline.Graph_Persistence.neo4j_writer.upsert_run`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests`, `Pipeline.GENERAL.tests.test_neo4j_writer.Neo4jWriterTests.test_reusable_identity_aligns_to_another_paper_and_rewires_relationships`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py:772-817`

```
def upsert_run(run_dir, parsed: dict, entities: list[dict], relationships: list[dict]) -> dict
```

Purpose. Validates ontology types, predicates, domains, ranges, and uniqueness before idempotently writing the paper-only evidence graph to local Neo4j.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `runtimeerror`, `valueerror`, `_post`, `_statement`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `RuntimeError`, `ValueError`, `_post`, `_statement`, `align_reusable_entity_ids`, `append`, `apply_ontology_labels`, `build_entity_rows`, `casefold`, `defaultdict`

Called by. `Pipeline.GENERAL.run.Handler`, `Pipeline.GENERAL.run.Handler.do_POST`, `Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction`, `Pipeline.Node_Pipeline.orchestrator.run_extraction`, `Pipeline.GENERAL.tools.rematerialize_node_run.main`

SOURCE `Pipeline/Graph_Persistence/neo4j_writer.py:820-1080`

GENERAL

Pipeline/GENERAL/node_only_server.py

```
class NodeOnlyHandler(base.Handler)
```

Purpose. Groups the state and behavior for `NodeOnlyHandler`.

Mechanics. Branches on validation or availability conditions; delegates to `do_get`, `graph_summary`, `json_read`, `read_bytes`.

Calls. `do_GET`, `graph_summary`, `json_read`, `read_bytes`, `send_bytes`, `send_json`, `super`, `unquote`, `urlparse`

SOURCE `Pipeline/GENERAL/node_only_server.py:15-42`

```
def do_GET(self) -> None
```

Purpose. Handles an HTTP request.

Mechanics. Branches on validation or availability conditions; delegates to `graph_summary`, `json_read`, `read_bytes`, `send_bytes`.

Calls. `do_GET`, `graph_summary`, `json_read`, `read_bytes`, `send_bytes`, `send_json`, `super`, `unquote`, `urlparse`

Called by. `Pipeline.GENERAL.node_only_server.NodeOnlyHandler`

SOURCE `Pipeline/GENERAL/node_only_server.py:16-42`

```
def main() -> None
```

Purpose. Provides the command-line or server entry point.

Mechanics. Branches on validation or availability conditions; delegates to `threadinghttpserver`, `load_existing_jobs`, `serve_forever`, `server_close`.

Calls. `ThreadingHTTPServer`, `load_existing_jobs`, `serve_forever`, `server_close`

SOURCE `Pipeline/GENERAL/node_only_server.py:45-55`

Pipeline/GENERAL/run.py

```
def advance_stage_reports(current_rows: list[dict], stage: str, message: str, value: int, report: dict | None=None) -> list[dict]
```

Purpose. Keep completed/running stage state accurate for live and finished runs.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `index`, `values`; returns a derived value or artifact.

Calls. `index`, `values`

Called by. `Pipeline.GENERAL.run.launch`, `Pipeline.GENERAL.run.launch.progress`

SOURCE `Pipeline/GENERAL/run.py:43-59`

def utcnow() -> str

Purpose. Implements the internal utcnow operation.

Mechanics. Delegates to isoformat, now; returns a derived value or artifact.

Calls. isoformat, now

Called by. Pipeline.GENERAL.run.launch

SOURCE Pipeline/GENERAL/run.py:62-63

def safe_name(value: str) -> str

Purpose. Implements the internal safe name operation.

Mechanics. Delegates to path, strip, sub; returns a derived value or artifact.

Calls. Path, strip, sub

Called by. Pipeline.GENERAL.run.launch, Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.send_bytes, Pipeline.GENERAL.run.Handler.do_POST

SOURCE Pipeline/GENERAL/run.py:66-69

def load_existing_jobs() -> None

Purpose. Reads and normalizes persisted state.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to get, glob, json_read, mkdir.

Calls. get, glob, json_read, mkdir

Called by. Pipeline.GENERAL.node_only_server.main, Pipeline.GENERAL.run.main

SOURCE Pipeline/GENERAL/run.py:72-83

def job_payload(job_id: str, *, details: bool=False) -> dict

Purpose. Returns lightweight live job state or, after completion, hydrates the response with review queues, entities, relationships, questions, validation, and download metadata.

Mechanics. Branches on validation or availability conditions; delegates to get, is_file, json_read, jsonl_read; returns a derived value or artifact.

Calls. get, is_file, json_read, jsonl_read

Called by. Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_GET

SOURCE Pipeline/GENERAL/run.py:86-108

def launch(filename: str, pdf_bytes: bytes) -> dict

Purpose. Creates a run ID, stores the uploaded PDF, registers job state, and starts the canonical orchestrator on a background thread.

Mechanics. Branches on validation or availability conditions; delegates to thread, advance_stage_reports, get, mkdir; returns a derived value or artifact.

Calls. Thread, advance_stage_reports, get, mkdir, now, print_exc, run_extraction, safe_name, stable_id, start

Called by. Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_POST

SOURCE Pipeline/GENERAL/run.py:111-148

def progress(stage: str, message: str, value: int, report: dict | None=None) -> None

Purpose. Implements the internal progress operation.

Mechanics. Delegates to advance_stage_reports, get, update.

Calls. advance_stage_reports, get, update

Called by. Pipeline.Edge_Extraction.extractor.run_relationship_extraction, Pipeline.Node_Pipeline.adjudication.optional_ilm_review, Pipeline.Node_Pipeline.node_only_runner.run_node_only_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction, Pipeline.Node_Pipeline.orchestrator.run_extraction.emit, Pipeline.Node_Pipeline.reference_resolution.optional_ilm_reference_resolution

SOURCE Pipeline/GENERAL/run.py:125-131

def worker() -> None

Purpose. Implements the internal worker operation.

Mechanics. Branches on validation or availability conditions; delegates to print_exc, run_extraction, type, update.

Calls. print_exc, run_extraction, type, update

SOURCE Pipeline/GENERAL/run.py:133-145

class Handler(BaseHTTPRequestHandler)

Purpose. Groups the state and behavior for Handler.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to bytesio, bytesparser, path, valueerror; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. BytesIO, BytesParser, Path, ValueError, ZipFile, any, apply_node_review, apply_reference_resolution, apply_resolution, decode

SOURCE Pipeline/GENERAL/run.py:151-421

def log_message(self, fmt: str, *args: object) -> None

Purpose. Implements the internal log message operation.

Mechanics. Delegates to log_date_time_string.

Calls. log_date_time_string

SOURCE Pipeline/GENERAL/run.py:154-155

def send_json(self, payload: object, status: int=200) -> None

Purpose. Formats and sends a response.

Mechanics. Delegates to dumps, encode, end_headers, send_header.

Calls. dumps, encode, end_headers, send_header, send_response, write

Called by. Pipeline.GENERAL.node_only_server.NodeOnlyHandler, Pipeline.GENERAL.node_only_server.NodeOnlyHandler.do_GET, Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_GET, Pipeline.GENERAL.run.Handler.do_POST

SOURCE Pipeline/GENERAL/run.py:157-164

def send_bytes(self, data: bytes, content_type: str, filename: str | None=None) -> None

Purpose. Formats and sends a response.

Mechanics. Branches on validation or availability conditions; delegates to end_headers, safe_name, send_header, send_response.

Calls. end_headers, safe_name, send_header, send_response, write

Called by. Pipeline.GENERAL.node_only_server.NodeOnlyHandler, Pipeline.GENERAL.node_only_server.NodeOnlyHandler.do_GET, Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_GET

SOURCE Pipeline/GENERAL/run.py:166-173

def read_json(self) -> dict

Purpose. Loads serialized input.

Mechanics. Branches on validation or availability conditions; delegates to valueerror, decode, get, loads; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. ValueError, decode, get, loads, read

Called by. Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_POST, Pipeline.GENERAL.tools.evaluate_fixed_corpus_run.main, Pipeline.GENERAL.tools.evaluate_regression_corpus.latest_run, Pipeline.GENERAL.tools.evaluate_regression_corpus.evaluate_case, Pipeline.GENERAL.tools.evaluate_regression_corpus.main

SOURCE Pipeline/GENERAL/run.py:175-179

def do_GET(self) -> None

Purpose. Handles an HTTP request.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to bytesio, path, zipfile, any.

Calls. BytesIO, Path, ZipFile, any, end_headers, fullmatch, get, getvalue, group, groups

Called by. Pipeline.GENERAL.node_only_server.NodeOnlyHandler, Pipeline.GENERAL.node_only_server.NodeOnlyHandler.do_GET

SOURCE Pipeline/GENERAL/run.py:181-270

def do_POST(self) -> None

Purpose. Handles an HTTP request.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to bytesparser, valueerror, apply_node_review, apply_reference_resolution; fails explicitly when an invariant is broken.

Calls. BytesParser, ValueError, apply_node_review, apply_reference_resolution, apply_resolution, encode, evaluate_question_answerability, fullmatch, get, get_filename

SOURCE Pipeline/GENERAL/run.py:272-421

def main() -> None

Purpose. Provides the command-line or server entry point.

Mechanics. Branches on validation or availability conditions; delegates to `threadinghttpserver`, `load_existing_jobs`, `serve_forever`, `server_close`.

Calls. `ThreadingHTTPServer`, `load_existing_jobs`, `serve_forever`, `server_close`

SOURCE `Pipeline/GENERAL/run.py:424-434`

Sector Discovery

Pipeline/Sector_Discovery/discovery.py

class GroundedCandidate

Purpose. Groups the state and behavior for `GroundedCandidate`.

Mechanics. Delegates to `dataclass`.

Calls. `dataclass`

Called by. `Pipeline.Sector_Discovery.discovery._cluster_system_candidates`, `Pipeline.Sector_Discovery.discovery._agent_candidates`, `Pipeline.Sector_Discovery.discovery._model_candidates`, `Pipeline.Sector_Discovery.discovery._profile_pattern_candidates`

SOURCE `Pipeline/Sector_Discovery/discovery.py:13-20`

def _normalized(value: str) -> str

Purpose. Implements the internal normalized operation.

Mechanics. Delegates to `casefold`, `join`, `split`, `sub`; returns a derived value or artifact.

Calls. `casefold`, `join`, `split`, `sub`

Called by. `Pipeline.Sector_Discovery.discovery._descriptor_tokens`, `Pipeline.Sector_Discovery.discovery._cluster_system_candidates`, `Pipeline.Sector_Discovery.discovery._agent_candidates`, `Pipeline.Sector_Discovery.discovery._valid_model_surface`, `Pipeline.Sector_Discovery.discovery._model_candidates`, `Pipeline.Sector_Discovery.discovery._profile_canonical`, `Pipeline.Sector_Discovery.discovery._canonicalize_counted_datasets`

SOURCE `Pipeline/Sector_Discovery/discovery.py:23-24`

def _stable_id(prefix: str, *values: object) -> str

Purpose. Implements the internal stable id operation.

Mechanics. Iterates over input records; delegates to `encode`, `hexdigest`, `join`, `sha256`; returns a derived value or artifact.

Calls. `encode`, `hexdigest`, `join`, `sha256`

Called by. `Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor`, `Pipeline.Grammatical_Parsing.candidates.SpacyPredicateTripleExtractor._analyze_doc`, `Pipeline.Sector_Discovery.discovery._cluster_system_candidates`, `Pipeline.Sector_Discovery.discovery._agent_candidates`, `Pipeline.Sector_Discovery.discovery._model_candidates`, `Pipeline.Sector_Discovery.discovery._canonicalize_counted_datasets`, `Pipeline.Sector_Discovery.discovery._profile_pattern_candidates`

SOURCE `Pipeline/Sector_Discovery/discovery.py:27-29`

def load_sector_profile(path: Path) -> dict

Purpose. Reads and normalizes persisted state.

Mechanics. Branches on validation or availability conditions; delegates to `valueerror`, `get`, `join`, `loads`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `ValueError`, `get`, `join`, `loads`, `read_text`

SOURCE `Pipeline/Sector_Discovery/discovery.py:32-39`

def _descriptor_tokens(value: str, profile: dict) -> set[str]

Purpose. Implements the internal descriptor tokens operation.

Mechanics. Iterates over input records; delegates to `_normalized`, `get`, `split`; returns a derived value or artifact.

Calls. `_normalized`, `get`, `split`

Called by. `Pipeline.Sector_Discovery.discovery._cluster_system_candidates`

SOURCE `Pipeline/Sector_Discovery/discovery.py:74-76`

def _cluster_system_candidates(rows: list[dict], document_id: str, profile: dict) -> dict[str, list[GroundedCandidate]]

Purpose. Implements the internal cluster system candidates operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to groundedcandidate, _descriptor_tokens, _normalized, _stable_id; returns a derived value or artifact.

Calls. GroundedCandidate, _descriptor_tokens, _normalized, _stable_id, append, casefold, defaultdict, endswith, find, finditer

Called by. Pipeline.Sector_Discovery.discovery.discover_document_candidates

SOURCE Pipeline/Sector_Discovery/discovery.py:79-131

def find(index: int) -> int

Purpose. Searches indexed or source records for matches.

Mechanics. Iterates over input records; returns a derived value or artifact.

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor,
Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor._candidate_from_proposal,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._sentence_candidates,
Pipeline.Node_Pipeline.canonicalization.UnionFind, Pipeline.Node_Pipeline.canonicalization.UnionFind.union,
Pipeline.Node_Pipeline.canonicalization.materialize_merges

SOURCE Pipeline/Sector_Discovery/discovery.py:97-101

def union(left: int, right: int) -> None

Purpose. Implements the internal union operation.

Mechanics. Branches on validation or availability conditions; delegates to find.

Calls. find

Called by. Pipeline.Node_Pipeline.canonicalization.materialize_merges, Pipeline.Sector_Discovery.discovery._cluster_system_candidates

SOURCE Pipeline/Sector_Discovery/discovery.py:103-106

def _discover_agent_aliases(rows: list[dict]) -> dict[tuple[str, str], str]

Purpose. Implements the internal discover agent aliases operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, casefold, defaultdict, finditer; returns a derived value or artifact.

Calls. append, casefold, defaultdict, finditer, group, items, search

Called by. Pipeline.Sector_Discovery.discovery._agent_candidates, Pipeline.GENERAL.tests.test_generalization_repairs.AgentEnumerationRepairTests,
Pipeline.GENERAL.tests.test_generalization_repairs.AgentEnumerationRepairTests.test_acronym_plus_ordinary_words_do_not_become_agents

SOURCE Pipeline/Sector_Discovery/discovery.py:134-161

def _agent_candidates(rows: list[dict], document_id: str) -> dict[str, list[GroundedCandidate]]

Purpose. Implements the internal agent candidates operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to groundedcandidate, _discover_agent_aliases, _normalized, _stable_id; returns a derived value or artifact.

Calls. GroundedCandidate, _discover_agent_aliases, _normalized, _stable_id, append, casefold, defaultdict, end, finditer, get

Called by. Pipeline.Sector_Discovery.discovery.discover_document_candidates,
Pipeline.GENERAL.tests.test_generalization_repairs.AgentEnumerationRepairTests,
Pipeline.GENERAL.tests.test_generalization_repairs.AgentEnumerationRepairTests.test_acronym_plus_ordinary_words_do_not_become_agents,
Pipeline.GENERAL.tests.test_generalization_repairs.AgentEnumerationRepairTests.test_explicit_local_identifier_family_is_retained

SOURCE Pipeline/Sector_Discovery/discovery.py:163-176

def _valid_model_surface(surface: str, *, allow_uppercase: bool=False) -> bool

Purpose. Implements the internal valid model surface operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _normalized, any, fullmatch, isdigit; returns a derived value or artifact.

Calls. _normalized, any, fullmatch, isdigit, isupper, search

Called by. Pipeline.Sector_Discovery.discovery._model_candidates

SOURCE Pipeline/Sector_Discovery/discovery.py:179-195

```
def _model_candidates(rows: list[dict], document_id: str, profile: dict) -> dict[str, list[GroundedCandidate]]
```

Purpose. Implements the internal model candidates operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to groundedcandidate, _normalized, _stable_id, _valid_model_surface; returns a derived value or artifact.

Calls. GroundedCandidate, _normalized, _stable_id, _valid_model_surface, abs, add, append, defaultdict, end, escape

Called by. Pipeline.Sector_Discovery.discovery.discover_document_candidates,

Pipeline.GENERAL.tests.test_ner_generalization.OntologyGuidedNERGeneralizationTests, Pipeline.GENERAL.tests.test_ner_generalization.OntologyGuidedNERGeneralizationTests.test_common_water_and_chemical_names_are_not_ai_models,

Pipeline.GENERAL.tests.test_ner_generalization.OntologyGuidedNERGeneralizationTests.test_named_model_near_model_cue_is_retained, Pipeline.GENERAL.tests.test_ner_generalization.OntologyGuidedNERGeneralizationTests.test_uppercase_scientific_models_near_model_cues_are_retained

SOURCE Pipeline/Sector_Discovery/discovery.py:204-241

```
def _profile_canonical(label: str, surface: str) -> str
```

Purpose. Implements the internal profile canonical operation.

Mechanics. Branches on validation or availability conditions; delegates to _normalized, get, group, search; returns a derived value or artifact.

Calls. _normalized, get, group, search, sub

Called by. Pipeline.Sector_Discovery.discovery._profile_pattern_candidates

SOURCE Pipeline/Sector_Discovery/discovery.py:245-271

```
def _canonicalize_counted_datasets(result: dict[str, list[GroundedCandidate]], document_id: str) -> None
```

Purpose. Implements the internal canonicalize counted datasets operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _normalized, _stable_id, append, casefold.

Calls. _normalized, _stable_id, append, casefold, defaultdict, enumerate, group, items, match, replace

Called by. Pipeline.Sector_Discovery.discovery._profile_pattern_candidates

SOURCE Pipeline/Sector_Discovery/discovery.py:274-291

```
def _profile_pattern_candidates(rows: list[dict], document_id: str, profile: dict) -> dict[str, list[GroundedCandidate]]
```

Purpose. Implements the internal profile pattern candidates operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to groundedcandidate, _canonicalize_counted_datasets, _normalized, _profile_canonical; returns a derived value or artifact.

Calls. GroundedCandidate, _canonicalize_counted_datasets, _normalized, _profile_canonical, _stable_id, any, append, defaultdict, end, finditer

Called by. Pipeline.Sector_Discovery.discovery.discover_document_candidates

SOURCE Pipeline/Sector_Discovery/discovery.py:294-328

```
def discover_document_candidates(rows: Iterable[dict], document_id: str, profile: dict) -> dict[str, list[GroundedCandidate]]
```

Purpose. Discover exact, ontology-typed spans without relying on a previously seen lexicon entry.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _agent_candidates, _cluster_system_candidates, _model_candidates, _profile_pattern_candidates; returns a derived value or artifact.

Calls. _agent_candidates, _cluster_system_candidates, _model_candidates, _profile_pattern_candidates, add, append, defaultdict, items

Called by. Pipeline.GENERAL.tests.test_sector_discovery.SectorDiscoveryTests,

Pipeline.GENERAL.tests.test_sector_discovery.SectorDiscoveryTests.flatten

SOURCE Pipeline/Sector_Discovery/discovery.py:330-348

Knowledge Graph Core

Pipeline/Knowledge_Graph_Core/edge_extractor.py

```
class EdgeExtractor
```

Purpose. LLM-only semantic relationship extraction over three-sentence windows.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _assertion_from_candidate, _candidate_from_proposal, _context_windows, _metadata_assertion; returns a derived value or artifact.

Calls. _assertion_from_candidate, _candidate_from_proposal, _context_windows, _metadata_assertion, _node_catalog, _unique_exact_node_match, _unknown_node, all, any, append

Called by. Pipeline.Edge_Extraction.extractor.run_relationship_extraction, Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline,

Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests,

Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_edge_extractor_uses_llm_selected_existing_nodes_and_all_context,

Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_edge_extractor_audits_unseen_endpoint_without_creating_edge

SOURCE Pipeline/Knowledge_Graph_Core/edge_extractor.py:10-524

```
def __init__(self, ontology: Ontology, relationship_judge=None, config: dict | None=None) -> None
```

Purpose. Implements the internal init operation.

Mechanics. Delegates to get.

Calls. get

Called by. Pipeline.Knowledge_Graph_Core.Ilm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.Ilm_reference_judge.MandatoryLLMReferenceJudge.__init__, Pipeline.Knowledge_Graph_Core.Ilm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.Ilm_relationship_judge.MandatoryLLMRelationshipJudge.__init__, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases

SOURCE Pipeline/Knowledge_Graph_Core/edge_extractor.py:18-32

```
def extract(self, parsed: dict, mentions: list[dict]) -> tuple[list[dict], list[dict]]
```

Purpose. Locates and materializes evidence-backed records.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _assertion_from_candidate, _candidate_from_proposal, _context_windows, _metadata_assertion; returns a derived value or artifact.

Calls. _assertion_from_candidate, _candidate_from_proposal, _context_windows, _metadata_assertion, _node_catalog, _unknown_node, append, casefold, enumerate, extract

Called by. Pipeline.Edge_Extraction.extractor.run_relationship_extraction, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_ilm_judges_candidates_and_discovers_out_of_lexicon_node, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_edge_extractor_uses_ilm_selected_existing_nodes_and_all_context, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_edge_extractor_audits_unseen_endpoint_without_creating_edge

SOURCE Pipeline/Knowledge_Graph_Core/edge_extractor.py:34-146

```
def _context_windows(parsed: dict) -> tuple[dict[str, dict], list[dict]]
```

Purpose. Implements the internal context windows operation.

Mechanics. Iterates over input records; delegates to append, enumerate, get, range; returns a derived value or artifact.

Calls. append, enumerate, get, range

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/edge_extractor.py:149-183

```
def _node_catalog(mentions: list[dict]) -> list[dict]
```

Purpose. Implements the internal node catalog operation.

Mechanics. Iterates over input records; delegates to get; returns a derived value or artifact.

Calls. get

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/edge_extractor.py:186-192

```
def _candidate_from_proposal(self, parsed: dict, window: dict, proposal: dict, proposal_number: int, mention_index: dict[str, dict], sentence_index: dict[str, dict], audit: dict) -> dict
```

Purpose. Implements the internal candidate from proposal operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _unique_exact_node_match, all, any, find; returns a derived value or artifact.

Calls. _unique_exact_node_match, all, any, find, get, join, stable_id, valid_relationship, values

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/edge_extractor.py:194-362

```
def _unique_exact_node_match(text: object, proposed_label: object, mention_index: dict[str, dict]) -> dict | None
```

Purpose. Implements the internal unique exact node match operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to casefold, get, iter, next; returns a derived value or artifact.

Calls. casefold, get, iter, next, strip, values

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor._candidate_from_proposal

SOURCE Pipeline/Knowledge_Graph_Core/edge_extractor.py:365-383

def _unknown_node(candidate: dict, proposal: dict, role: str) -> dict

Purpose. Implements the internal unknown node operation.

Mechanics. Delegates to get, stable_id; returns a derived value or artifact.

Calls. get, stable_id

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/edge_extractor.py:386-412

def _assertion_from_candidate(candidate: dict) -> dict

Purpose. Implements the internal assertion from candidate operation.

Mechanics. Delegates to join, round, stable_id; returns a derived value or artifact.

Calls. join, round, stable_id

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/edge_extractor.py:415-460

def _metadata_assertion(predicate: str, source: dict, target: dict, quote: str) -> dict

Purpose. Implements the internal metadata assertion operation.

Mechanics. Delegates to stable_id; returns a derived value or artifact.

Calls. stable_id

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/edge_extractor.py:463-524

Pipeline/Knowledge_Graph_Core/embeddings.py

class SentenceEmbeddingBackend

Purpose. Lazy local sentence-transformer backend used by entity resolution.

Mechanics. Branches on validation or availability conditions; delegates to runtimeerror, sentencetransformer, encode; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. RuntimeError, SentenceTransformer, encode

Called by. Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver, Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver.__init__

SOURCE Pipeline/Knowledge_Graph_Core/embeddings.py:9-33

def __init__(self, model_name: str=DEFAULT_MODEL, batch_size: int=64, model=None)

Purpose. Implements the internal init operation.

Mechanics. Branches on validation or availability conditions; delegates to runtimeerror, sentencetransformer; fails explicitly when an invariant is broken.

Calls. RuntimeError, SentenceTransformer

Called by. Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge.__init__, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.__init__, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases

SOURCE Pipeline/Knowledge_Graph_Core/embeddings.py:12-24

def encode(self, texts: Sequence[str])

Purpose. Implements the internal encode operation.

Mechanics. Returns a derived value or artifact.

Calls. encode

Called by. Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.send_json, Pipeline.GENERAL.run.Handler.do_POST, Pipeline.Grammatical_Parsing.candidates._stable_id, Pipeline.Grammatical_Parsing.candidates.SciBERTOntologyTyper, Pipeline.Grammatical_Parsing.candidates.SciBERTOntologyTyper.__init__, Pipeline.Grammatical_Parsing.candidates.SciBERTOntologyTyper.type_arguments

SOURCE Pipeline/Knowledge_Graph_Core/embeddings.py:26-33

Pipeline/Knowledge_Graph_Core/entity_resolution.py

def _tokens(value: str) -> set[str]

Purpose. Implements the internal tokens operation.

Mechanics. Delegates to casefold, findall; returns a derived value or artifact.

Calls. casefold, findall

Called by. Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver, Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver.calibrate

SOURCE Pipeline/Knowledge_Graph_Core/entity_resolution.py:15-16

def _acronym(value: str) -> str

Purpose. Implements the internal acronym operation.

Mechanics. Iterates over input records; delegates to casefold, findall, join; returns a derived value or artifact.

Calls. casefold, findall, join

Called by. Pipeline.Knowledge_Graph_Core.entity_resolution._acronym_match

SOURCE Pipeline/Knowledge_Graph_Core/entity_resolution.py:19-22

def _acronym_match(left: str, right: str) -> bool

Purpose. Implements the internal acronym match operation.

Mechanics. Delegates to _acronym, casefold, sub; returns a derived value or artifact.

Calls. _acronym, casefold, sub

Called by. Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver, Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver.resolve

SOURCE Pipeline/Knowledge_Graph_Core/entity_resolution.py:25-28

def _embedding_text(mention: dict) -> str

Purpose. Implements the internal embedding text operation.

Mechanics. Returns a derived value or artifact.

Called by. Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver, Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver.resolve

SOURCE Pipeline/Knowledge_Graph_Core/entity_resolution.py:31-32

class EntityResolver

Purpose. Groups the state and behavior for EntityResolver.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to sentenceembeddingbackend, valueerror, _acronym_match, _decision; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. SentenceEmbeddingBackend, ValueError, _acronym_match, _decision, _embedding_text, _tokens, append, arrange, argpartition, asarray

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_resolution_never_auto_merges, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_embedding_resolution_finds_semantic_match_without_shared_name_tokens

SOURCE Pipeline/Knowledge_Graph_Core/entity_resolution.py:35-206

def __init__(self, config: dict, embedding_backend=None)

Purpose. Implements the internal init operation.

Mechanics. Delegates to sentenceembeddingbackend, get.

Calls. SentenceEmbeddingBackend, get

Called by. Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge.__init__, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.__init__, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases

SOURCE Pipeline/Knowledge_Graph_Core/entity_resolution.py:36-41

def resolve(self, mentions: list[dict]) -> tuple[list[dict], dict]

Purpose. Maps ambiguous or duplicate records to explicit decisions.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to valueerror, _acronym_match, _decision, _embedding_text; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. ValueError, _acronym_match, _decision, _embedding_text, append, arrange, argpartition, asarray, calibrate, clip

Called by. Pipeline.GENERAL.run.Handler, Pipeline.GENERAL.run.Handler.do_GET, Pipeline.GENERAL.run.Handler.do_POST, Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline, Pipeline.Node_Pipeline.lexicon.load_prior_node_catalog, Pipeline.GENERAL.tools.export_node_results_report.main, Pipeline.GENERAL.tools.rematerialize_node_run.main

SOURCE Pipeline/Knowledge_Graph_Core/entity_resolution.py:43-102


```
def _decision(self, left: dict, right: dict, label: str, embedding_score: float, acronym_same: bool, exact_name: bool) -> dict
```

Purpose. Implements the internal decision operation.

Mechanics. Branches on validation or availability conditions; delegates to round, stable_id; returns a derived value or artifact.

Calls. round, stable_id

Called by. Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver, Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver.resolve

SOURCE Pipeline/Knowledge_Graph_Core/entity_resolution.py:104-160

```
def calibrate(decisions: list[dict]) -> dict
```

Purpose. Implements the internal calibrate operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _tokens, append, casefold, range; returns a derived value or artifact.

Calls. _tokens, append, casefold, range, round, strip

Called by. Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver, Pipeline.Knowledge_Graph_Core.entity_resolution.EntityResolver.resolve

SOURCE Pipeline/Knowledge_Graph_Core/entity_resolution.py:163-206

Pipeline/Knowledge_Graph_Core/llm_node_judge.py

```
class LLMNodeJudgeError(RuntimeError)
```

Purpose. Groups the state and behavior for LLMNodeJudgeError.

Mechanics. Encapsulates one small, reusable operation.

Called by. Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge.__init__, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge.verify_runtime, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge.judge, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._send, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._output_text

SOURCE Pipeline/Knowledge_Graph_Core/llm_node_judge.py:54-55

```
class MandatoryLLMNodeJudge
```

Purpose. Groups the state and behavior for MandatoryLLMNodeJudge.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to llmnodejudgeerror, request, _align_and_validate_span, _judge_batch; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. LLMNodeJudgeError, Request, _align_and_validate_span, _judge_batch, _output_text, _request_payload, _response_complete, _response_id, _schema, _send

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline, Pipeline.Node_Pipeline.adjudication.optional_llm_review, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_llm_judge_is_mandatory_without_credential, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_ollama_provider_requires_no_api_key, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_ollama_node_judge_expands_context_only_for_oversized_payloads, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_ollama_node_judge_splits_candidate_heavy_sentence_and_merges_audits

SOURCE Pipeline/Knowledge_Graph_Core/llm_node_judge.py:61-828

```
def __init__(self, ontology: Ontology, config: dict, cache_dir: Path, *, api_key: str | None=None, transport: Transport | None=None) -> None
```

Purpose. Implements the internal init operation.

Mechanics. Branches on validation or availability conditions; delegates to llmnodejudgeerror, casefold, encode, get; fails explicitly when an invariant is broken.

Calls. LLMNodeJudgeError, casefold, encode, get, getenv, hexdigest, mkdir, rstrip, sha256

Called by. Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge.__init__, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.__init__, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases

SOURCE Pipeline/Knowledge_Graph_Core/llm_node_judge.py:62-99

def verify_runtime(self) -> None

Purpose. Fail before parsing when the configured real model runtime is unavailable.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to llmnodejudgeerror, request, decode, get; fails explicitly when an invariant is broken.

Calls. LLMNodeJudgeError, Request, decode, get, loads, read, urlopen

Called by. Pipeline.Edge_Extraction.extractor.run_relationship_extraction, Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline, Pipeline.Node_Pipeline.adjudication.optional_llm_review, Pipeline.Node_Pipeline.reference_resolution.optional_llm_reference_resolution, Pipeline.GENERAL.tools.smoke_test_deepseek.main

SOURCE Pipeline/Knowledge_Graph_Core/llm_node_judge.py:101-123

def judge(self, sentence_records: list[dict]) -> dict[str, dict]

Purpose. Implements the internal judge operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to llmnodejudgeerror, _judge_batch, _split_oversized_sentence, append; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. LLMNodeJudgeError, _judge_batch, _split_oversized_sentence, append, extend, get, range, setdefault

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract, Pipeline.Node_Pipeline.adjudication.optional_llm_review, Pipeline.Node_Pipeline.reference_resolution.optional_llm_reference_resolution, Pipeline.GENERAL.tools.smoke_test_deepseek.main, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_ollama_node_judge_splits_candidate_heavy_sentence_and_merges_audits

SOURCE Pipeline/Knowledge_Graph_Core/llm_node_judge.py:125-160

def _split_oversized_sentence(self, sentence: dict) -> list[dict]

Purpose. Split only candidate-heavy sentences that exceed normal num_ctx.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _request_payload, _schema, get, items; returns a derived value or artifact.

Calls. _request_payload, _schema, _split_oversized_sentence, get, items

Called by. Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge.judge

SOURCE Pipeline/Knowledge_Graph_Core/llm_node_judge.py:162-194

def _judge_batch(self, batch: list[dict], batch_number: int) -> tuple[dict, dict]

Purpose. Implements the internal judge batch operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to llmnodejudgeerror, _output_text, _request_payload, _response_complete; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. LLMNodeJudgeError, _output_text, _request_payload, _response_complete, _response_id, _schema, _send, _usage, _validate_batch_result, append

Called by. Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge.judge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge.judge

SOURCE Pipeline/Knowledge_Graph_Core/llm_node_judge.py:196-317

def _request_payload(self, user_payload: dict, schema: dict) -> dict

Purpose. Implements the internal request payload operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to dumps, get; returns a derived value or artifact.

Calls. dumps, get

Called by. Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._split_oversized_sentence, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch

SOURCE Pipeline/Knowledge_Graph_Core/llm_node_judge.py:319-379

def _send(self, payload: dict) -> dict

Purpose. Formats and sends a response.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `llmnodejudgeerror`, `request`, `_response_complete`, `decode`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `LLMNodeJudgeError`, `Request`, `_response_complete`, `decode`, `dumps`, `encode`, `get`, `loads`, `range`, `read`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`,
`Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_node_judge.py:381-423`

def _response_complete(self, response: dict) -> bool

Purpose. Implements the internal response complete operation.

Mechanics. Delegates to `get`; returns a derived value or artifact.

Calls. `get`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`,
`Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._send`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_node_judge.py:425-426`

def _output_text(response: dict) -> str

Purpose. Implements the internal output text operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `llmnodejudgeerror`, `get`, `isinstance`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `LLMNodeJudgeError`, `get`, `isinstance`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`,
`Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_node_judge.py:429-443`

def _response_id(response: dict) -> str

Purpose. Implements the internal response id operation.

Mechanics. Branches on validation or availability conditions; delegates to `dumps`, `encode`, `get`, `hexdigest`; returns a derived value or artifact.

Calls. `dumps`, `encode`, `get`, `hexdigest`, `sha256`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`,
`Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_node_judge.py:446-452`

def _usage(response: dict) -> dict

Purpose. Implements the internal usage operation.

Mechanics. Branches on validation or availability conditions; delegates to `get`, `isinstance`; returns a derived value or artifact.

Calls. `get`, `isinstance`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`,
`Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_node_judge.py:455-465`

```
def _validate_batch_result(self, parsed: dict, batch: list[dict]) -> tuple[list[dict], list[dict], list[dict]]
```

Purpose. Checks invariants and reports or rejects violations.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `llmnodejudgeerror`, `_align_and_validate_span`, `add`, `append`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `LLMNodeJudgeError`, `_align_and_validate_span`, `add`, `append`, `casefold`, `extend`, `get`, `isinstance`, `setdefault`, `startswith`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`,
`Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_node_judge.py:467-710`

```
def _align_and_validate_span(text: str, row: dict, sentence_id: str) -> dict | None
```

Purpose. Implements the internal align and validate span operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `llmnodejudgeerror`, `abs`, `escape`, `finditer`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `LLMNodeJudgeError`, `abs`, `escape`, `finditer`, `start`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`,
`Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._validate_batch_result`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_node_judge.py:713-740`

```
def _schema(self, batch: list[dict] | None=None) -> dict
```

Purpose. Implements the internal schema operation.

Mechanics. Iterates over input records; delegates to `get`; returns a derived value or artifact.

Calls. `get`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`,
`Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._split_oversized_sentence`,
`Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`,
`Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_node_judge.py:742-828`

Pipeline/Knowledge_Graph_Core/llm_reference_judge.py

```
class MandatoryLLMReferenceJudge(MandatoryLLMNodeJudge)
```

Purpose. Contextual SAME/DIFFERENT/UNCERTAIN adjudication for mention pairs.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `llmnodejudgeerror`, `__init__`, `_judge_batch`, `_output_text`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `LLMNodeJudgeError`, `__init__`, `_judge_batch`, `_output_text`, `_request_payload`, `_response_complete`, `_response_id`, `_schema`, `_send`, `_usage`

Called by. `Pipeline.Node_Pipeline.reference_resolution.optional_llm_reference_resolution`,
`Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests`,
`Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_reference_judge_sends_contextual_pair_and_returns_same`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_reference_judge.py:33-177`

```
def __init__(self, ontology: Ontology, config: dict, cache_dir: Path, *, api_key: str | None=None, transport=None) -> None
```

Purpose. Implements the internal init operation.

Mechanics. Delegates to `deepcopy`, `encode`, `get`, `hexdigest`.

Calls. `__init__`, `deepcopy`, `encode`, `get`, `hexdigest`, `sha256`, `super`

Called by. `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`,
`Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.__init__`, `Pipeline.Node_Pipeline.lexicon.Lexicon`,
`Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_reference_judge.py:36-51`

def judge(self, pairs: list[dict]) -> dict[str, dict]

Purpose. Implements the internal judge operation.

Mechanics. Iterates over input records; delegates to `_judge_batch`, `append`, `range`; returns a derived value or artifact.

Calls. `_judge_batch`, `append`, `range`

Called by. `Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor`, `Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract`, `Pipeline.Node_Pipeline.adjudication.optional_llm_review`, `Pipeline.Node_Pipeline.reference_resolution.optional_llm_reference_resolution`, `Pipeline.GENERAL.tools.smoke_test_deepseek.main`, `Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests`, `Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_ollama_node_judge_splits_candidate_heavy_sentence_and_merges_audits`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_reference_judge.py`:53-62

def _judge_batch(self, batch: list[dict], batch_number: int) -> tuple[dict, dict]

Purpose. Implements the internal judge batch operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `llmnodejudgeerror`, `_output_text`, `_request_payload`, `_response_complete`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `LLMNodeJudgeError`, `_output_text`, `_request_payload`, `_response_complete`, `_response_id`, `_schema`, `_send`, `_usage`, `_validate_result`, `dumps`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`, `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge.judge`, `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`, `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge.judge`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_reference_judge.py`:64-105

def _request_payload(self, user_payload: dict, schema: dict) -> dict

Purpose. Implements the internal request payload operation.

Mechanics. Branches on validation or availability conditions; delegates to `dumps`, `get`; returns a derived value or artifact.

Calls. `dumps`, `get`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`, `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._split_oversized_sentence`, `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch`, `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`, `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch`, `Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`, `Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_reference_judge.py`:107-141

def _validate_result(parsed: dict, batch: list[dict]) -> None

Purpose. Checks invariants and reports or rejects violations.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `llmnodejudgeerror`, `get`, `isinstance`; fails explicitly when an invariant is broken.

Calls. `LLMNodeJudgeError`, `get`, `isinstance`

Called by. `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`, `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_reference_judge.py`:144-152

def _schema(pair_count: int) -> dict

Purpose. Implements the internal schema operation.

Mechanics. Returns a derived value or artifact.

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`, `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._split_oversized_sentence`, `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch`, `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`, `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch`, `Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`, `Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_reference_judge.py`:155-177

Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py

class MandatoryLLMRelationshipJudge(MandatoryLLMNodeJudge)

Purpose. Generate endpoints and predicates, then judge them in one structured call.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `llmnodejudgeerror`, `__init__`, `_eligible_predicates`, `_extract_batch`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `LLMNodeJudgeError`, `__init__`, `_eligible_predicates`, `_extract_batch`, `_fallback_audit`, `_normalize_result`, `_output_text`, `_request_payload`, `_response_complete`, `_response_id`

Called by. `Pipeline.Edge_Extraction.extractor.run_relationship_extraction`, `Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline`, `Pipeline.GENERAL.tools.smoke_test_deepseek.main`, `Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests`, `Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_relationship_llm_selects_only_candidate_permitted_predicate`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py`:37-326

def __init__(self, ontology: Ontology, config: dict, cache_dir: Path, *, api_key: str | None=None, transport=None) -> None

Purpose. Implements the internal init operation.

Mechanics. Delegates to `deepcopy`, `encode`, `get`, `hexdigest`.

Calls. `__init__`, `deepcopy`, `encode`, `get`, `hexdigest`, `sha256`, `super`

Called by. `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`, `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge.__init__`, `Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`, `Pipeline.Node_Pipeline.lexicon.Lexicon`, `Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py`:40-57

def extract(self, windows: list[dict], accepted_nodes: list[dict]) -> dict[str, dict]

Purpose. Locates and materializes evidence-backed records.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `_extract_batch`, `_fallback_audit`, `append`, `items`; returns a derived value or artifact.

Calls. `_extract_batch`, `_fallback_audit`, `append`, `items`, `range`

Called by. `Pipeline.Edge_Extraction.extractor.run_relationship_extraction`, `Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor`, `Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract`, `Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline`, `Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests`, `Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_llm_judges_candidates_and_discovers_out_of_lexicon_node`, `Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_edge_extractor_uses_llm_selected_existing_nodes_and_all_context`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py`:59-76

def _extract_batch(self, batch: list[dict], accepted_nodes: list[dict], batch_number: int) -> tuple[dict, dict]

Purpose. Locates and materializes evidence-backed records.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to `llmnodejudgeerror`, `_eligible_predicates`, `_normalize_result`, `_output_text`; fails explicitly when an invariant is broken; returns a derived value or artifact.

Calls. `LLMNodeJudgeError`, `_eligible_predicates`, `_normalize_result`, `_output_text`, `_request_payload`, `_response_complete`, `_response_id`, `_schema`, `_send`, `_usage`

Called by. `Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`, `Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.extract`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py`:78-141

def _request_payload(self, user_payload: dict, schema: dict) -> dict

Purpose. Implements the internal request payload operation.

Mechanics. Branches on validation or availability conditions; delegates to `dumps`, `get`; returns a derived value or artifact.

Calls. `dumps`, `get`

Called by. `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge`, `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._split_oversized_sentence`, `Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch`, `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge`, `Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch`, `Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge`, `Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch`

SOURCE `Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py`:143-177

```
def _normalize_result(parsed: dict, batch: list[dict]) -> tuple[dict, list[dict]]
```

Purpose. Converts text into a stable comparison form.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to append, get, isinstance, items; returns a derived value or artifact.

Calls. append, get, isinstance, items

Called by. Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch, Pipeline.GENERAL.tests.test_generalization_repairs.RelationshipFaultIsolationTests, Pipeline.GENERAL.tests.test_generalization_repairs.RelationshipFaultIsolationTests.test_unknown_non_object_and_missing_windows_degrade_to_empty

SOURCE Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py:180-208

```
def _fallback_audit(self, batch: list[dict], accepted_nodes: list[dict], batch_number: int, exc: Exception) -> dict
```

Purpose. Implements the internal fallback audit operation.

Mechanics. Iterates over input records; delegates to splitlines, type; returns a derived value or artifact.

Calls. splitlines, type

Called by. Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.extract

SOURCE Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py:210-237

```
def _eligible_predicates(self, accepted_nodes: list[dict]) -> list[str]
```

Purpose. Implements the internal eligible predicates operation.

Mechanics. Iterates over input records; delegates to any, compatible_class, get, items; returns a derived value or artifact.

Calls. any, compatible_class, get, items

Called by. Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch

SOURCE Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py:239-246

```
def _schema(self, windows: list[dict] | int, accepted_nodes: list[dict] | None=None, eligible_predicates: list[str] | None=None) -> dict
```

Purpose. Implements the internal schema operation.

Mechanics. Iterates over input records; delegates to get, isinstance; returns a derived value or artifact.

Calls. get, isinstance

Called by. Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._split_oversized_sentence, Pipeline.Knowledge_Graph_Core.llm_node_judge.MandatoryLLMNodeJudge._judge_batch, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge._judge_batch, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge._extract_batch

SOURCE Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py:248-326

Pipeline/Knowledge_Graph_Core/models.py

```
def stable_id(prefix: str, *parts: object) -> str
```

Purpose. Implements the internal stable id operation.

Mechanics. Iterates over input records; delegates to encode, hexdigest, join, sha256; returns a derived value or artifact.

Calls. encode, hexdigest, join, sha256

Called by. Pipeline.Edge_Extraction.materialization.materialize_canonical_relationships, Pipeline.Edge_Extraction.observation_mentions.assemble_observation_mentions, Pipeline.GENERAL.run.launch, Pipeline.Graph_Persistence.neo4j_writer.build_entity_rows, Pipeline.Graph_Persistence.neo4j_writer._legacy_upsert_run, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/models.py:10-12

```
def sha256_bytes(data: bytes) -> str
```

Purpose. Implements the internal sha256 bytes operation.

Mechanics. Delegates to hexdigest, sha256; returns a derived value or artifact.

Calls. hexdigest, sha256

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline

SOURCE Pipeline/Knowledge_Graph_Core/models.py:15-16

def sha256_file(path: Path) -> str

Purpose. Implements the internal sha256 file operation.

Mechanics. Iterates over input records; delegates to hexdigest, iter, open, read; returns a derived value or artifact.

Calls. hexdigest, iter, open, read, sha256, update

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline

SOURCE Pipeline/Knowledge_Graph_Core/models.py:19-24

def normalize_name(text: str) -> str

Purpose. Converts text into a stable comparison form.

Mechanics. Delegates to casefold, join, replace, split; returns a derived value or artifact.

Calls. casefold, join, replace, split, sub

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._metadata_mention,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._sentence_candidates,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._mention_from_candidate,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._mention_from_discovery,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._new_class_review,
Pipeline.Knowledge_Graph_Core.review_queues.build_review_queues

SOURCE Pipeline/Knowledge_Graph_Core/models.py:27-30

def write_json(path: Path, payload: object) -> None

Purpose. Serializes output deterministically.

Mechanics. Delegates to dumps, mkdir, write_text.

Calls. dumps, mkdir, write_text

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline

SOURCE Pipeline/Knowledge_Graph_Core/models.py:33-35

def write_jsonl(path: Path, records: Iterable[dict]) -> None

Purpose. Serializes output deterministically.

Mechanics. Iterates over input records; delegates to dumps, mkdir, open, write.

Calls. dumps, mkdir, open, write

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline

SOURCE Pipeline/Knowledge_Graph_Core/models.py:38-42

def read_jsonl(path: Path) -> list[dict]

Purpose. Loads serialized input.

Mechanics. Iterates over input records; delegates to loads, read_text, splitlines, strip; returns a derived value or artifact.

Calls. loads, read_text, splitlines, strip

Called by. Pipeline.GENERAL.tools.evaluate_regression_corpus.evaluate_case, Pipeline.GENERAL.tools.export_node_results_report.main

SOURCE Pipeline/Knowledge_Graph_Core/models.py:45-46

Pipeline/Knowledge_Graph_Core/node_extractor.py

class SpanCandidate

Purpose. Groups the state and behavior for SpanCandidate.

Mechanics. Delegates to dataclass.

Calls. dataclass

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._sentence_candidates

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:65-70

class NodeExtractor

Purpose. Groups the state and behavior for NodeExtractor.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to spancandidate, _attributes, _candidate_for_prompt, _decision_record; returns a derived value or artifact.

Calls. SpanCandidate, _attributes, _candidate_for_prompt, _decision_record, _definition, _mention_from_candidate, _mention_from_discovery, _mention_key, _metadata_mention, _new_class_review

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_llm_judges_candidates_and_discovers_out_of_lexicon_node

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:73-431

def __init__(self, ontology: Ontology, lexicon_path: Path, config: dict, llm_judge: MandatoryLLMNodeJudge) -> None

Purpose. Implements the internal init operation.

Mechanics. Iterates over input records; delegates to compile, escape, items, loads.

Calls. compile, escape, items, loads, read_text

Called by. Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge.__init__, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.__init__, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:74-91

def extract(self, parsed: dict, pipeline_document_id: str) -> tuple[list[dict], list[dict], list[dict], list[dict]]

Purpose. Locates and materializes evidence-backed records.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _candidate_for_prompt, _decision_record, _mention_from_candidate, _mention_from_discovery; returns a derived value or artifact.

Calls. _candidate_for_prompt, _decision_record, _mention_from_candidate, _mention_from_discovery, _mention_key, _metadata_mention, _new_class_review, _sentence_candidates, _split_authors, add

Called by. Pipeline.Edge_Extraction.extractor.run_relationship_extraction, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract, Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_llm_judges_candidates_and_discovers_out_of_lexicon_node, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_edge_extractor_uses_llm_selected_existing_nodes_and_all_context

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:93-198

def _split_authors(authors: str | None) -> list[str]

Purpose. Divides input into traceable units.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to split, strip, sub; returns a derived value or artifact.

Calls. split, strip, sub

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:201-205

def _metadata_mention(self, document_id: str, label: str, text: str, field: str, confidence: float, attributes: dict) -> dict

Purpose. Implements the internal metadata mention operation.

Mechanics. Delegates to _definition, normalize_name, stable_id; returns a derived value or artifact.

Calls. _definition, normalize_name, stable_id

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:207-221

def _sentence_candidates(self, document_id: str, sentence: dict, paragraph: dict, section: dict) -> tuple[list[dict], list[dict]]

Purpose. Implements the internal sentence candidates operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to spancandidate, append, class_scores, end; returns a derived value or artifact.

Calls. SpanCandidate, append, class_scores, end, find, finditer, get, group, items, match

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:223-294

def _candidate_for_prompt(candidate: dict) -> dict

Purpose. Implements the internal candidate for prompt operation.

Mechanics. Returns a derived value or artifact.

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:297-305

def _decision_record(judgment: dict, audit: dict, candidate: dict) -> dict

Purpose. Implements the internal decision record operation.

Mechanics. Delegates to get; returns a derived value or artifact.

Calls. get

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:308-318

def _mention_from_candidate(self, candidate: dict, judgment: dict, decision: dict) -> dict

Purpose. Implements the internal mention from candidate operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to _attributes, any, append, class_scores; returns a derived value or artifact.

Calls. _attributes, any, append, class_scores, next, normalize_name, sort, stable_id, update

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:320-351

def _mention_from_discovery(self, document_id: str, context: dict, discovery: dict, audit: dict) -> dict

Purpose. Implements the internal mention from discovery operation.

Mechanics. Delegates to class_scores, get, normalize_name, stable_id; returns a derived value or artifact.

Calls. class_scores, get, normalize_name, stable_id

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:353-387

def _new_class_review(document_id: str, context: dict, proposal: dict, audit: dict) -> dict

Purpose. Implements the internal new class review operation.

Mechanics. Delegates to get, normalize_name, stable_id; returns a derived value or artifact.

Calls. get, normalize_name, stable_id

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:390-413

def _mention_key(mention: dict) -> tuple[str, int, int, str]

Purpose. Implements the internal mention key operation.

Mechanics. Delegates to get; returns a derived value or artifact.

Calls. get

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor, Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor.extract

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:416-419

def _attributes(self, label: str, surface: str, sentence: str) -> dict

Purpose. Implements the internal attributes operation.

Mechanics. Branches on validation or availability conditions; delegates to group, search; returns a derived value or artifact.

Calls. group, search

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._mention_from_candidate

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:421-428

def _definition(self, surface: str, label: str, context: str) -> str

Purpose. Implements the internal definition operation.

Mechanics. Returns a derived value or artifact.

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._metadata_mention

SOURCE Pipeline/Knowledge_Graph_Core/node_extractor.py:430-431

Pipeline/Knowledge_Graph_Core/ontology.py

class Ontology

Purpose. Groups the state and behavior for Ontology.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to add, ancestors, compatible_class, cosine_vectors; returns a derived value or artifact.

Calls. add, ancestors, compatible_class, cosine_vectors, extend, get, idf_for, join, loads, pop

Called by. Pipeline.Edge_Extraction.extractor.run_relationship_extraction, Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline, Pipeline.Node_Pipeline.adjudication.optional_llm_review, Pipeline.Node_Pipeline.reference_resolution.optional_llm_reference_resolution, Pipeline.GENERAL.tools.smoke_test_deepseek.main, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.setUpClass

SOURCE Pipeline/Knowledge_Graph_Core/ontology.py:9-58

def __init__(self, path: Path)

Purpose. Implements the internal init operation.

Mechanics. Iterates over input records; delegates to get, idf_for, join, loads.

Calls. get, idf_for, join, loads, read_text, vector

Called by. Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge, Pipeline.Knowledge_Graph_Core.llm_reference_judge.MandatoryLLMReferenceJudge.__init__, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.__init__, Pipeline.Node_Pipeline.lexicon.Lexicon, Pipeline.Node_Pipeline.lexicon.Lexicon.add_reviewed_aliases

SOURCE Pipeline/Knowledge_Graph_Core/ontology.py:10-27

def ancestors(self, label: str) -> set[str]

Purpose. Implements the internal ancestors operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to add, extend, get, pop; returns a derived value or artifact.

Calls. add, extend, get, pop

Called by. Pipeline.Knowledge_Graph_Core.ontology.Ontology, Pipeline.Knowledge_Graph_Core.ontology.Ontology.compatible_class

SOURCE Pipeline/Knowledge_Graph_Core/ontology.py:29-38

def compatible_class(self, actual: str, allowed: list[str]) -> bool

Purpose. Implements the internal compatible class operation.

Mechanics. Delegates to ancestors; returns a derived value or artifact.

Calls. ancestors

Called by. Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge, Pipeline.Knowledge_Graph_Core.llm_relationship_judge.MandatoryLLMRelationshipJudge.eligible_predicates, Pipeline.Knowledge_Graph_Core.ontology.Ontology, Pipeline.Knowledge_Graph_Core.ontology.Ontology.valid_relationship

SOURCE Pipeline/Knowledge_Graph_Core/ontology.py:40-41

def valid_relationship(self, predicate: str, source_label: str, target_label: str) -> bool

Purpose. Implements the internal valid relationship operation.

Mechanics. Branches on validation or availability conditions; delegates to compatible_class, get; returns a derived value or artifact.

Calls. compatible_class, get

Called by. Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor.extract, Pipeline.Knowledge_Graph_Core.edge_extractor.EdgeExtractor._candidate_from_proposal, Pipeline.Knowledge_Graph_Core.validation.validate_run, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_domain_range_accepts_subclass, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_domain_range_rejects_invalid_direction

SOURCE Pipeline/Knowledge_Graph_Core/ontology.py:43-50

```
def class_scores(self, text: str) -> list[dict]
```

Purpose. Implements the internal class scores operation.

Mechanics. Iterates over input records; delegates to cosine_vectors, round, vector, zip; returns a derived value or artifact.

Calls. cosine_vectors, round, vector, zip

Called by. Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._sentence_candidates,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._mention_from_candidate,
Pipeline.Knowledge_Graph_Core.node_extractor.NodeExtractor._mention_from_discovery

SOURCE Pipeline/Knowledge_Graph_Core/ontology.py:52-58

Pipeline/Knowledge_Graph_Core/pipeline.py

```
def _slug(value: str) -> str
```

Purpose. Implements the internal slug operation.

Mechanics. Delegates to casefold, strip, sub; returns a derived value or artifact.

Calls. casefold, strip, sub

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline

SOURCE Pipeline/Knowledge_Graph_Core/pipeline.py:27-29

```
def run_pipeline(papers_dir: Path, output_dir: Path, ontology_path: Path, lexicon_path: Path, config_path: Path, parser_project: Path, *, limit: int | None=None, reuse_parsed: bool=False, progress_callback: Callable[[str, dict], None] | None=None) -> dict
```

Purpose. An alternate shared-core batch runner for a directory of PDFs. It is not the workbench's canonical single-paper path, but it reuses parser, judges, resolver, validation, and artifact contracts.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to counter, edgeextractor, entityresolver, mandatoryllmnodejudge; returns a derived value or artifact.

Calls. Counter, EdgeExtractor, EntityResolver, MandatoryLLMNodeJudge, MandatoryLLMRelationshipJudge, NodeExtractor, Ontology, _slug, all, append

SOURCE Pipeline/Knowledge_Graph_Core/pipeline.py:32-409

```
def notify(event: str, **payload: object) -> None
```

Purpose. Implements the internal notify operation.

Mechanics. Branches on validation or availability conditions; delegates to progress_callback.

Calls. progress_callback

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline

SOURCE Pipeline/Knowledge_Graph_Core/pipeline.py:44-46

Pipeline/Knowledge_Graph_Core/review_queues.py

```
def build_review_queues(decisions: list[dict]) -> dict[str, list[dict]]
```

Purpose. Assembles a derived artifact.

Mechanics. Iterates over input records; delegates to append, defaultdict, items, normalize_name; returns a derived value or artifact.

Calls. append, defaultdict, items, normalize_name, sort, split, values

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline, Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests,
Pipeline.Knowledge_Graph_Core.tests.test_pipeline.PipelineUnitTests.test_review_queue_collapses_repeated_name_pairs

SOURCE Pipeline/Knowledge_Graph_Core/review_queues.py:8-38

Pipeline/Knowledge_Graph_Core/similarity.py

```
def features(text: str) -> Counter[str]
```

Purpose. Implements the internal features operation.

Mechanics. Iterates over input records; delegates to counter, casefold, findall, join; returns a derived value or artifact.

Calls. Counter, casefold, findall, join, range

Called by. Pipeline.Knowledge_Graph_Core.similarity.idf_for, Pipeline.Knowledge_Graph_Core.similarity.vector

SOURCE Pipeline/Knowledge_Graph_Core/similarity.py:11-19

```
def idf_for(texts: list[str]) -> dict[str, float]
```

Purpose. Implements the internal idf for operation.

Mechanics. Iterates over input records; delegates to counter, features, items, log; returns a derived value or artifact.

Calls. Counter, features, items, log, update

Called by. Pipeline.Knowledge_Graph_Core.ontology.Ontology, Pipeline.Knowledge_Graph_Core.ontology.Ontology.__init__, Pipeline.Knowledge_Graph_Core.similarity.cosine, Pipeline.Knowledge_Graph_Core.similarity.matrix_vectors

SOURCE Pipeline/Knowledge_Graph_Core/similarity.py:22-28

```
def vector(text: str, idf: dict[str, float]) -> dict[str, float]
```

Purpose. Implements the internal vector operation.

Mechanics. Iterates over input records; delegates to features, get, items, values; returns a derived value or artifact.

Calls. features, get, items, values

Called by. Pipeline.Knowledge_Graph_Core.ontology.Ontology, Pipeline.Knowledge_Graph_Core.ontology.Ontology.__init__, Pipeline.Knowledge_Graph_Core.ontology.Ontology.class_scores, Pipeline.Knowledge_Graph_Core.similarity.cosine, Pipeline.Knowledge_Graph_Core.similarity.matrix_vectors

SOURCE Pipeline/Knowledge_Graph_Core/similarity.py:31-34

```
def cosine_vectors(left: dict[str, float], right: dict[str, float]) -> float
```

Purpose. Implements the internal cosine vectors operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to sqrt, values; returns a derived value or artifact.

Calls. sqrt, values

Called by. Pipeline.Knowledge_Graph_Core.ontology.Ontology, Pipeline.Knowledge_Graph_Core.ontology.Ontology.class_scores, Pipeline.Knowledge_Graph_Core.similarity.cosine

SOURCE Pipeline/Knowledge_Graph_Core/similarity.py:37-44

```
def cosine(left: str, right: str) -> float
```

Purpose. Implements the internal cosine operation.

Mechanics. Delegates to cosine_vectors, idf_for, vector; returns a derived value or artifact.

Calls. cosine_vectors, idf_for, vector

Called by. Pipeline.Knowledge_Graph_Core.validation.validate_run

SOURCE Pipeline/Knowledge_Graph_Core/similarity.py:47-49

```
def matrix_vectors(texts: list[str]) -> list[dict[str, float]]
```

Purpose. Implements the internal matrix vectors operation.

Mechanics. Iterates over input records; delegates to idf_for, vector; returns a derived value or artifact.

Calls. idf_for, vector

SOURCE Pipeline/Knowledge_Graph_Core/similarity.py:52-54

Pipeline/Knowledge_Graph_Core/validation.py

```
def validate_run(output_dir: Path, ontology: Ontology, config: dict, documents: list[dict], mentions: list[dict], assertions: list[dict], resolutions: list[dict]) -> dict
```

Purpose. Checks invariants and reports or rejects violations.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to counter, add, append, cosine; returns a derived value or artifact.

Calls. Counter, add, append, cosine, defaultdict, get, is_file, items, round, valid_relationship

Called by. Pipeline.Knowledge_Graph_Core.pipeline.run_pipeline

SOURCE Pipeline/Knowledge_Graph_Core/validation.py:29-192

Top-level

Pipeline/check_environment.py

def package_version(distribution: str) -> str | None

Purpose. Implements the internal package version operation.

Mechanics. Branches on validation or availability conditions; delegates to version; returns a derived value or artifact.

Calls. version

Called by. Pipeline.check_environment.main

SOURCE Pipeline/check_environment.py:19-23

def url_available(url: str, timeout: float=3.0) -> tuple[bool, str]

Purpose. Implements the internal url available operation.

Mechanics. Branches on validation or availability conditions; delegates to isinstance, urlopen; returns a derived value or artifact.

Calls. isinstance, urlopen

Called by. Pipeline.check_environment.main

SOURCE Pipeline/check_environment.py:26-34

def ollama_models(base_url: str) -> tuple[bool, set[str], str]

Purpose. Implements the internal ollama models operation.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to decode, get, isinstance, loads; returns a derived value or artifact.

Calls. decode, get, isinstance, loads, read, rstrip, urlopen

Called by. Pipeline.check_environment.main

SOURCE Pipeline/check_environment.py:37-49

def main() -> int

Purpose. Provides the command-line or server entry point.

Mechanics. Iterates over input records; branches on validation or availability conditions; delegates to argumentparser, add_argument, append, exists; returns a derived value or artifact.

Calls. ArgumentParser, add_argument, append, exists, glob, load, loads, ollama_models, package_version, parse_args

SOURCE Pipeline/check_environment.py:52-127

18. Browser JavaScript function reference

function \$(selector, root = document)

Purpose. Returns the first DOM element matching a selector.

SOURCE Pipeline/GENERAL/static/app.js:1

function \$\$ (selector, root = document)

Purpose. Returns all DOM elements matching a selector as an iterable collection.

SOURCE Pipeline/GENERAL/static/app.js:2

function esc(value = "")

Purpose. Implements the browser-side esc behavior.

SOURCE Pipeline/GENERAL/static/app.js:4

function toast(message)

Purpose. Implements the browser-side toast behavior.

SOURCE Pipeline/GENERAL/static/app.js:8

function api(path, options = {})

Purpose. Calls a workbench JSON endpoint and normalizes HTTP errors.

SOURCE Pipeline/GENERAL/static/app.js:16

function chooseFile(file)

Purpose. Implements the browser-side chooseFile behavior.

SOURCE Pipeline/GENERAL/static/app.js:23

function stageIndex(stage)

Purpose. Implements the browser-side stageIndex behavior.

SOURCE Pipeline/GENERAL/static/app.js:31

function stageOutput(output = {})

Purpose. Implements the browser-side stageOutput behavior.

SOURCE Pipeline/GENERAL/static/app.js:35

function updateRun(job)

Purpose. Applies current job state to progress cards and status text.

SOURCE Pipeline/GENERAL/static/app.js:42

function begin(path, options = {})

Purpose. Starts the upload/extraction interaction and initializes progress UI.

SOURCE Pipeline/GENERAL/static/app.js:68

function pollJob()

Purpose. Polls the server until the run completes or fails.

SOURCE Pipeline/GENERAL/static/app.js:84

function metric(value, label, blue = false)

Purpose. Implements the browser-side metric behavior.

SOURCE Pipeline/GENERAL/static/app.js:109

function renderResults(job)

Purpose. Coordinates all completed-run renderers.

SOURCE Pipeline/GENERAL/static/app.js:113

function renderQuestions(result)

Purpose. Implements the browser-side renderQuestions behavior.

SOURCE Pipeline/GENERAL/static/app.js:164

function gateLabel(value)

Purpose. Implements the browser-side gateLabel behavior.

SOURCE Pipeline/GENERAL/static/app.js:177

function renderRelationships(rows)

Purpose. Builds relationship and assertion views.

SOURCE Pipeline/GENERAL/static/app.js:191

function renderNodes(entities, mentions)

Purpose. Builds the canonical node table and evidence affordances.

SOURCE Pipeline/GENERAL/static/app.js:215

function draw()

Purpose. Implements the browser-side draw behavior.

SOURCE Pipeline/GENERAL/static/app.js:220

function highlightedQuote(text, start, end)

Purpose. Implements the browser-side highlightedQuote behavior.

SOURCE Pipeline/GENERAL/static/app.js:242

function renderEvidence(entity, mentionMap)

Purpose. Implements the browser-side renderEvidence behavior.

SOURCE Pipeline/GENERAL/static/app.js:247

function normalized(value = "")

Purpose. Implements the browser-side normalized behavior.

SOURCE Pipeline/GENERAL/static/app.js:279

function renderSimilar(pairs)

Purpose. Renders possible-duplicate pair decisions.

SOURCE Pipeline/GENERAL/static/app.js:281

function renderReferenceReview(rows)

Purpose. Renders antecedent choices for ambiguous vague references.

SOURCE Pipeline/GENERAL/static/app.js:311

function resolveReference(card, decision)

Purpose. Posts a selected antecedent for a vague mention.

SOURCE Pipeline/GENERAL/static/app.js:337

function resolvePair(card, decision)

Purpose. Posts a duplicate-pair merge/keep-separate decision.

SOURCE Pipeline/GENERAL/static/app.js:359

function renderReview(rows)

Purpose. Renders node accept/reject controls.

SOURCE Pipeline/GENERAL/static/app.js:378

function submitNodeReview(card, decision)

Purpose. Posts a human node decision and refreshes downstream results.

SOURCE Pipeline/GENERAL/static/app.js:397

function renderDownloads(files)

Purpose. Implements the browser-side renderDownloads behavior.

SOURCE Pipeline/GENERAL/static/app.js:420

function activateTab(name)

Purpose. Implements the browser-side activateTab behavior.

SOURCE Pipeline/GENERAL/static/app.js:452

function loadHistory()

Purpose. Implements the browser-side loadHistory behavior.

SOURCE Pipeline/GENERAL/static/app.js:457

function refreshGraphHealth()

Purpose. Fetches Neo4j counts for the dashboard.

SOURCE Pipeline/GENERAL/static/app.js:472

function init()

Purpose. Wires DOM events, loads history, and initializes the workbench.

SOURCE Pipeline/GENERAL/static/app.js:489

19. Tests and maintenance tools index

These 240 definitions are not part of a normal run. They document intended behavior, guard regressions, inspect corpora, rematerialize reviewed runs, and export reports. Test names are deliberately descriptive: they are executable design claims.

Pipeline/GENERAL/tests/test_author_metadata.py:8	AuthorMetadataTests	Regression check for AuthorMetadataTests.
Pipeline/GENERAL/tests/test_author_metadata.py:9	AuthorMetadataTests.test_affiliation_markers_do_not_become_people	Regression check for affiliation markers do not become people.
Pipeline/GENERAL/tests/test_generalization_repairs.py:14	AgentEnumerationRepairTests	Regression check for AgentEnumerationRepairTests.
Pipeline/GENERAL/tests/test_generalization_repairs.py:15	AgentEnumerationRepairTests.test_acronym_plus_ordinary_words_do_not_become_agents	Regression check for acronym plus ordinary words do not become agents.
Pipeline/GENERAL/tests/test_generalization_repairs.py:23	AgentEnumerationRepairTests.test_explicit_local_identifier_family_is_retained	Regression check for explicit local identifier family is retained.
Pipeline/GENERAL/tests/test_generalization_repairs.py:35	RelationshipFaultIsolationTests	Regression check for RelationshipFaultIsolationTests.
Pipeline/GENERAL/tests/test_generalization_repairs.py:37	RelationshipFaultIsolationTests._batch	Regression check for batch.
Pipeline/GENERAL/tests/test_generalization_repairs.py:43	RelationshipFaultIsolationTests.test_unknown_non_object_and_missing_windows_degrade_to_empty	Regression check for unknown non object and missing windows degrade to empty.
Pipeline/GENERAL/tests/test_generalization_repairs.py:57	RelationshipFaultIsolationTests.test_schema_allows_empty_endpoints_and_empty_relationship_lists	Regression check for schema allows empty endpoints and empty relationship lists.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:15	FakeTyper	Regression check for FakeTyper.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:16	FakeTyper.type_arguments	Regression check for type arguments.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:37	FakeEncoder	Regression check for FakeEncoder.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:38	FakeEncoder.encode	Regression check for encode.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:53	AmbiguousFakeTyper	Regression check for AmbiguousFakeTyper.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:54	AmbiguousFakeTyper.type_arguments	Regression check for type arguments.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:72	GrammaticalCandidateTests	Regression check for GrammaticalCandidateTests.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:74	GrammaticalCandidateTests.parsed	Regression check for parsed.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:85	GrammaticalCandidateTests.parser	Regression check for parser.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:91	GrammaticalCandidateTests.test_complete_pos_parse_and_predicate_first_triples	Regression check for complete pos parse and predicate first triples.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:130	GrammaticalCandidateTests.test_conjoined_predicates_inherit_the_subject	Regression check for conjoined predicates inherit the subject.

Location	Test Name	Description
Pipeline/GENERAL/tests/test_grammatical_candidates.py:147	GrammaticalCandidateTests.test_pretrained_scibert_similarity_assigns_provisional_type	Regression check for pretrained scibert similarity assigns provisional type.
Pipeline/GENERAL/tests/test_grammatical_candidates.py:164	GrammaticalCandidateTests.test_ambiguous_scibert_type_does_not_remove_grammar_candidate	Regression check for ambiguous scibert type does not remove grammar candidate.
Pipeline/GENERAL/tests/test_neo4j_writer.py:10	Neo4jWriterTests	Regression check for Neo4jWriterTests.
Pipeline/GENERAL/tests/test_neo4j_writer.py:11	Neo4jWriterTests.test_reusable_identity_allows_to_another_paper_and_rewires_relationships	Regression check for reusable identity aligns to another paper and rewires relationships.
Pipeline/GENERAL/tests/test_neo4j_writer.py:42	Neo4jWriterTests.test_mentions_are_embedded_on_one_canonical_ontology_node	Regression check for mentions are embedded on one canonical ontology node.
Pipeline/GENERAL/tests/test_neo4j_writer.py:71	Neo4jWriterTests.test_document_metadata_mentions_use_exact_field_provenance	Regression check for document metadata mentions use exact field provenance.
Pipeline/GENERAL/tests/test_neo4j_writer.py:91	Neo4jWriterTests.test_duplicate_canonical_nodes_are_rejected	Regression check for duplicate canonical nodes are rejected.
Pipeline/GENERAL/tests/test_neo4j_writer.py:99	Neo4jWriterTests.test_scoped_observations_may_have_the_same_display_value	Regression check for scoped observations may have the same display value.
Pipeline/GENERAL/tests/test_neo4j_writer.py:107	Neo4jWriterTests.test_provenance_classes_cannot_become_instance_nodes	Regression check for provenance classes cannot become instance nodes.
Pipeline/GENERAL/tests/test_neo4j_writer.py:113	Neo4jWriterTests.test_concrete_ontology_labels_are_applied_to_canonical_entities	Regression check for concrete ontology labels are applied to canonical entities.
Pipeline/GENERAL/tests/test_neo4j_writer.py:128	Neo4jWriterTests.test_unknown_entity_class_is_rejected_before_neo4j_write	Regression check for unknown entity class is rejected before neo4j write.
Pipeline/GENERAL/tests/test_neo4j_writer.py:132	Neo4jWriterTests.test_relationship_domain_and_range_are_enforced	Regression check for relationship domain and range are enforced.
Pipeline/GENERAL/tests/test_neo4j_writer.py:151	Neo4jWriterTests.test_undefined_relationship_is_rejected	Regression check for undefined relationship is rejected.
Pipeline/GENERAL/tests/test_ner_generalization.py:16	OntologyGuidedNERGeneralizationTests	Regression check for OntologyGuidedNERGeneralizationTests.
Pipeline/GENERAL/tests/test_ner_generalization.py:17	OntologyGuidedNERGeneralizationTests.test_author_name_ending_in_rag_is_not_an_ai_system	Regression check for author name ending in rag is not an ai system.
Pipeline/GENERAL/tests/test_ner_generalization.py:21	OntologyGuidedNERGeneralizationTests.test_true_uppercase_rag_product_is_retained	Regression check for true uppercase rag product is retained.
Pipeline/GENERAL/tests/test_ner_generalization.py:25	OntologyGuidedNERGeneralizationTests.test_common_water_and_chemical_names_are_not_ai_models	Regression check for common water and chemical names are not ai models.
Pipeline/GENERAL/tests/test_ner_generalization.py:35	OntologyGuidedNERGeneralizationTests.test_named_model_near_model_cue_is_retained	Regression check for named model near model cue is retained.
Pipeline/GENERAL/tests/test_ner_generalization.py:40	OntologyGuidedNERGeneralizationTests.test_uppercase_scientific_models_near_model_cues_are_retained	Regression check for uppercase scientific models near model cues are retained.
Pipeline/GENERAL/tests/test_ner_generalization.py:49	OntologyGuidedNERGeneralizationTests.test_water_assets_and_named_role_compounds_are_typed	Regression check for water assets and named role compounds are typed.
Pipeline/GENERAL/tests/test_node_pipeline.py:32	NodePipelineTests	Regression check for NodePipelineTests.

Source	Destination	Notes
Pipeline/GENERAL/tests/test_node_pipeline.py:33	NodePipelineTests.test_nested_candidates_are_retained_for_high_recall_adjudication	Regression check for nested candidates are retained for high recall adjudication.
Pipeline/GENERAL/tests/test_node_pipeline.py:48	NodePipelineTests.test_explicit_introduction_discovers_unseen_system_names	Regression check for explicit introduction discovers unseen system names.
Pipeline/GENERAL/tests/test_node_pipeline.py:56	NodePipelineTests.test_scibert_svo_arguments_are_reviewable_and_lexicon_stays_trusted	Regression check for scibert svo arguments are reviewable and lexicon stays trusted.
Pipeline/GENERAL/tests/test_node_pipeline.py:84	NodePipelineTests.test_only_metadata_and_unambiguous_lexicon_matches_are_trusted	Regression check for only metadata and unambiguous lexicon matches are trusted.
Pipeline/GENERAL/tests/test_node_pipeline.py:100	NodePipelineTests.test_ambiguous_persistent_alias_requires_adjudication	Regression check for ambiguous persistent alias requires adjudication.
Pipeline/GENERAL/tests/test_node_pipeline.py:117	NodePipelineTests.test_llm_accept_review_and_reject_are_all_audited	Regression check for llm accept review and reject are all audited.
Pipeline/GENERAL/tests/test_node_pipeline.py:120	NodePipelineTests.test_llm_accept_review_and_reject_are_all_audited.candidate	Regression check for candidate.
Pipeline/GENERAL/tests/test_node_pipeline.py:148	NodePipelineTests.test_llm_accept_review_and_reject_are_all_audited.Judge	Regression check for Judge.
Pipeline/GENERAL/tests/test_node_pipeline.py:154	NodePipelineTests.test_llm_accept_review_and_reject_are_all_audited.Judge.verify_runtime	Regression check for verify runtime.
Pipeline/GENERAL/tests/test_node_pipeline.py:157	NodePipelineTests.test_llm_accept_review_and_reject_are_all_audited.Judge.judge	Regression check for judge.
Pipeline/GENERAL/tests/test_node_pipeline.py:187	NodePipelineTests.test_missing_sentence_id_is_rejected_before_llm	Regression check for missing sentence id is rejected before llm.
Pipeline/GENERAL/tests/test_node_pipeline.py:196	NodePipelineTests.test_missing_sentence_id_is_rejected_before_llm.Judge	Regression check for Judge.
Pipeline/GENERAL/tests/test_node_pipeline.py:199	NodePipelineTests.test_missing_sentence_id_is_rejected_before_llm.Judge.verify_runtime	Regression check for verify runtime.
Pipeline/GENERAL/tests/test_node_pipeline.py:210	NodePipelineTests.test_hallucinated_canonical_name_is_rejected	Regression check for hallucinated canonical name is rejected.
Pipeline/GENERAL/tests/test_node_pipeline.py:225	NodePipelineTests.test_hallucinated_canonical_name_is_rejected.Judge	Regression check for Judge.
Pipeline/GENERAL/tests/test_node_pipeline.py:228	NodePipelineTests.test_hallucinated_canonical_name_is_rejected.Judge.verify_runtime	Regression check for verify runtime.
Pipeline/GENERAL/tests/test_node_pipeline.py:231	NodePipelineTests.test_hallucinated_canonical_name_is_rejected.Judge.judge	Regression check for judge.
Pipeline/GENERAL/tests/test_node_pipeline.py:247	NodePipelineTests.test_trusted_exact_span_removes_redundant_review_candidate	Regression check for trusted exact span removes redundant review candidate.
Pipeline/GENERAL/tests/test_node_pipeline.py:260	NodePipelineTests.test_human_accept_adds_only_sentence_grounding_review_node	Regression check for human accept adds only sentence grounded review node.
Pipeline/GENERAL/tests/test_node_pipeline.py:292	NodePipelineTests.test_mas_suffix_does_not_turn_thomas_into_ai_system	Regression check for mas suffix does not turn thomas into ai system.
Pipeline/GENERAL/tests/test_node_pipeline.py:296	NodePipelineTests.test_generic_the_benchmark_is_not_a_dataset	Regression check for generic the benchmark is not a dataset.
Pipeline/GENERAL/tests/test_node_pipeline.py:300	NodePipelineTests.test_rag_abbreviation_normalizes_to_method	Regression check for rag abbreviation normalizes to method.

Location	Test Name	Test Description
Pipeline/GENERAL/tests/test_node_pipeline.py:305	NodePipelineTests.test_provider_model_repairs_spacing_around_hyphens	Regression check for provider model repairs spacing around hyphens.
Pipeline/GENERAL/tests/test_node_pipeline.py:313	NodePipelineTests.test_llm_based_adjective_is_not_an_ai_system	Regression check for llm based adjective is not an ai system.
Pipeline/GENERAL/tests/test_node_pipeline.py:318	NodePipelineTests.test_definite_framework_after_explicit_introduction_auto_resolves	Regression check for definite framework after explicit introduction auto resolves.
Pipeline/GENERAL/tests/test_node_pipeline.py:344	NodePipelineTests.test_that_complementizer_is_not_resolved_as_that_model	Regression check for that complementizer is not resolved as that model.
Pipeline/GENERAL/tests/test_node_pipeline.py:353	NodePipelineTests.test_model_variant_suffix_is_not_swallowed_by_base_lexicon_entry	Regression check for model variant suffix is not swallowed by base lexicon entry.
Pipeline/GENERAL/tests/test_node_pipeline.py:360	NodePipelineTests.test_equal_looking_statistics_remain_occurrence_scoped	Regression check for equal looking statistics remain occurrence scoped.
Pipeline/GENERAL/tests/test_node_pipeline.py:374	NodePipelineTests.test_same_named_entity_from_lexicon_and_discovery_is_one_node	Regression check for same named entity from lexicon and discovery is one node.
Pipeline/GENERAL/tests/test_node_pipeline.py:394	NodePipelineTests.test_unmerged_entity_keeps_its_persistent_identifier	Regression check for unmerged entity keeps its persistent identifier.
Pipeline/GENERAL/tests/test_node_pipeline.py:406	NodePipelineTests.reference_fixture	Regression check for reference fixture.
Pipeline/GENERAL/tests/test_node_pipeline.py:430	NodePipelineTests.test_context_crosses_paragraphs_but_not_sections	Regression check for context crosses paragraphs but not sections.
Pipeline/GENERAL/tests/test_node_pipeline.py:442	NodePipelineTests.test_distinct_authors_in_one_metadata_field_are_not_deduplicated	Regression check for distinct authors in one metadata field are not deduplicated.
Pipeline/GENERAL/tests/test_node_pipeline.py:443	NodePipelineTests.test_distinct_authors_in_one_metadata_field_are_not_deduplicated.author	Regression check for author.
Pipeline/GENERAL/tests/test_node_pipeline.py:452	NodePipelineTests.test_vague_model_is_not_a_node_candidate	Regression check for vague model is not a node candidate.
Pipeline/GENERAL/tests/test_node_pipeline.py:460	NodePipelineTests.test_unique_compatible_vague_reference_is_resolved_to_existing_node	Regression check for unique compatible vague reference is resolved to existing node.
Pipeline/GENERAL/tests/test_node_pipeline.py:472	NodePipelineTests.test_contextual_llm_same_maps_vague_mention_to_specific_node	Regression check for contextual llm same maps vague mention to specific node.
Pipeline/GENERAL/tests/test_node_pipeline.py:477	NodePipelineTests.test_contextual_llm_same_maps_vague_mention_to_specific_node.SameJudge	Regression check for SameJudge.
Pipeline/GENERAL/tests/test_node_pipeline.py:478	NodePipelineTests.test_contextual_llm_same_maps_vague_mention_to_specific_node.SameJudge.__init__	Regression check for init.
Pipeline/GENERAL/tests/test_node_pipeline.py:485	NodePipelineTests.test_contextual_llm_same_maps_vague_mention_to_specific_node.SameJudge.verify_runtime	Regression check for verify runtime.
Pipeline/GENERAL/tests/test_node_pipeline.py:488	NodePipelineTests.test_contextual_llm_same_maps_vague_mention_to_specific_node.SameJudge.judge	Regression check for judge.
Pipeline/GENERAL/tests/test_node_pipeline.py:513	NodePipelineTests.test_contextual_llm_uncertain_keeps_pair_for_human_review	Regression check for contextual llm uncertain keeps pair for human review.
Pipeline/GENERAL/tests/test_node_pipeline.py:519	NodePipelineTests.test_contextual_llm_uncertain_keeps_pair_for_human_review.UncertainJudge	Regression check for UncertainJudge.

Source	Destination	Notes
Pipeline/GENERAL/tests/test_node_pipeline.py:522	NodePipelineTests.test_contextual_llm_uncertain_keeps_pair_for_human_review.UncertainJudge.verify_runtime	Regression check for verify runtime.
Pipeline/GENERAL/tests/test_node_pipeline.py:525	NodePipelineTests.test_contextual_llm_uncertain_keeps_pair_for_human_review.UncertainJudge.judge	Regression check for judge.
Pipeline/GENERAL/tests/test_node_pipeline.py:541	NodePipelineTests.test_ambiguous_vague_reference_requires_human_review	Regression check for ambiguous vague reference requires human review.
Pipeline/GENERAL/tests/test_node_pipeline.py:550	NodePipelineTests.test_weak_definite_reference_is_not_automatically_resolved	Regression check for weak definite reference is not automatically resolved.
Pipeline/GENERAL/tests/test_node_pipeline.py:557	NodePipelineTests.test_agentic_system_reference_prefers_ai_system_over_nearby_tool	Regression check for agentic system reference prefers ai system over nearby tool.
Pipeline/GENERAL/tests/test_node_pipeline.py:593	NodePipelineTests.test_unresolvable_vague_reference_is_ignored	Regression check for unresolvable vague reference is ignored.
Pipeline/GENERAL/tests/test_node_pipeline.py:598	NodePipelineTests.test_human_reference_choice_adds_traceable_mention_without_aliases_pollution	Regression check for human reference choice adds traceable mention without alias pollution.
Pipeline/GENERAL/tests/test_node_pipeline.py:617	NodePipelineTests.test_different_explicit_versions_are_not_duplicate_candidates	Regression check for different explicit versions are not duplicate candidates.
Pipeline/GENERAL/tests/test_node_pipeline.py:625	NodePipelineTests.test_lead_verb_is_not_a_contaminant	Regression check for lead verb is not a contaminant.
Pipeline/GENERAL/tests/test_node_pipeline.py:630	NodePipelineTests.test_epanet_is_a_tool_not_a_standard	Regression check for epanet is a tool not a standard.
Pipeline/GENERAL/tests/test_node_pipeline.py:636	NodePipelineTests.test_possessive_system_names_do_not_create_fragment_nodes	Regression check for possessive system names do not create fragment nodes.
Pipeline/GENERAL/tests/test_node_pipeline.py:643	NodePipelineTests.test_named_agents_and_models_are_specific_candidates	Regression check for named agents and models are specific candidates.
Pipeline/GENERAL/tests/test_node_pipeline.py:652	NodePipelineTests.test_challenge_inflections_share_one_canonical_name	Regression check for challenge inflections share one canonical name.
Pipeline/GENERAL/tests/test_question_evaluation.py:10	QuestionEvaluationTests	Regression check for QuestionEvaluationTests.
Pipeline/GENERAL/tests/test_question_evaluation.py:11	QuestionEvaluationTests.test_question_requires_graph_fact_and_exact_evidence	Regression check for question requires graph fact and exact evidence.
Pipeline/GENERAL/tests/test_relationship_pipeline.py:13	RelationshipPipelineTests	Regression check for RelationshipPipelineTests.
Pipeline/GENERAL/tests/test_relationship_pipeline.py:15	RelationshipPipelineTests.parsed	Regression check for parsed.
Pipeline/GENERAL/tests/test_relationship_pipeline.py:24	RelationshipPipelineTests.test_quantitative_result_creates_condition_aware_observation	Regression check for quantitative result creates condition aware observation.
Pipeline/GENERAL/tests/test_relationship_pipeline.py:37	RelationshipPipelineTests.test_figure_number_is_not_misread_as_hour_measurement	Regression check for figure number is not misread as hour measurement.
Pipeline/GENERAL/tests/test_relationship_pipeline.py:41	RelationshipPipelineTests.test_existing_observation_is_not_duplicated	Regression check for existing observation is not duplicated.
Pipeline/GENERAL/tests/test_relationship_pipeline.py:51	RelationshipPipelineTests.test_canonical_relationship_aggregates_distinct_assertion_support	Regression check for canonical relationship aggregates distinct assertion support.

Location	Description	Notes
Pipeline/GENERAL/tests/test_required_ner.py:8	RequiredNERTests	Regression check for RequiredNERTests.
Pipeline/GENERAL/tests/test_required_ner.py:10	RequiredNERTests.parsed	Regression check for parsed.
Pipeline/GENERAL/tests/test_required_ner.py:29	RequiredNERTests.config	Regression check for config.
Pipeline/GENERAL/tests/test_required_ner.py:38	RequiredNERTests.test_required_ner_materializes_exact_span_audit	Regression check for required ner materializes exact span audit.
Pipeline/GENERAL/tests/test_required_ner.py:55	RequiredNERTests.test_untrusted_generator_cannot_bypass_adjudication	Regression check for untrusted generator cannot bypass adjudication.
Pipeline/GENERAL/tests/test_required_ner.py:70	RequiredNERTests.test_required_ner_cannot_be_disabled	Regression check for required ner cannot be disabled.
Pipeline/GENERAL/tests/test_required_ner.py:76	RequiredNERTests.test_required_ner_blocks_suspicious_early_parse_cutoff	Regression check for required ner blocks suspicious early parse cutoff.
Pipeline/GENERAL/tests/test_sector_discovery.py:28	source_rows	Regression check for source rows.
Pipeline/GENERAL/tests/test_sector_discovery.py:33	SectorDiscoveryTests	Regression check for SectorDiscoveryTests.
Pipeline/GENERAL/tests/test_sector_discovery.py:34	SectorDiscoveryTests.flatten	Regression check for flatten.
Pipeline/GENERAL/tests/test_sector_discovery.py:38	SectorDiscoveryTests.test_descriptive_systems_cluster_and_spaced_agents_propagate	Regression check for descriptive systems cluster and spaced agents propagate.
Pipeline/GENERAL/tests/test_sector_discovery.py:51	SectorDiscoveryTests.test_false_models_are_rejected_but_enumerated_models_are_kept	Regression check for false models are rejected but enumerated models are kept.
Pipeline/GENERAL/tests/test_sector_discovery.py:58	SectorDiscoveryTests.test_profile_types_and_counted_datasets_are_canonicalized	Regression check for profile types and counted datasets are canonicalized.
Pipeline/GENERAL/tests/test_semantic_completeness.py:14	SemanticCompletenessTests	Regression check for SemanticCompletenessTests.
Pipeline/GENERAL/tests/test_semantic_completeness.py:15	SemanticCompletenessTests.test_missing_expected_core_nodes_is_reported_as_incomplete	Regression check for missing expected core nodes is reported as incomplete.
Pipeline/GENERAL/tests/test_semantic_completeness.py:25	SemanticCompletenessTests.test_grounded_connected_core_with_traceability_is_complete	Regression check for grounded connected core with traceability is complete.
Pipeline/GENERAL/tools/evaluate_fixed_corpus_run.py:17	main	Provides the command-line or server entry point.
Pipeline/GENERAL/tools/evaluate_regression_corpus.py:15	read_json	Loads serialized input.
Pipeline/GENERAL/tools/evaluate_regression_corpus.py:19	read_jsonl	Loads serialized input.
Pipeline/GENERAL/tools/evaluate_regression_corpus.py:25	normalized	Implements the internal normalized operation.
Pipeline/GENERAL/tools/evaluate_regression_corpus.py:29	latest_run	Implements the internal latest run operation.
Pipeline/GENERAL/tools/evaluate_regression_corpus.py:41	evaluate_case	Computes a result from already-materialized evidence.
Pipeline/GENERAL/tools/evaluate_regression_corpus.py:95	main	Provides the command-line or server entry point.

Source	Destination	Notes
Pipeline/GENERAL/tools/export_node_results_report.py:8	read_jsonl	Loads serialized input.
Pipeline/GENERAL/tools/export_node_results_report.py:12	cell	Implements the internal cell operation.
Pipeline/GENERAL/tools/export_node_results_report.py:16	main	Provides the command-line or server entry point.
Pipeline/GENERAL/tools/rematerialize_node_run.py:23	internal_candidate	Implements the internal internal candidate operation.
Pipeline/GENERAL/tools/rematerialize_node_run.py:45	RecordedJudge	Groups the state and behavior for RecordedJudge.
Pipeline/GENERAL/tools/rematerialize_node_run.py:46	RecordedJudge.__init__	Implements the internal init operation.
Pipeline/GENERAL/tools/rematerialize_node_run.py:54	RecordedJudge.verify_runtime	Implements the internal verify runtime operation.
Pipeline/GENERAL/tools/rematerialize_node_run.py:57	RecordedJudge.judge	Implements the internal judge operation.
Pipeline/GENERAL/tools/rematerialize_node_run.py:85	main	Provides the command-line or server entry point.
Pipeline/GENERAL/tools/run_waterrag_quick_test.py:16	main	Provides the command-line or server entry point.
Pipeline/GENERAL/tools/run_waterrag_quick_test.py:21	main.progress	Implements the internal progress operation.
Pipeline/GENERAL/tools/smoke_test_deepseek.py:19	main	Provides the command-line or server entry point.
Pipeline/GENERAL/tools/verify_corpus.py:15	main	Provides the command-line or server entry point.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:25	PipelineUnitTests	Regression check for PipelineUnitTests.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:27	PipelineUnitTests.setUpClass	Regression check for setUpClass.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:31	PipelineUnitTests.test_llm_judge_is_mandatory_without_credential	Regression check for llm judge is mandatory without credential.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:38	PipelineUnitTests.test_ollama_provider_requires_no_api_key	Regression check for ollama provider requires no api key.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:46	PipelineUnitTests.test_ollama_node_judge_expands_context_only_for_oversized_payloads	Regression check for ollama node judge expands context only for oversized payloads.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:65	PipelineUnitTests.test_ollama_node_judge_splits_candidate_heavy_sentence_and_merges_audits	Regression check for ollama node judge splits candidate heavy sentence and merges audits.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:75	PipelineUnitTests.test_ollama_node_judge_splits_candidate_heavy_sentence_and_merges_audits.transport	Regression check for transport.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:118	PipelineUnitTests.test_reference_judge_sends_contextual_pair_and_returns_same	Regression check for reference judge sends contextual pair and returns same.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:123	PipelineUnitTests.test_reference_judge_sends_contextual_pair_and_returns_same.transport	Regression check for transport.

File Name	Test Name	Description
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:159	PipelineUnitTests.test_lexicon_contains_only_specific_anchors	Regression check for lexicon contains only specific anchors.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:173	PipelineUnitTests.test_llm_judges_candidates_and_discovers_out_of_lexicon_node	Regression check for llm judges candidates and discovers out of lexicon node.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:174	PipelineUnitTests.test_llm_judges_candidates_and_discovers_out_of_lexicon_node.transport	Regression check for transport.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:243	PipelineUnitTests.test_missing_llm_candidate_judgment_follows_reject_policy	Regression check for missing llm candidate judgment follows reject policy.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:244	PipelineUnitTests.test_missing_llm_candidate_judgment_follows_reject_policy.incomplete_transport	Regression check for incomplete transport.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:262	PipelineUnitTests.test_llm_discovery_must_match_exact_source_span	Regression check for llm discovery must match exact source span.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:263	PipelineUnitTests.test_llm_discovery_must_match_exact_source_span.invented_span_transport	Regression check for invented span transport.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:290	PipelineUnitTests.test_llm_invalid_span_can_be_configured_to_fail_closed	Regression check for llm invalid span can be configured to fail closed.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:291	PipelineUnitTests.test_llm_invalid_span_can_be_configured_to_fail_closed.invented_span_transport	Regression check for invented span transport.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:315	PipelineUnitTests.test_llm_real_text_with_wrong_offset_is_aligned_and_audited	Regression check for llm real text with wrong offset is aligned and audited.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:316	PipelineUnitTests.test_llm_real_text_with_wrong_offset_is_aligned_and_audited.wrong_offset_transport	Regression check for wrong offset transport.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:343	PipelineUnitTests.test_stable_id_is_deterministic	Regression check for stable id is deterministic.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:346	PipelineUnitTests.test_missing_llm_candidate_judgment_is_quarantined_for_review	Regression check for missing llm candidate judgment is quarantined for review.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:350	PipelineUnitTests.test_missing_llm_candidate_judgment_is_quarantined_for_review.transport	Regression check for transport.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:384	PipelineUnitTests.test_malformed_structured_json_is_retrieved	Regression check for malformed structured json is retrieved.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:389	PipelineUnitTests.test_malformed_structured_json_is_retrieved.transport	Regression check for transport.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:420	PipelineUnitTests.test_name_normalization	Regression check for name normalization.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:423	PipelineUnitTests.test_domain_range_accepts_subclass	Regression check for domain range accepts subclass.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:427	PipelineUnitTests.test_domain_range_rejects_invalid_direction	Regression check for domain range rejects invalid direction.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:431	PipelineUnitTests._relationship_fixture	Regression check for relationship fixture.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:453	PipelineUnitTests.test_edge_extractor_uses_llm_selected_existing_nodes_and_all_context	Regression check for edge extractor uses llm selected existing nodes and all context.

Source	Destination	Notes
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:454	PipelineUnitTests.test_edge_extractor_uses_llm_selected_existing_nodes_and_all_context.ContextJudge	Regression check for ContextJudge.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:457	PipelineUnitTests.test_edge_extractor_uses_llm_selected_existing_nodes_and_all_context.ContextJudge.extract	Regression check for extract.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:490	PipelineUnitTests.test_edge_extractor_audits_unseen_endpoint_without_creating_edge	Regression check for edge extractor audits unseen endpoint without creating edge.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:491	PipelineUnitTests.test_edge_extractor_audits_unseen_endpoint_without_creating_edge.UnknownNodeJudge	Regression check for UnknownNodeJudge.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:494	PipelineUnitTests.test_edge_extractor_audits_unseen_endpoint_without_creating_edge.UnknownNodeJudge.extract	Regression check for extract.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:522	PipelineUnitTests.test_relationship_llm_selects_only_candidate_permitted_predicate	Regression check for relationship llm selects only candidate permitted predicate.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:523	PipelineUnitTests.test_relationship_llm_selects_only_candidate_permitted_predicate.transport	Regression check for transport.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:544	PipelineUnitTests.test_resolution_never_auto_merges	Regression check for resolution never auto merges.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:545	PipelineUnitTests.test_resolution_never_auto_merges.FakeEmbeddingBackend	Regression check for FakeEmbeddingBackend.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:549	PipelineUnitTests.test_resolution_never_auto_merges.FakeEmbeddingBackend.encode	Regression check for encode.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:575	PipelineUnitTests.test_embedding_resolution_finds_semantic_match_without_shared_name_tokens	Regression check for embedding resolution finds semantic match without shared name tokens.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:576	PipelineUnitTests.test_embedding_resolution_finds_semantic_match_without_shared_name_tokens.FakeEmbeddingBackend	Regression check for FakeEmbeddingBackend.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:580	PipelineUnitTests.test_embedding_resolution_finds_semantic_match_without_shared_name_tokens.FakeEmbeddingBackend.encode	Regression check for encode.
Pipeline/Knowledge_Graph_Core/tests/test_pipeline.py:608	PipelineUnitTests.test_review_queue_collapses_repeated_name_pairs	Regression check for review queue collapses repeated name pairs.
Pipeline/Paper_Parsing/tests/test_text_utils.py:20	TextUtilitiesTest	Regression check for TextUtilitiesTest.
Pipeline/Paper_Parsing/tests/test_text_utils.py:21	TextUtilitiesTest.test_dehyphenates_line_break	Regression check for dehyphenates line break.
Pipeline/Paper_Parsing/tests/test_text_utils.py:24	TextUtilitiesTest.test_preserves_real_compound_at_line_break	Regression check for preserves real compound at line break.
Pipeline/Paper_Parsing/tests/test_text_utils.py:27	TextUtilitiesTest.test_repairs_short_hyphenated_line_fragment	Regression check for repairs short hyphenated line fragment.
Pipeline/Paper_Parsing/tests/test_text_utils.py:30	TextUtilitiesTest.test_repairs_unicode_hyphenated_line_fragment	Regression check for repairs unicode hyphenated line fragment.
Pipeline/Paper_Parsing/tests/test_text_utils.py:33	TextUtilitiesTest.test_normalizes_grammatical_seams	Regression check for normalizes grammatical seams.

Source	Destination	Notes
Pipeline/Paper_Parsing/tests/test_text_utils.py:39	TextUtilitiesTest.test_normalizes_unicode_compound_hyphen	Regression check for normalizes unicode compound hyphen.
Pipeline/Paper_Parsing/tests/test_text_utils.py:42	TextUtilitiesTest.test_preserves_scientific_abbreviation	Regression check for preserves scientific abbreviation.
Pipeline/Paper_Parsing/tests/test_text_utils.py:46	TextUtilitiesTest.test_preserves_abbreviation_capitalization	Regression check for preserves abbreviation capitalization.
Pipeline/Paper_Parsing/tests/test_text_utils.py:50	TextUtilitiesTest.test_incomplete_fragment_is_not_a_sentence	Regression check for incomplete fragment is not a sentence.
Pipeline/Paper_Parsing/tests/test_text_utils.py:53	TextUtilitiesTest.test_layout_change_does_not_cut_unfinished_sentence	Regression check for layout change does not cut unfinished sentence.
Pipeline/Paper_Parsing/tests/test_text_utils.py:68	TextUtilitiesTest.test_citation_after_period_is_terminal	Regression check for citation after period is terminal.
Pipeline/Paper_Parsing/tests/test_text_utils.py:71	TextUtilitiesTest.test_citation_fragment_remains_in_sentence_coverage	Regression check for citation fragment remains in sentence coverage.
Pipeline/Paper_Parsing/tests/test_text_utils.py:75	TextUtilitiesTest.test_numbered_prose_is_not_a_heading	Regression check for numbered prose is not a heading.
Pipeline/Paper_Parsing/tests/test_text_utils.py:78	TextUtilitiesTest.test_styled_numbered_section_is_a_heading	Regression check for styled numbered section is a heading.
Pipeline/Paper_Parsing/tests/test_text_utils.py:81	TextUtilitiesTest.test_plain_numbered_section_is_a_heading	Regression check for plain numbered section is a heading.
Pipeline/Paper_Parsing/tests/test_text_utils.py:84	TextUtilitiesTest.test_numbered_prose_and_chart_labels_are_not_headings	Regression check for numbered prose and chart labels are not headings.
Pipeline/Paper_Parsing/tests/test_text_utils.py:90	TextUtilitiesTest.test_equation_fragment_is_not_narrative	Regression check for equation fragment is not narrative.
Pipeline/Paper_Parsing/tests/test_text_utils.py:93	TextUtilitiesTest.test_cid_formula_encoding_is_detected	Regression check for cid formula encoding is detected.
Pipeline/Paper_Parsing/tests/test_text_utils.py:96	TextUtilitiesTest.test_short_scientific_sentence_is_narrative	Regression check for short scientific sentence is narrative.
Pipeline/Paper_Parsing/tests/test_text_utils.py:99	TextUtilitiesTest.test_detects_acs_est_profile	Regression check for detects acs est profile.
Pipeline/Paper_Parsing/tests/test_text_utils.py:103	TextUtilitiesTest.test_detects_water_research_profile	Regression check for detects water research profile.
Pipeline/Paper_Parsing/tests/test_text_utils.py:107	TextUtilitiesTest.test_detects_frontiers_profile	Regression check for detects frontiers profile.
Pipeline/Paper_Parsing/tests/test_text_utils.py:111	TextUtilitiesTest.test_detects_ieee_arxiv_profile_and_repairs_small_caps	Regression check for detects ieee arxiv profile and repairs small caps.
Pipeline/Paper_Parsing/tests/test_text_utils.py:116	TextUtilitiesTest.test_detects_all_caps_conference_preprint_profile	Regression check for detects all caps conference preprint profile.
Pipeline/Paper_Parsing/tests/test_text_utils.py:124	TextUtilitiesTest.test_recognizes_numbered_introduction_heading	Regression check for recognizes numbered introduction heading.
Pipeline/Paper_Parsing/tests/test_text_utils.py:130	TextUtilitiesTest.test_numbered_references_heading_is_back_matter	Regression check for numbered references heading is back matter.
Pipeline/Paper_Parsing/tests/test_text_utils.py:133	TextUtilitiesTest.test_reference_table_header_before_midpoint_is_not_back_matter	Regression check for reference table header before midpoint is not back matter.
Pipeline/Paper_Parsing/tests/test_text_utils.py:137	TextUtilitiesTest.test_late_bold_references_is_back_matter	Regression check for late bold references is back matter.

Source	Description	Notes
Pipeline/Paper_Parsing/tests/test_text_utils.py:141	TextUtilitiesTest.test_column_gap_splits_words_before_line_interleaving	Regression check for column gap splits words before line interleaving.
Pipeline/Paper_Parsing/tests/test_text_utils.py:150	TextUtilitiesTest.test_cleans_preprint_author_markers	Regression check for cleans preprint author markers.
Pipeline/Paper_Parsing/tests/test_text_utils.py:158	TextUtilitiesTest.test_bold_preprint_front_matter_fallback_and_reference_stop	Regression check for bold preprint front matter fallback and reference stop.
Pipeline/Paper_Parsing/tests/test_text_utils.py:175	TextUtilitiesTest.test_wrapped_smaller_font_author_line_is_retained	Regression check for wrapped smaller font author line is retained.
Pipeline/Paper_Parsing/tests/test_text_utils.py:189	TextUtilitiesTest.test_same_baseline_author_fragments_are_joined	Regression check for same baseline author fragments are joined.
Pipeline/Paper_Parsing/tests/test_text_utils.py:204	TextUtilitiesTest.test_affiliation_lines_are_not_appended_to_authors	Regression check for affiliation lines are not appended to authors.
Pipeline/Paper_Parsing/tests/test_text_utils.py:219	TextUtilitiesTest.test_first_page_introduction_is_retained_after_abstract	Regression check for first page introduction is retained after abstract.
Pipeline/Paper_Parsing/tests/test_text_utils.py:234	TextUtilitiesTest.test_unlabelled_water_research_introduction_is_synthesized	Regression check for unlabelled water research introduction is synthesized.
Pipeline/Paper_Parsing/tests/test_text_utils.py:251	TextUtilitiesTest.test_centered_abstract_body_is_retained	Regression check for centered abstract body is retained.
Pipeline/Paper_Parsing/tests/test_text_utils.py:263	TextUtilitiesTest.test_frontiers_unlabelled_abstract_is_synthesized	Regression check for frontiers unlabelled abstract is synthesized.
Pipeline/Paper_Parsing/tests/test_text_utils.py:276	TextUtilitiesTest.test_wrapped_styled_heading_is_joined	Regression check for wrapped styled heading is joined.
Pipeline/Paper_Parsing/tests/test_text_utils.py:285	TextUtilitiesTest.test_bare_number_is_not_a_profile_heading	Regression check for bare number is not a profile heading.
Pipeline/Paper_Parsing/tests/test_text_utils.py:288	TextUtilitiesTest.test_body_line_with_changed_style_does_not_extend_heading	Regression check for body line with changed style does not extend heading.
Pipeline/Paper_Parsing/tests/test_text_utils.py:296	TextUtilitiesTest.test_acs_inline_subsection_body_is_split_and_retained	Regression check for acs inline subsection body is split and retained.
Pipeline/Paper_Parsing/tests/test_text_utils.py:308	TextUtilitiesTest.test_author_selection_prefers_names_over_email_and_banner	Regression check for author selection prefers names over email and banner.
Pipeline/Paper_Parsing/tests/test_text_utils.py:320	TextUtilitiesTest.test_author_selection_joins_columns_and_wrapped_surnames	Regression check for author selection joins columns and wrapped surnames.
Pipeline/Paper_Parsing/tests/test_text_utils.py:345	TextUtilitiesTest.test_author_selection_separates_full_names_on_wrapped_lines	Regression check for author selection separates full names on wrapped lines.
Pipeline/Paper_Parsing/tests/test_text_utils.py:357	TextUtilitiesTest.test_infers_ruled_column_table	Regression check for infers ruled column table.
Pipeline/Paper_Parsing/tests/test_text_utils.py:368	TextUtilitiesTest.test_builds_section_hierarchy_and_inherits_pages	Regression check for builds section hierarchy and inherits pages.
Pipeline/Paper_Parsing/tools/audit_corpus.py:26	_sha256	Implements the internal sha256 operation.
Pipeline/Paper_Parsing/tools/audit_corpus.py:34	_safe_stem	Implements the internal safe stem operation.
Pipeline/Paper_Parsing/tools/audit_corpus.py:39	_flatten_paragraphs	Implements the internal flatten paragraphs operation.
Pipeline/Paper_Parsing/tools/audit_corpus.py:47	_issues	Implements the internal issues operation.
Pipeline/Paper_Parsing/tools/audit_corpus.py:101	main	Provides the command-line or server entry point.

Source	Definition	Intent
Pipeline/Paper_Parsing/tools/audit_json.py:11	main	Provides the command-line or server entry point.
Pipeline/Paper_Parsing/tools/inspect_pdf_lines.py:19	main	Provides the command-line or server entry point.

20. Fast index: module ownership

Pipeline/Edge_Extraction/common.py	2	relationship windows, observations, canonical edge materialization
Pipeline/Edge_Extraction/extractor.py	1	relationship windows, observations, canonical edge materialization
Pipeline/Edge_Extraction/materialization.py	1	relationship windows, observations, canonical edge materialization
Pipeline/Edge_Extraction/observation_mentions.py	1	relationship windows, observations, canonical edge materialization
Pipeline/GENERAL/node_only_server.py	3	workbench HTTP server, static UI, configuration
Pipeline/GENERAL/run.py	16	workbench HTTP server, static UI, configuration
Pipeline/Grammatical_Parsing/candidates.py	22	spaCy analysis, SVO triples, SciBERT typing
Pipeline/Graph_Persistence/neo4j_writer.py	12	ontology validation and Neo4j upsert
Pipeline/Knowledge_Graph_Core/edge_extractor.py	10	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/embeddings.py	3	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/entity_resolution.py	9	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/llm_node_judge.py	16	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/llm_reference_judge.py	7	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/llm_relationship_judge.py	9	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/models.py	7	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/node_extractor.py	15	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/ontology.py	6	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/pipeline.py	3	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/review_queues.py	1	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/similarity.py	6	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Knowledge_Graph_Core/validation.py	1	shared judges/models, alternate batch pipeline, resolver and validation
Pipeline/Node_Pipeline/adjudication.py	4	candidate boundary, adjudication, references, canonicalization, review rebuilds
Pipeline/Node_Pipeline/candidate_generation.py	6	candidate boundary, adjudication, references, canonicalization, review rebuilds

Module	Dependencies	Pipeline responsibility
Pipeline/Node_Pipeline/canonicalization.py	8	candidate boundary, adjudication, references, canonicalization, review rebuilds
Pipeline/Node_Pipeline/common.py	9	candidate boundary, adjudication, references, canonicalization, review rebuilds
Pipeline/Node_Pipeline/contracts.py	3	candidate boundary, adjudication, references, canonicalization, review rebuilds
Pipeline/Node_Pipeline/lexicon.py	9	candidate boundary, adjudication, references, canonicalization, review rebuilds
Pipeline/Node_Pipeline/ner_boundary.py	1	candidate boundary, adjudication, references, canonicalization, review rebuilds
Pipeline/Node_Pipeline/node_only_runner.py	1	candidate boundary, adjudication, references, canonicalization, review rebuilds
Pipeline/Node_Pipeline/orchestrator.py	2	Public entry point or environment check
Pipeline/Node_Pipeline/reference_resolution.py	8	candidate boundary, adjudication, references, canonicalization, review rebuilds
Pipeline/Node_Pipeline/review_workflows.py	3	candidate boundary, adjudication, references, canonicalization, review rebuilds
Pipeline/Node_Pipeline/span_detection.py	7	candidate boundary, adjudication, references, canonicalization, review rebuilds
Pipeline/Paper_Parsing/models.py	2	PDF layout, narrative hierarchy, source evidence
Pipeline/Paper_Parsing/pdf_parser.py	6	PDF layout, narrative hierarchy, source evidence
Pipeline/Paper_Parsing/pipeline.py	21	PDF layout, narrative hierarchy, source evidence
Pipeline/Paper_Parsing/profiles.py	5	PDF layout, narrative hierarchy, source evidence
Pipeline/Paper_Parsing/text_utils.py	9	PDF layout, narrative hierarchy, source evidence
Pipeline/Question_Evaluation/evaluator.py	2	graph-only answerability
Pipeline/Sector_Discovery/discovery.py	16	document-grounded exact-span candidate discovery
Pipeline/Validation/generalization_diagnostics.py	5	semantic completeness and generalization diagnostics
Pipeline/Validation/semantic_completeness.py	3	semantic completeness and generalization diagnostics
Pipeline/check_environment.py	4	Public entry point or environment check

You do not need to memorize 300 functions.

Own the boundaries: evidence -> candidate -> judgment -> mention -> entity; assertion -> relationship; review -> deterministic rebuild; validation -> ontology-safe persistence.



Use the source references in this guide to answer the next question by tracing the actual code, not by guessing.